OCS
V100R002C02LG0253

Call Detail Record Reference

Issue 11

Date 2015-10-14

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About This Document

Purpose

This document describes the CDRs that generated by the OCS service.

Intended Audience

This document is intended for:

- Service operators
- Debugging engineers
- Maintenance engineers

Symbol Conventions

The symbols that may be found in this document are defined as follows.

Symbol	Description
DANGER	Indicates an imminently hazardous situation which, if not avoided, will result in death or serious injury.
WARNING	Indicates a potentially hazardous situation which, if not avoided, could result in death or serious injury.
CAUTION	Indicates a potentially hazardous situation which, if not avoided, may result in minor or moderate injury.
NOTICE	Indicates a potentially hazardous situation which, if not avoided, could result in equipment damage, data loss, performance deterioration, or unanticipated results. NOTICE is used to address practices not related to personal injury.
NOTE	Calls attention to important information, best practices and tips. NOTE is used to address information not related to personal injury, equipment damage, and environment

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Symbol	Description
	deterioration.

Change History

Issue 11 (2015-10-14)

The ninth release of OCS V100R002C02LG0253. Compared with the previous version, update in this version is as follows:

Added the fields **PDN Connection Id**, **Serving-Node-Type**, **Start-Time**, **Stop-Time** and **Diagnostics** for **data CDR** under **CBP Generated CDR** .

Issue 10(2015-05-07)

The ninth release of OCS V100R002C02LG0248. Compared with the previous version, update in this version is as follows:

Added the **OperatorId** for **mon CDR** under **CBP Generated CDR** .

Issue 09 (2015-02-12)

The ninth release of OCS V100R002C02LG0244. Compared with the previous version, update in this version is as follows:

Added the **BalExt CDR** under **CBP Generated CDR** .

Added the **OperateResult** 59 in **w_kpi** CDR.

Issue 08 (2014-12-05)

The eighth release of OCS V100R002C02LG0241. Compared with the previous version, update in this version is as follows:

Description of **CallingCellId** is modified in **VOU** CDR.

Issue 07 (2014-10-17)

The seventh release of OCS V100R002C02LG0240. Compared with the previous version, update in this version is as follows:

Added the **DTDiscount** and **DTCellId** fields in **REC**, **SMS**, and **DATA** CDR.

Issue 06 (2014-07-10)

The sixth release of OCS V100R002C02LG0236SPC002. Compared with the previous version, update in this version is as follows:

Added the **abvou** CDR.

Issue 05 (2014-01-21)

The fifth release of OCS V100R002C02LG0236. Compared with the previous version, update in this version is as follows:

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Add the **OperateResult** 49, 50 ,51, 52 and 53 in **w_kpi** CDR.

Issue 04 (2013-11-30)

The fourth release of OCS V100R002C02LG0235. Compared with the previous version, update in this version is as follows:

Modify the value range from **Char[20]** to **Char[24]** of **SMSID** parameter in **sms** CDR.

Issue 03 (2013-10-15)

The third release of OCS V100R002C02LG0233SPC007. Compared with the previous version, update in this version is as follows:

Add the field **DDSLocationNumber** in **rec** CDR.

Issue 02 (2013-08-30)

The second release of OCS V100R002C02LG0233SPC001. Compared with the previous version, updates in this version are as follows:

- Change the generation scenario in the **clr** CDR.
- Delete the note for **CallDuration** parameter in **w_abr**, **rec** and **w_rec** CDRs.
- Delete the value range for **IMEI** parameter in **rec**, **sms**, **mms** and **data** CDRs.

Issue 01 (2013-07-30)

The first release of OCS V100R002C02LG0233.

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1 CDR Overview

About This Chapter

This section describes the basic concepts and terms related to call detail records (CDRs). System generates CDR instantly at the end of the service usage.

The following lists the contents of this chapter.

1.1 Basic Concepts

This section describes the basic concepts and terms related to call detail records (CDRs).

1.2 CDR Storage Directories

The CDR storage directories indicate where a CDR generation node such as the CBP stores CDR files. It is recommended to store CDRs in different directories according to the CDR generation condition.

1.3 Maximum CDR File Size

The maximum CDR file size is used to limit the size of CDR files generated by the SCP and CBP CDR Generation nodes.

1.4 CDR-Processing Mechanism

The mechanism for processing CDRs and the related components.

1.5 CDR Types

This section describes the types of CDRs generated by the CBP, the SCP and the MDSP in the OCS service, and the service scenarios of each type of CDRs.

1.6 CDR Types in Service Application Scenarios

The section describes the types of CDRs generated in service scenarios.

1.7 Conventions

The section describes the conventions in this document.

1.1 Basic Concepts

This section describes the basic concepts and terms related to call detail records (CDRs).

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OCS

The online charging system (OCS) implements the convergence of the following items: Online charging and offline charging Voice service charging and data service charging Prepaid subscriber management and postpaid subscriber management. The OCS provides functions such as rating, subscriber balance management, and real-time monitor. The OCS is composed of the convergent billing point (CBP), business management point (BMP), and service control point (SCP).

CBP

The CBP is designed to provide the following capabilities:

- Prepaid and convergent charging capability of multiple networks and services
- CDR processing
- Configuring and managing charging data

After the charging is complete, the CBP system generates the CDR.

SCP

In the OCS system, the SCP provides call access and call control functions.

In the OCS system, the SCP provides service control and service processing functions, including:

- Call access and signaling control
- Service logic management
- Basic data operation and processing
- Data verification
- Operation maintenance management

The SCP is connected to the BMP for data synchronization. The SCP communicates with the uniform voucher center (UVC) for recharging. The SCP communicates with the intelligent peripheral (IP) to obtain the voice resources needed by the subscriber terminal. In addition, the SCP is connected to the CBP to send messages to the CBP for charging and rating. The SCP provides the interactive voice response (IVR) process through which subscribers can recharge accounts, query for balances, and change passwords.

CDR

A CDR refers to the information about a charging event collected in a certain format. The information includes call start time, call duration, and transferred data amount. The information is used for charging and settlement. It is also provided for subscribers as the service usage records.

The system generates a separate CDR for each party of a charging event. In this case, the system does not consider whether the party pays for a partial charge or the entire charge. In other words, a charging event can have one or more CDRs, for example, when the call duration is long or when there is more than one charged party.

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RBI

The recorded bill interface (RBI) is the channel for transferring files between the CBP and an external element. The RBI can split, compress, and filter collected CDR files. It can also transfer CDRs of the CBP to the charging center.

Mediation

The Mediation collects CDRs generated by each network elements. In the OCS service, the Mediation preprocesses the CDR files transferred by the RBI and uploads them to the Billing system.

Billing

In the OCS service, the Mediation preprocesses the CDR files transferred by the RBI and uploads them to the Billing system for CDR sorting, storage, accumulation, and summary.

CDR File Name Prefix

A CDR file name prefix identifies a CDR file corresponding to a type of charging events. For example, for a CBP-generated CDR, the system administrator can define conditions for classifying charge CDRs and the mapping CDR file name prefix in the CBP.

The following takes the rec CDR on the CBP to describe the naming format of the CDR prefixes on the CBP:

- The prefix of the normal CDRs of prepaid subscribers is rec.
- The prefix of the failure CDRs is f_rec or f_orec.
- The prefix of the error CDRs is e_rec or e_orec.
- The prefix of the CDRs of test subscribers is trec, regardless of the subscriber type (prepaid) and processing result (normal, failure, or error).

NOTE

This document uses the prefixes of the normal CDRs of prepaid subscribers as examples to describe the formats of the CDRs generated by the CBP.

The naming format of the CDRs generated by the SCP is different from the naming format of the CDRs generated by the CBP.

CDR File Name Suffix

A CDR file name suffix is associated with the file format in which the CDR file is stored.

For example, the file name suffix of a CBP-generated CDR can be configured based on the service requirement. The suffix can have a maximum of seven characters.

Offline CDR

A charging mechanism in which the charging information does not affect the service providing in real time. The offline charging is based on CDR charging, that is, the system performs rating and fee deduction based on the collected original call records of subscribers.

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Online CDR

A charging mechanism in which the account balance is monitored online and in real time. When the account balance of a subscriber is used up, the system prevents the subscriber from using services. In this case, the risk of owing fees is reduced.

1.2 CDR Storage Directories

The CDR storage directories indicate where a CDR generation node such as the CBP stores CDR files. It is recommended to store CDRs in different directories according to the CDR generation condition.

1.2.1 CBP-generated CDR storage directories

CDR storage directories

The directory for storing general CDRs must be configured based on the actual situation of the current network. The path to CDRs varies according to subscribers. The default directories are as follows:

- Prepaid subscribers
  - Path to normal CDRs: **${CBP_CDRPATH}/output/pps/normal**
  - Path to failure CDRs: **${CBP_CDRPATH}/output/pps/fail**
  - Path to error CDRs: **${CBP_CDRPATH}/output/pps/error**
- Test subscribers

  Path to various CDRs: **${CBP_CDRPATH}/output/test**. All the normal, failure, and error CDRs of test subscribers are stored in this path.

The temp and bak directories are created in the preceding paths to store the temporary and backup CDRs respectively. For example, **${CBP_CDRPATH}/output/pps/normal/temp** and **${CBP_CDRPATH}/output/pps/normal/bak**.

NOTE

- Temporary directory: It is the directory for storing temporary CDR files before the CDR files meet the switching conditions.
- Backup directory: After obtaining a CDR from the destination directory where the CBP stores CDRs, the RBI backs up the CDR in the backup directory.

Paths to Offline CDRs for Re-rating

The paths to the offline CDRs for re-rating are as follows:

- Path to international roaming offline CDRs: **${CBP_CDRPATH}/input/ir**

  The **voice**, **sms** and **data** directories in this path contain the subdirectories of **src**, **bak** and **err**.
- Path to SCP passby CDRs: **${CBP_CDRPATH}/input/scpabr**

  The **voice** and **data** directories in this path contain the subdirectories of **src**, **bak** and **err**.
- Input path to MDSP passby offline CDRs for re-rating: **${CBP_CDRPATH}/input/mdsp**

  The directories that are categorized by offline CDRs type in this path contain the subdirectories of **src**, **bak** and **err**.

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- Path to IMS offline CDRs for re-rating: **${CBP_CDRPATH}/input/ims**

  The **voice** and **data** directories in this path contain the subdirectories of **src**, **bak** and **err**.

Paths to CDRs Generated After Offline CDRs are Re-rated

The paths to the CDRs that are generated after offline CDRs are re-rated are as follows:

- Path to the CDRs that are generated after the international roaming offline CDRs are rated: **${CBP_CDRPATH}/output/offline/ir**

  The **voice**, **sms** and **data** directories in this path contain the **temp** and **bak** subdirectories.
- Path to the CDRs that are generated after the SCP passby CDRs are re-rated: **${CBP_CDRPATH}/output/offline/scpabr**

  The **voice** and **data** directories in this path contain the **temp** and **bak** subdirectories.
- Path to the CDRs that are generated after the MDSP passby offline CDRs are re-rated: **${CBP_CDRPATH}/output/offline/mdsp**

  The directories that are categorized by data service type in this path contain the **temp** and **bak** subdirectories.
- Path to the CDRs that are generated after the IMS offline CDRs are re-rated: **${CBP_CDRPATH}/output/offline/ims**

  The **temp** and **bak** directories exist in this path.

NOTE

The re-rating of offline CDRs has a special requirement. The ***.err** CDRs generated for the CDRs that fail to be re-rated are all stored in the **error** directory. No **fail** directory is involved.

Configuration for the storage directories and switching conditions of CBP-generated CDR

The directory for storing CBP-generated CDR files can be configured on CBP. Before specifying the directory for storing CDR files, you need to set the environment variables of the CDR directory: **${CBP_CDRPATH}**. Here, **${CBP_CDRPATH}** refers to the installation directory of the CBP and can be configured in the **.cshrc** file.

A CDR file is moved from the temporary directory to the destination directory as long as either of the following conditions is met:

- The CDR file size reaches the upper limit.
- The duration for writing the CDR file reaches the maximum duration.

For details, see 1.3 Maximum CDR File Size.

NOTE

The CDR files are switched at 00:00:00 without conditions.

1.2.2 SCP-generated CDR storage directories

CDR storage directories

The description of the SCP-generated CDR storage directories are as follows:

- Destination directory: **${TELLIN_DIR}/sms/record**

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It is the destination directory for storing CDR files after the CDR files meet the switching conditions.

- Temporary directory: **${TELLIN_DIR}/sms/record/recbuf**

  It is the directory for storing temporary CDR files before the CDR files meet the switching conditions.

- Backup directory: **${TELLIN_DIR}/sms/record/backup**

  After obtaining a CDR from the destination directory where the SCP stores CDRs, the system backs up the CDR in the backup directory. The CDR can be viewed to locate problems.

Here, **${TELLIN_DIR}** refers to the installation directory of the SCP.

Configuration for the storage directories and switching conditions of SCP-generated CDR

The directory for storing CDR files cannot be configure. The values listed precedently are fixed.

The description of the SCP-generated CDR file switching conditions is as follows:

A CDR file in the temporary directory is moved to the destination directory if either of the following conditions is met:

- The CDR file size reaches the upper limit.
- The duration for writing the CDR file reaches the maximum duration.

  For details, see 1.3 Maximum CDR File Size.

NOTE

The CDR files are not switched automatically at 00:00:00.

1.2.3 BMP-generated CDR storage directories

CDR storage directories

The description of the BMP-generated CDR storage directories are as follows:

- Destination directory: **${HOME}/cdr/output**

  It is the destination directory for storing CDR files after the CDR files meet the switching conditions.

- Temporary directory: **${HOME}/cdr/output/temp**

  It is the directory for storing temporary CDR files before the CDR files meet the switching conditions.

- Backup directory: **${HOME}/cdr/output/bak**

  After obtaining a CDR from the destination directory where the BMP stores CDRs, the system backs up the CDR in the backup directory. The CDR can be viewed to locate problems.

Here, **${HOME}** refers to the installation directory of the BMP.

Switching conditions of BMP generated CDR

The description of the BMP-generated CDR file switching conditions is as follows:

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A CDR file in the temporary directory is moved to the destination directory if either of the following conditions is met:

- The CDR file size reaches the upper limit.
- The duration for writing the CDR file reaches the maximum duration.

  For details, see 1.3 Maximum CDR File Size.

1.3 Maximum CDR File Size

The maximum CDR file size is used to limit the size of CDR files generated by the SCP and CBP CDR Generation nodes.

Usually, the maximum size of a CDR file is 10MB or it can contain a maximum of 5,000 records.

- The size of a SCP-generated CDR file is set by using the **CRTRECSIZE** field in the **nodesyspara.cfg** file in the following directory:

  ${TELLIN_DIR}/config

  For example:

  ```
  [COMMON]
  ...
  CRTRECTIME = 600    // Maximum interval for storing the CDR file [600,86400 seconds]
  CRTRECSIZE = 5000    // Maximum size of CDR file [1,10000KB]
  ```

  In this example, the maximum size of a CDR file is set to 5,000 KB and the maximum duration for writing a CDR file is set to 600 seconds. Thus, a CDR file is moved from the temporary directory to the destination one if either of the following conditions is met:

  – The CDR file size is equal to or greater than 5,000 KB.

  – The maximum duration for writing the CDR file is equal to or greater than 600 seconds.
- The size of a CBP-generated CDR file can be not configured.

1.4 CDR-Processing Mechanism

The mechanism for processing CDRs and the related components.

1.4.1 CBP-generated CDR Processing Process

Components Related to CDRs Generated by the CBP

Figure 1-1 shows the components for processing CDRs generated by CBP.

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Figure 1-1 Relationships between components for processing CDRs generated by the CBP

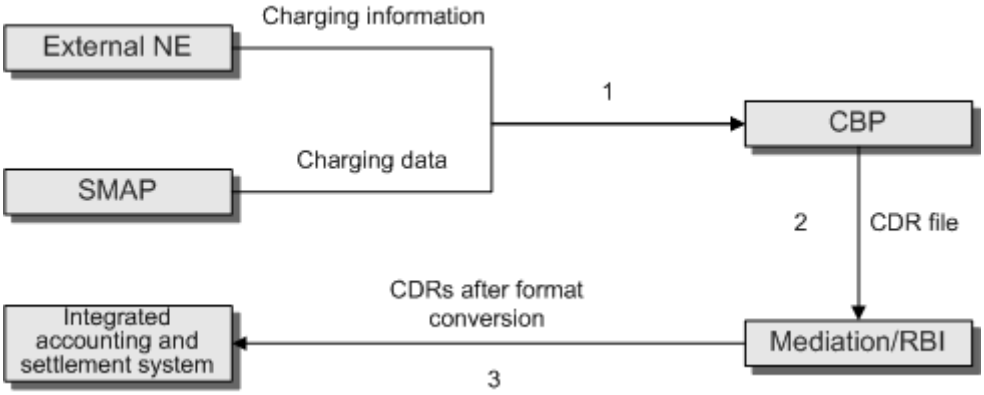

Table 1-1 describes the process in which components process CDRs generated by the CBP.

Table 1-1 Process of processing CDRs generated by the CBP

No.	Description
1	The charging information sent from external NEs and charging data configured on the SMAP are loaded on the CBP for charging.
2	Internal CDR files generated after the charging on the CBP are sent to the Mediation/RBI for format conversion through file interfaces. The Mediation/RBI converts internal CDR formats into external CDR formats. According to configurations, the RBI regularly obtains CDRs after conversion from the output directory of the Mediation/RBI.
3	The Mediation/RBI sends the CDR files after format conversion to the integrated accounting and settlement system for further processing.

CDR Generation Process of the CBP

The CBP generates CDRs when a charging event ends. The CBP finishes charging by interacting with an external NE once or more times. After a charging event ends, the CBP generates a CDR file and stores the file in a specified directory. The process of generating CDRs is complete.

The CBP can process charging requests in messages or CDRs. The details are as follows:

- Messages: The CBP analyzes charging requests according to the Diameter messages that are transferred between external NEs and the CBP, performs charging, and generates CDR files.
- CDRs: The CBP analyzes charging information in the offline CDRs that are transferred between external NEs and the CBP, performs charging, and generates CDR files.

The processes of processing charging requests in messages and CDRs are similar.

The following takes the charging request initiated by the SCP as an example to describe the CDR generation process:

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1. Set the conditions for generating CDR files on the CBP.
2. The SCP requests the CBP to perform charging based on the charging request initiated by a subscriber.
3. The CBP interacts with the SCP, charges the subscriber for services in real time based on the charging information and charging data configured on the SMAP, and generates CDR files after the charging.

Process of Processing CDRs Generated by the CBP

After CDRs are generated, the CDR dump service scans the CDR generation directory. If CDR files meet specified conditions, the service stores the files in the database so that other NEs can query the files.

After the CDRs generated by the CBP are converted into external CDR files, the integrated accounting and settlement system processes the files.

1.4.2 SCP-generated CDR Processing Process

Components Related to CDRs Generated by the SCP

Figure 1-2 shows the components for the SCP interacts normally with the CBP, a CDR generated by the SCP has no direct impact on a CDR generated by the CBP.

Figure 1-2 Relationships between components for processing CDRs generated by the SCP

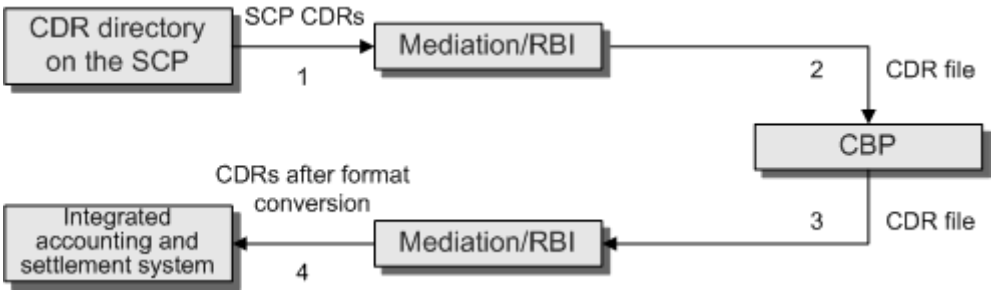

Table 1-2 describes the process in which components process CDRs generated by the SCP.

Table 1-2 Process of processing CDRs generated by the SCP

No.	Description
1	The Mediation/RBI obtains the CDRs periodically from the destination directory according to configuration, and then transfers them to the CBP.
2	The CBP generates a CDR after charging.
3	The Mediation/RBI obtains CDRs periodically from the CBP according to configuration.
4	The Mediation/RBI preprocesses the CDR files and uploads them to the accounting and settlement system for further processing.

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CDR Generation Process of the SCP

CDRs generated by the SCP are similar to CDR logs.

- If the interaction between the SCP and CBP is normal, the CDRs generated by the SCP are used for reconciliation, query, and problem locating, and do not affect the CDRs generated by the CBP.
- When an exception occurs during the interaction between the SCP and CBP in a calling, called, or forwarding flow, the SCP generates a CDR of abnormal call fee deduction if the system allows the subscriber to go on using the call service. Then the CBP deducts the fee that has not been deducted based on the CDR.

Process of Processing CDRs Generated by the SCP

- In call processes including the ones of a calling party, a called party, and a forwarding party, when an exception occurs during the interaction between the SCP and CBP, the SCP generates a CDR of abnormal call fee deduction if the system allows the party to go on using the call service. Then the CBP deducts the fee after the normal time based on the CDR.
- In the process of using the SM service, when an exception occurs during the interaction between the SCP and CBP, the SCP generates an SM offline CDR if the system allows the user to go on using the SM service. The exception can be a time-out disconnection between the SCP and CBP or a system error. In this case, the CBP obtains CDRs periodically and deducts the fee based on the CDR.

1.4.3 BMP-generated CDR Processing Process

Components Related to CDRs Generated by the BMP

Figure 1-3 shows the components for processing CDRs generated by the BMP.

Figure 1-3 Relationships between components for processing CDRs generated by the BMP

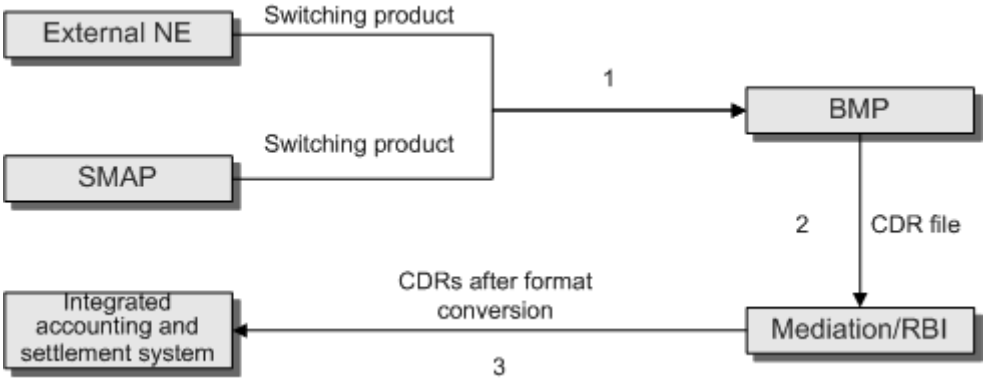

Table 1-3 describes the process for processing a BMP-generated CDR is as follows:

Table 1-3 Process of processing normal CDRs generated by the BMP

No.	Description
1	The BMP generates CDRs.

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No.	Description
2	The Mediation/RBI obtains CDRs from the BMP periodically according to configuration and transfers them to the Mediation/RBI.
3	The accounting and settlement system obtains CDR files and processes the files for further processing.

1.4.4 CDR Processing Flow

Generating a CDR

This document describes the SCP and CBP-generated CDRs. Their functions are as follows:

- A CBP-generated CDR is transferred from the Mediation to the Billing system. The CDR is used to generate a detailed bill or a bill of the accumulated charge.
- A SCP-generated CDR can be regarded as a CDR log.
  - When the SCP interacts normally with the CBP, a CDR generated by the SCP has no direct impact on a CDR generated by the CBP.
  - In call processes including the ones of a calling party, a called party, and a forwarding party, when an exception occurs during the interaction between the SCP and CBP, the SCP generates a CDR of abnormal call fee deduction if the system allows the party to go on using the call service. Then the CBP deducts the fee after the normal time based on the CDR. For the CDR-processing mechanism, see section 1.4.2 SCP-generated CDR Processing Process.
  - In the process of using the SM service, when an exception occurs during the interaction between the SCP and CBP, the SCP generates an SM offline CDR if the system allows the user to go on using the SM service. The exception can be a time-out disconnection between the SCP and CBP or a system error. In this case, the CBP obtains CDRs periodically and deducts the fee based on the CDR. For the CDR-processing mechanism, see section 1.4.2 SCP-generated CDR Processing Process.

The CBP system supports the charging requests in the message and the CDR modes. It also generates matching CDR files. The CDR-processing processes are similar for the two modes. The charging request initiated by the SCP is taken as an example in this document.

The CBP system generates a CDR when a charging event ends. For prepaid charging events, the processing mechanisms for writing CDRs are same. The CBP system completes charging after one or more attempts to interact with an external network element in the Diameter message or CDR mode. After a charging event ends, the CBP system writes the CDR file.

The process for generating a CDR file is as follows:

1. Set the conditions for generating CDR files on the CBP.
2. The SCP requests the CBP to charge based on the charging event initiated by a user.
3. The CBP interacts with the SCP and completes the charging.
4. The CBP generates the CDR file.

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Processing a CDR

After a charging event ends, the CBP system writes the CDR records and stores them in the specified directory. Thus, the process for writing a CDR is complete.

The charge bill dumping service then scans the CDR storage directories. If CDR files meet the required conditions, the service stores them in the database so that other network elements can use the CDR files.

1.5 CDR Types

This section describes the types of CDRs generated by the CBP, the SCP and the MDSP in the OCS service, and the service scenarios of each type of CDRs.

1.5.1 CBP-Generated CDR

In the OCS service, CBP-generated CDRs are classified based on the CDR file name prefixes. For details, see Table 1-4.

Table 1-4 CBP_Generated CDR type list

CDR File Name Prefix	CDR Description
rec	The **rec** CDR records the information about the call.
sms	The **sms** CDR records the information about the short message.
mms	The **mms** CDR records the information about the multimedia message.
data	The **data** CDR records the basic traffic of the data service.
vou	The **vou** CDR records the information about the recharge or bonus.
mon	The **mon** CDR records the information about the monthly fee and the bonus.
com	The **com** CDR records the general information.
mgr	The **mgr** CDR records the information about management charging.
clr	The **clr** CDR records the information about clearing the accounts of subscribers.

1.5.2 SCP-Generated CDR

In the OCS service, SCP-generated CDRs are classified based on the CDR file name prefixes. For details, see Table 1-5.

Table 1-5 SCP _Generated CDR type list

CDR File Name Prefix	CDR Description

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CDR File Name Prefix	CDR Description
w_rec	The **w_rec** CDR records the call information about subscribers.
w_abr	The system records the **w_abr** CDR after sending a request to the CBE and errors occur on the CBE.
w_vou	The system records the **w_vou** CDR after setting common recharge cards and sending a confirmation request.
w_abu	The system records the **w_abu** CDR after setting the state of common recharge cards and sending a request to the CBE to confirm the recharge, but errors occur on the CBE.
w_smr	The system records the **w_smr** CDR is recorded when the subscriber information does not exist, but the SMS transfer is allowed.
w_smo	When an error occurs in the CBE, the **w_smo** CDR is recorded if an SMS can be sent when deducting the fee fails. The CBE rates and deducts fees based on the CDR after the CBE recovers.

1.5.3 BMP-Generated CDR

In the OCS service, BMP-generated CDRs are classified based on the CDR file name prefixes. For details, see Table 1-6.

Table 1-6 BMP_Generated CDR type list

CDR File Name Prefix	CDR Description
fns	The **fns** CDR records the information when a subscriber changes or adds an FN number.
prod	The **prod** CDR records the information about a new subscription, validity period extension during product renewal, or validity period adjustment.

1.6 CDR Types in Service Application Scenarios

The section describes the types of CDRs generated in service scenarios.

Table 1-7 List of CDR generation scenarios

Service Application Scenario Category	Scenario Description	CDR Type	CDR Format Reference

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Service Application Scenario Category	Scenario Description	CDR Type	CDR Format Reference
CBP-generated CDR			
Call	In the OCS service, when subscriber A calls subscriber B and subscriber B hangs up after answering the call for certain duration, an rec CDR is generated. The CDR records the information about the call.	rec CDR	3.1.2 rec CDR
Recharge	A vou CDR is generated in the following scenarios: • When an OCS subscriber uses the call service for the first time or activates the MSISDN through the IVR recharge, the subscriber can obtain a certain bonus. • When an OCS subscriber uses a recharge card or cash for recharge, the system provides a bonus. • The system successfully recharges an account in advance. • An OCS subscriber successfully recharges an account in advance through the IVR management process.	vou CDR	3.1.6 vou CDR
Short Message	When OCS subscriber A sends a short message to subscriber B, and the system successfully deducts the fee from the account of subscriber A, an sms CDR is generated.	sms CDR	3.1.3 sms CDR
Month rent	When an OCS subscriber subscribes to a package and the system successfully deducts the monthly fee from the account of the subscriber, an Mon monthly billing CDR is generated.	mon CDR	3.1.7 mon CDR
SCP-generated CDR			
Call	A w_rec CDR is generated in the following case: In the OCS service, • When subscriber A makes a voice call, the SCP generates a w_rec CDR for the calling party after the call. • When subscriber B answers a voice	w_rec CDR	3.2.2 w_rec CDR

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Service Application Scenario Category	Scenario Description	CDR Type	CDR Format Reference
	call, the SCP generates a w_rec CDR for the called party after the call.		
Recharge	A w_vou CDR is generated in the following case: In the OCS service, when subscriber A recharge accounts using ordinary recharge cards, the SCP generates a w_vou CDR after the successful recharging.	w_vou CDR	3.2.4 w_vou CDR
SM	A w_smo CDR is generated in the following case: In the OCS service, when a subscriber succeeds in sending short messages, but the CBP fails to deduct fees due to system error, the SCP generates a w_smo CDR. The CBE rates and deducts fees based on the CDR after the CBE recovers.	w_smo CDR	3.2.7 w_smo CDR

1.7 Conventions

The section describes the conventions in this document.

This document covers only the CBP CDR and SCP CDR processing flow and mechanism.

NOTE

The number of CDR fields will increase in later versions. The new fields will be added to the end of the CDR. When the cooperation of a third party is required, do not limit the number of CDR fields.

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2 General Description of CDR Formats

About This Chapter

The section describes the formats of CDR file names.

The following lists the contents of this chapter.

2.1 CDR File Name Format

The section describes the formats of CDR file names.

2.2 Specifications of CDR Formats

The section describes the specifications of CDR formats.

2.1 CDR File Name Format

The section describes the formats of CDR file names.

2.1.1 CBP-Generated CDR

In the OCS service, CDRs generated by the CBP can be regarded as CDR logs. The CBP CDR file name is in the following format:

{CDR prefix}+{YYYYMMDD}+{_}+{CBP node number}+{_}+{SN}+{CDR suffix}

Table 2-1 describes fields in a CBP-generated CDR file name.

Table 2-1 Description of fields in a CBP-generated CDR file name

Field	Description
CDR prefix	Indicates the prefix of a CDR file name and refers to the type of the CDR. For example, the prefix **rec** indicates that a CDR is a call CDR. CDRs can be distinguished based on CDR file name prefixes. For details, see 1.5.1 CBP-Generated CDR .
YYYYMMDD	Indicates the date when a CDR file is generated.
_	Indicates an underline.

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Field	Description
CBP node number	Indicates the ID of the CBP. The ID is a numeric string of three digits. When the ID contains less than three digits, add 0 to the beginning of the ID. Value range: 001-999
SN	Indicates the serial number of a CDR file. The serial number is obtained from the serial number generation table in the system memory database. The serial number is a numeric string of six digits. Value range: 000001-999999 **NOTE** There is no duplicate serial number on the current day. Serial numbers are consecutive, and they are also generated for empty files.
CDR suffix	Indicates the extension of a CDR file name. The extension is **.unl** by default.

Example of a rec CDR file name: rec20030501_101_000006.unl. This CDR file name indicates that the CBP with the node number 101 generated the sixth rec CDR file on May 1, 2003.

2.1.2 SCP-Generated CDR

In the OCS service, CDRs generated by the SCP can be regarded as CDR logs. The SCP CDR file name is in the following format:

{CDR prefix}+{YYYYMMDD}+{_}+{SCPID}+{BB}+{CC}+{_}+{Index number}+{CDR suffix}

Table 2-2 describes fields in a SCP-generated CDR file name.

Table 2-2 Description of fields in a SCP-generated CDR file name

Field	Description
CDR prefix	Indicates the prefix of a CDR file name and refers to the type of the CDR. For example, the prefix w_rec indicates that a CDR is a call CDR. CDRs can be distinguished based on CDR file name prefixes. For details, see 1.5.2 SCP-Generated CDR .
YYYYMMDD	Indicates the date when a CDR file is generated.
_	Indicates an underline.
SCPID	Indicates the ID of the SCP that generates a CDR file. The serial number is a numeric string of three digits. Value range: 001-999
BB	Indicates the ID of a package.

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Field	Description
	The value of this field is always **00**.
CC	Indicates the number of the SCF. The SCF number is a numeric string of two digits. Value range: 01-99
Index number	Indicates the serial number of a CDR file in a day. The serial number is a numeric string of four digits. Value range: 0000-9999
CDR suffix	Indicates the extension of a CDR file name. The extension is **.unl** by default.

Example of an SCP-generated CDR file name: w_abr20020501_1020001_0001.unl

It indicates that the SCF with the process ID 0001 in the SCP with the node number 102 generated the second w_abr CDR file on May 1, 2002.

2.1.3 BMP-Generated CDR

In the OCS service, CDRs generated by the BMP can be regarded as CDR logs. The BMP CDR file name is in the following format:

{CDR prefix}+{YYYYMMDDHHMMSS}+{_}+<{BMPNodeID}>+{_}+<{SEQ}>+{CDR suffix}

Table 2-3 describes fields in a BMP-generated CDR file name.

Table 2-3 Description of fields in a BMP-generated CDR file name

Field	Description
CDR prefix	Indicates the prefix of a CDR file name and refers to the type of the CDR.
YYYYMMDDHHMMSS	Indicates the time when a CDR file is generated. • YYYY: Year, for example, 2007 • MM: Month, for example, 05 • DD: Day, for example, 29 • HH: Hour, for example, 10 • MM: Minute, for example, 58 • SS: Second, for example, 59
_	Indicates an underline.
BMPNodeID	Indicates the ID of a BMP node. The serial number is a numeric string of three digits. Value range: 001-999
_	Indicates an underline.

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Field	Description
SEQ	Indicates the sequence number of a file. The serial number is a numeric string of six digits. Value range: 000001-999999 **NOTE** There is no duplicate serial number on the current day. Serial numbers are consecutive, and they are also generated for empty files.
CDR suffix	Indicates the extension of a CDR file name. The extension is **.unl** by default.

Example of a BMP-generated CDR file name: fns20070529105859_180_000006.unl

This CDR file name indicates that the BMP with the node number 180 generated the sixth fns CDR file on 10:58:59, May 29, 2007.

2.2 Specifications of CDR Formats

The section describes the specifications of CDR formats.

2.2.1 CDR File Content Format

The description of a CDR file content format is as follows:

- The file is a text file.
- Fields in the file are separated by vertical bars (|).
- The values of certain fields can be null, which appears as "||" in the CDR records.
- Each record occupies one line.
- Each record ends with a carriage return character.
- Data encoded with ASCII characters.

The following is the example of a CDR file:

0|13300123301|20070407231445|3|1|1|755|101|0|10000|46020571234|||||

2.2.2 CDR Field Type

Table 2-4 lists the type of each field in a CDR. Field types are described according to the definition of data types in the JDK.

Table 2-4 CDR field types and description

Data Type	Description
Integer	Indicates an integer.
Char	Indicates a character string. Numbers enclosed in [] indicates the length of a character string. The expression of the character string length can be classified into

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Data Type	Description
	four types: • Range type Char [20]: Indicates a character string of 0 to 20 characters. • Unfixed length type Char[]: Indicates no special requirements for the length of a character string.

2.2.3 Time Format

Table 2-5 lists the requirements for the format of time in a CDR.

Table 2-5 Time format requirements

Time	Format	Example
Year	YYYY	2006
Month	MM	04
Date	DD	25
Hour	hh	11
Minute	mm	42
Second	ss	23

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3 CDR Format Details

About This Chapter

The section describes the file name format of CBP and SCP CDRs of each type, CBP and SCP CDRs storage directories and CBP and SCP CDRs generation scenarios.

NOTE

This chapter describes the service CDR formats and lists all possible CDR fields and field values. The current site may not use all these fields and values in the service CDRs. If the value of a field in a CDR is null or the field does not cover all values in the actual situation, it means that the current site does not use the field or all values of the field.

The following lists the contents of this chapter.

3.1 CBP-Generated CDR

The section describes the file name format of CBP-generated CDRs of each type, CBP-generated CDR storage directories and CBP-generated CDR generation scenarios.

3.2 SCP-Generated CDR

The section describes the file name format of SCP-generated CDRs of each type, SCP-generated CDR storage directories and SCP-generated CDR generation scenarios.

3.3 BMP-Generated CDR

The section describes the file name format of BMP-generated CDR of each type, BMP-generated CDR storage directories and BMP-generated CDR generation scenarios.

3.1 CBP-Generated CDR

The section describes the file name format of CBP-generated CDRs of each type, CBP-generated CDR storage directories and CBP-generated CDR generation scenarios.

3.1.1 General Description

The directory for storing general CDRs must be configured based on the actual situation of the current network. For description about the CDR storage directories, see 1.2 CDR Storage Directories.

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Result code of CBP-generated CDRs

Table 3-1 lists the error code of CBP-generated CDRs.

Table 3-1 Result code of CBP-generated CDRs

Result code	Description
102022024	The short message is sent successfully and refund does not need to be performed.
102022027	The maximum recharge amount is exceeded.
102022028	The recharge amount exceeds the maximum amount.
102022046	A system error occurred during authentication.
102022051	Refund does not need to be performed.
102022056	When the fee for the short message is refunded, the history fee deduction record cannot be found.
102022057	The service has not been subscribed or the call is barred according to the age of the calling party.
102022058	The balance is insufficient.
102022060	The international gateway function is forbidden for an international roaming subscriber.
102022081	The balance of the transfer-out party is smaller than the amount to be transferred.
102022082	The maximum transfer times of the current month is exceeded.
102022083	The minimum transfer amount at a time is not reached.
102022084	The maximum transfer amount at a time is exceeded.
102022085	The minimum balance before balance transfer is not reached.
102022086	The minimum balance after balance transfer is not reached.
102022087	The maximum transfer amount of the current day is exceeded.
102022088	The maximum transfer amount of the current month is exceeded.
102022089	The maximum transfer times of the current day is exceeded.
102022090	The maximum account balance is exceeded.
102022092	The short message notification is sent, indicating the maximum number of short messages that can be sent in a cycle.

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Result code	Description
102022093	The latest activation time is over or the maximum validity period expires.
102022094	The transfer amount is not a multiple of the specified amount.
102022095	The account book type specified by the transfer-out party does not exist.
102022096	When the transfer-in and transfer-out functions are enabled for brands, authentication fails.
102022097	During the three-level authentication for enabling the transfer function, the authentication at the brand level fails.
102022098	During the three-level authentication for enabling the transfer function, the authentication at the sub-brand level fails.
102022099	During the three-level authentication for enabling the transfer function, the authentication at the subscriber level fails.
102022100	The transfer function is not subscribed according to sub-brands. Therefore, transfer-out authentication fails.
102022101	The transfer function is not subscribed according to sub-brands. Therefore, transfer-in authentication fails.
102022102	The account book type specified by the transfer-in party does not exist.
102022103	The site does not support off-net MMS.
102022104	The CUG function is completely disabled, so services are interrupted.
102022105	The group customer is locked.
102022106	The group member is locked.
102022109	The balance is insufficient to repay the loan.
102022111	The recharge package type does not match the recharge amount.
102022112	The service duration of the transfer-in party does not reach the minimum limit.
102022113	The service duration of the transfer-out party does not reach the minimum limit.
102023002	The recharge record does not exist.
102023003	Rollback is performed.
102023005	The balance of the account to be adjusted is insufficient.

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Result code	Description
102023008	**Initial** request message for recharge. The recharge card has been used.
102023009	**Event** query message for recharge. The recharge card has been used.
102023010	The session times out.
102023011	The maximum transfer amount of the current week is exceeded.
102023012	The maximum transfer times of the current week is exceeded.
102023013	During the three-level authentication for enabling the transfer function, the authentication at the transfer-in subscriber level fails.
102023014	During the three-level authentication for enabling the transfer function, the brand of the transfer-in subscriber does not support the account transfer function.
102023015	During the three-level authentication for enabling the transfer function, the sub-brand of the transfer-in subscriber does not support the account transfer function.
102023025	Forbid triggering service through normal child card.
102023038	The number of licenses is insufficient for activation.
102023064	The recharge card is not valid to recharge.
102029998	Authentication failed.
102022109	The balance is insufficient to pay pack the loan.
402020000-402029999	System error.

Account Types Supported by OCS System

Table 3-2 lists the account types supported by OCS.

Table 3-2 Details about account type

Account Type ID	Account Description
2000	Prepaid main fund subaccount.
2001	Bonus fund subaccount.
4001	Free call duration subaccount1.
4002	Free call duration subaccount2.

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Account Type ID	Account Description
4003	Free call duration subaccount3.
4004	Free short message subaccount1.
4005	Free short message subaccount2.
4006	Free short message subaccount3.
4007	Free multimedia messaging subaccount1.
4008	Free volume subaccount1.
4501	Free call duration subaccount4 for Bucket.
4502	Free call duration subaccount5 for Bucket.
4503	Free call duration subaccount6 for Bucket.
4504	Free short message subaccount4 for Bucket.
4505	Free short message subaccount5 for Bucket.
4506	Free short message subaccount6 for Bucket.
4507	Free multimedia messaging subaccount2 for Bucket.
4508	Free volume subaccount2 for Bucket.

3.1.2 rec CDR

Description

The rec CDRs are the CDRs for the voice call service. The CDRs record the following information in the voice call flow: the service scenario information reported by the call control module and the information generated by the charging modules, such as rating and fee deduction information.

Generation Scenario

When subscriber A calls subscriber B and subscriber B answers the call, the OCS system generates an rec CDR after the call.

NOTE

If the charging number has child-parent relations with other numbers, that is, the charging number and other numbers are child-parent cards, the charging information about other numbers is contained in the generated CDR of the charging party.

The child-parent cards refer to two or more numbers with one number being the parent card and other numbers being child cards. The parent card pays for the child cards and child cards can use the products that the parent card subscribes to.

CDR File Name Format

recYYYYMMDD_CBPNODE_Seqno.unl

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Example of an rec CDR file name: rec20071112_101_000151.unl

Format

Table 3-3 describes each field of rec CDR file.

Table 3-3 Format of rec CDR

SN	Field Name	Field Type	Description
1	SerialNo	Integer[8]	Indicates the serial number. It is generated by the system and the serial number is exclusive. The default value is **0**. Not null
2	SubSequence	Integer[4]	Indicates the sequence number when the bill is split for mid-night and end of the bill cycle. The default value is **0**.
3	TimeStamp	Char[14]	Indicates the timestamp of the bill item generation. Format: YYYYMMDDhhmmss Not null
4	ServiceKey	Integer[4]	Indicates the service key of the intelligent service to which the subscriber subscribes. The default value is **1**.
5	CallingPartyNumber	Char[64]	Indicates the calling number. The number format are as follows: • **Country code + Area number + PSTN** The number format is the format of the fixed-line phone number. • **Country code + MSISDN** The number format is the format of the mobile phone number. Value range: numerals within 0-9, *, #
6	CalledPartyNumber	Char[64]	Indicates the called number. The number format are as follows: • **Country code + Area number + PSTN** The number format is the format of the fixed-line phone number. • **Country code + MSISDN** The number format is the format of the mobile phone number. Value range: numerals within 0-9, *, #
7	CallingPartyIM	Char[64]	Indicates the IMSI of calling party.

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SN	Field Name	Field Type	Description
	SI		When the SCP transfers an IMSI, the value of this field is the IMSI transferred. When the SCP does not transfer an IMSI, the value of this field is the value of the corresponding field in the attribute table of the charged subscriber. The value is **Null** when the relevant value cannot be obtained. Value range: numerals within 0-9, *, #
8	CalledPartyIMSI	Char[64]	Indicates the IMSI of called party. When the SCP transfers an IMSI, the value of this field is the IMSI transferred. When the SCP does not transfer an IMSI, the value of this field is the value of the corresponding field in the attribute table of the charged subscriber. The value is **Null** when the relevant value cannot be obtained. Value range: numerals within 0-9, *, #
9	DialedNumber	Char[64]	Indicates the original number that the calling party dials in the calling process. It is valid in MO process. Value range: numerals within 0-9, *, #
10	OriginalCalledParty	Char[64]	Indicates the original called number in a forwarding process. Special number reroute. International number form in call forwarding, actual call number in special number reroute, filling in original called number when dialed through an international gateway office. Value range: numerals within 0-9, *, #
11	ServiceFlow	Char[1]	Indicates the call process indicator. The options are as follows: • 1: calling process • 2: called process • 3: forward flow Not null
12	CallForwardIndicator	Char[2]	Indicates the call forwarding flag. In MT: A->B->C, In MT process of B, it needs to know that whether B party transfers the call to C. Usually, the MT of B party is set free and all the cost will be charged in MF of B party in this situation. In MF: this field is related to the parameter of

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SN	Field Name	Field Type	Description
			"redirect reason" in IDP. And there are two features implemented by OCS will also be record here: • Forward to voicemail when subscriber in abnormal state • Forward to international destiny when on international roaming The options are as follows: • 00: No call forwarding • 10: Call forwarding conditional (none special) • 11: Call forward busy • 12: Call forward no answer • 13: Call forward unreachable • 14: Call forward power off • 20: Call forwarding unconditional (none special) • 21: Call forward to voicemail (OCS control) • 22: ICF call forward (OCS control)
13	CallingRoamInfo	Char[20]	Indicates the information used to analyze the roaming state of calling party. It can be the location number, MSC (mobile switching center) address, VLR (visitor location register) number, and so on. The actual value is related with the service configuration. Value range: numerals within 0-9, *, #
14	CallingCellID	Char[20]	Indicates the cell ID of the calling party, which is directly filled in according to the IDP report. Value range: numerals within 0-9, *, #
15	CalledRoamInfo	Char[20]	Indicates the information used to analyze the roaming state of called party. It can be the location number, MSC address, VLR number, and so on. The actual value is related with the service configuration. Value range: numerals within 0-9, *, #
16	CalledCellID	Char[20]	Indicates the cell ID of the called party, which is handled just as that for **CallingCellID**. Value range: numerals within 0-9, *, #
17	TimeStampOfSSP	Char[14]	Indicates the time stamp report from SSP in initial_IDP. If missing in charging interface, this field will be filled with the local time stamp of OCS.

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SN	Field Name	Field Type	Description
			Format: YYYYMMDDhhmmss
18	TimeZoneOfSSP	Integer[4]	Indicates the time zone report from SSP in initial_IDP. If missing in charging interface, this field will be filled with the time zone of the location of OCS, 15 min. per sector. The default value is **0**. Value range: -48-48 **NOTE** The value -48 is not included.
19	BearerCapability	Char[32]	Indicates the beared service or additional service. [1]: supplied service/supplemented service Others reserved The options of **[1]** are as follows: • 0: NormalCall • 8: VideoCall • 9: VoiceData • 10: VoiceFax
20	BearerCapability2	Char[32]	Reserved
21	ChargingTime	Char[14]	Indicates the time stamp of the charging beginning. Format: YYYYMMDDhhmmss Not null
22	WaitDuration	Integer[4]	Indicates the wait duration for the calling party. Unit: second The default value is **0**.
23	CallDuration	Integer[4]	Indicates the call duration. Unit: second The default value is **0**. Not null
24	TerminationReason	Integer[4]	Indicates the reason why the call cannot be connected. The options are as follows: • 0: unknown • 1: Hang up • 2: Insufficient balance • 3: Over maximum call duration • Other: reserved The default value is **0**.

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SN	Field Name	Field Type	Description
25	CallReferenceNumber	Char[128]	Indicates the call reference number. It is reported by the SSP and is used to synchronized CDRs generated on both the IN and the SSP. Value range: numerals within 0-9, *, #
26	IMEI	Char[24]	Indicates the IE containing the IMEISV of the ME in use by the served subscriber.
27	ChargeDuration	Integer[4]	Indicates the charge duration. Unit: second The default value is **0**. Not null
28	AccessPrefix	Char[20]	Indicates the special prefix dialed by a subscriber for making calls. For example, the prefix of IP calls, or the prefix of the international gateway. Value range: numerals within 0-9, *, #
29	RoutingPrefix	Char[20]	Indicates the route prefix for re-routing a call. Value range: numerals within 0-9, *, #
30	RedirectingPartyID	Char[20]	Indicates the redirection number in the forwarding process.
31	BrandID	Integer[4]	Indicates the brand ID of subscribers. The default value is **0**.
32	SubCOSID	Integer[4]	Indicates the class ID of subscribers. The default value is **0**.
33	ChargingPartyNumber	Char[64]	Indicates the telephone number of the charging party. The number is same as calling party number in MO and called party number in MT. There are several special situations. Like Child-Parent card feature, the calling party number is the child, the charging party number is the parent's number. This field is null when the service is paid only by account. The number format are as follows: • **Area number + PSTN** The number format is the format of the fixed-line telephone number. • **MSISDN** The number format is the format of the mobile phone number. Value range: numerals within 0-9, *, #

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SN	Field Name	Field Type	Description
34	ChargePartyIndicator	Char[1]	Indicates the charging party. The options are as follows: • 1: The calling party is charged • 2: The called party is charged • 3: Charge of third party • 4: Charge of parent & child card • 5: Charge of calling party & VPN
35	PayType	Char[1]	Indicates the pay type of subscribers. The options are as follows: • 0: Prepaid • 1: Postpaid
36	BillCycleID	Char[20]	Indicates the bill cycle ID. Format: YYYYMMTT The description of the format is as follows: • YYYY: indicates the year. • MM: indicates the month. • TT: indicates the type of a bill cycle. Value range: numerals within 0-9
37	ChargingType	Char[1]	Indicates the charging type. The options are as follows: • 0: V-H (caller visit to called location) • 1: H-H (caller location to called location) • 2: V-V (caller visit to called visit) • 3: H-V (caller location to called visit)
38	CallType	Char[1]	Indicates the analyzed call type according to the **ChargingType**. The options are as follows: • 0: Local call • 1: Toll call within charging area • 2: Toll call between charging area • 3: International call The wildcard value is **-1**.
39	RoamState	Char[1]	Indicates the roaming state of charging party. The options are as follows: • 0: No roaming • 1: Roaming within charging area • 2: Roaming out of charging area • 3: International Roaming

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SN	Field Name	Field Type	Description
40	ResultCode	Integer[4]	Indicates the internal result code. Value range: -2^{31}-2^{31}-1 The default value is **0**.
41	FPHPrefix	Char[20]	Indicates the prefix for called party payment. Value range: numerals within 0-9, *, #
42	CallingHomeCountryCode	Integer[4]	Indicates the home country code information of calling party. >=0, -1, 0 or 9999 means unknown. The default value is **0**.
43	CallingHomeAreaNumber	Integer[4]	Indicates the home area number of the calling party. >=0, -1, 0 or 9999 means unknown. The default value is **0**.
44	CallingHomeNetworkCode	Integer[4]	Indicates the home network code of the calling party. >=0, -1, 0 or 9999 means unknown. The default value is **0**.
45	CallingRoamCountryCode	Integer[4]	Indicates the roam country code of the calling party. >=0, -1, 0 or 9999 means unknown. The default value is **0**.
46	CallingRoamAreaNumber	Integer[4]	Indicates the roam area number of the calling party. >=0, -1, 0 or 9999 means unknown. The default value is **0**.
47	CallingRoamNetworkCode	Integer[4]	Indicates the roam network code of the calling party. >=0, -1, 0 or 9999 means unknown. The default value is **0**.
48	CalledHomeCountryCode	Integer[4]	Indicates the home country code of the called party. >=0, -1, 0 or 9999 means unknown. The default value is **0**.
49	CalledHomeAreaNumber	Integer[4]	Indicates the home area number of the called party. >=0, -1, 0 or 9999 means unknown. The default value is **0**.
50	CalledHomeNetworkCode	Integer[4]	Indicates the home network code of the called party.

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SN	Field Name	Field Type	Description
			>=0, -1, 0 or 9999 means unknown. The default value is **0**.
51	CalledRoamCountryCode	Integer[4]	Indicates the roam country code of the called party. >=0, -1, 0 or 9999 means unknown. The default value is **0**.
52	CalledRoamAreaNumber	Integer[4]	Indicates the roam area number of the called party. >=0, -1, 0 or 9999 means unknown. The default value is **0**.
53	CalledRoamNetworkCode	Integer[4]	Indicates the roam network code of the called party. >=0, -1, 0 or 9999 means unknown. The default value is **0**.
54	ProductID	Char[20]	Indicates the last effect rating product.
55	ServiceType	Char[32]	Indicates the service included in the product. They are predefined in OCS. The options are as follows: • 1st digit: Normal Voice tariff • 2nd digit: Familiarity number service • 3rd digit: Data service • 4th digit: VoIP/international carrier call • 5th digit: pre-IP call • 6th digit: International Call Forward • 7th digit: Call from Home Zone • 8th digit: UCB service • 9th digit: P_FPH service • 10th digit: Remote Access • 11th digit: Call from Special Zone • 12th digit: Calling Collect Call • 13th digit: Called Collect Call • 14th digit: Carrier Service Prefix • 15th digit: Calling Roam OutNet • 16th digit: Called Roam OutNet • 17th digit: Special number call • 18th digit: VoIP call • 19th digit: Fax Other: reserved The value range of each digit indicates whether the service is activated.

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SN	Field Name	Field Type	Description
			• 0: Not activated • 1: Activated Not null
56	HotLineIndicator	Integer[4]	Indicates the special number type, like free call number, emergency number, and so on. Segment from 0 to 1000 is reserved by OCS, from 1001 can be extended by operator according to special purpose. The options are as follows: • 0: Normal call • 1: ForbiddenNumber • 2: EmergencyNumber • 3: FreeNumber • 4: SpecialNumber • 8: CallCenterandAttandent • 9: Voice Mail • 16: Dunning long number The default value is **0**.
57	HomeZoneID	Integer[4]	Indicates the home zone ID. When a subscriber enjoys the home zone discount, the value of this field is the home zone ID of the area where the subscriber currently locates. The default value is **0**.
58	NPFlag	Char[1]	Indicates the flag of number portable. The options are as follows: • 0: Not NP • 1: NP out The default value is **0**.
59	NPPrefix	Char[6]	Indicates the number portability prefix of called party in the MO process or number portability prefix of the calling party in the MT process. Value range: numerals within 0-9, *, #
60	CallingGroupNo	Integer[4]	Indicates the group number of the calling party when enjoying CUG service. It is the group number related to the rating. The default value is **0**.
61	CalledGroupNo	Integer[4]	Indicates the group number of the called party when enjoying CUG or VPN service. It is the group number related to the rating. The default value is **0**.

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SN	Field Name	Field Type	Description
62	UserState	Char[10]	Indicates the life cycle state and the management state of a subscriber. The **UserState** parameter is a numeric string of seven digits, in the format CCMMMMM. The first and second digits C indicate the life cycle state of a subscriber. The first digit C indicates the state of a prepaid subscriber, and the second digit C indicates the state of a postpaid subscriber. The options are as follows: • 0: Idle • 1: Active • 2: Suspended • 3: Disable • 4: Pool • 5: Pool without activation The next five digits MMMMM indicate the management state of a subscriber. The options are as follows: Third digit: Whether the subscriber claims missing. • 0: no • 1: yes Fourth digit: Whether the subscriber suspends the service. • 0: non-suspend • 1: suspend • 2: suspend by operator (B1W) • 3: suspend by operator (B2W) Fifth digit: Whether the subscriber applies for suspension with the number being reserved. • 0: no • 1: yes Sixth digit: Whether the subscriber is blocked. • 0: no • 1: yes Seventh digit: Whether the subscriber is blacklisted. • 0: no • 1: yes
63	SubscriberID	Char[20]	Indicates the subscriber ID. Value range: characters in [0-9, a-z, A-Z]
64	OpposeNumber	Integer[4]	Indicates the network type of oppose side.

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SN	Field Name	Field Type	Description
	Type		The options are as follows: • 1: PSTN • 2: Mobile • 3: Native Unknown • 4: International Unknown • Other: to be extended The default value is **0**. The wildcard value is **-1**.
65	Reserved	Integer[4]	-
66	ChargeOfItemsAccounts	Integer[8]	Indicates the free call times used. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
67	ChargeOfDurationAccounts	Integer[8]	Indicates the total call time charge using reword resource. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
68	ChargeOfFundAccounts	Integer[8]	Indicates the total cost amount charging from fund account Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
69	ChargeFromPrepaid	Integer[8]	Indicates the charge amount from prepaid account. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
70	PrepaidBalance	Integer[8]	Indicates the balance of prepaid account. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
71	ChargeFromPostpaid	Integer[8]	Indicates the charge amount from postpaid account. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
72	PostPaidBalance	Integer[8]	Indicates the balance of postpaid available limitation.

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SN	Field Name	Field Type	Description
			Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
73	AccountID	Char[20]	Indicates the ID of the main account. Value range: characters in [0-9, a-z, A-Z]
74	AccountKey	Integer[8]	Indicates the key of the main account. Value range: -2^{63}-2^{63}-1 The default value is **0**.
75	CurrencyCode	Integer[4]	Currency code. Also, if this field missed or present be Zero, it means the system use local minimal currency as default, and no need to specify in CDR. The default value is **0**.
76	Reserved	Char[20]	-
77	AccountType1	Integer[4]	Indicates the account type of the first account. 0 means not used. Not null The default value is **0**.
78	FeeType1	Integer[4]	Indicates the fee type of the first account. 0 means not used. Not null The default value is **0**.
79	ChargeAmount 1	Integer[8]	Indicates the changing amount of the first account. 0 means not used. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
80	CurrentAcctAm ount1	Integer[8]	Indicates the current amount of the first account. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
81	AccountType2	Integer[4]	Indicates the account type of the second account. 0 means not used. Not null The default value is **0**.
82	FeeType2	Integer[4]	Indicates the fee type of the second account. 0 means not used.

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SN	Field Name	Field Type	Description
			Not null The default value is **0**.
83	ChargeAmount2	Integer[8]	Indicates the changing amount of the second account. 0 means not used. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
84	CurrentAcctAmount2	Integer[8]	Indicates the current amount of the second account. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
85	AccountType3	Integer[4]	Indicates the account type of the third account. 0 means not used. Not null The default value is **0**.
86	FeeType3	Integer[4]	Indicates the fee type of the third account. 0 means not used. Not null The default value is **0**.
87	ChargeAmount3	Integer[8]	Indicates the changing amount of the third account. 0 means not used. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
88	CurrentAcctAmount3	Integer[8]	Indicates the current amount of the third account. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
89	AccountType4	Integer[4]	Indicates the account type of the fourth account. 0 means not used. Not null The default value is **0**.
90	FeeType4	Integer[4]	Indicates the fee type of the fourth account. 0 means not used. Not null

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SN	Field Name	Field Type	Description
			The default value is **0**.
91	ChargeAmount4	Integer[8]	Indicates the changing amount of the fourth account. 0 means not used. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
92	CurrentAcctAmount4	Integer[8]	Indicates the current amount of the fourth account. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
93	AccountType5	Integer[4]	Indicates the account type of the fifth account. 0 means not used. Not null The default value is **0**.
94	FeeType5	Integer[4]	Indicates the fee type of the fifth account. 0 means not used. Not null The default value is **0**.
95	ChargeAmount5	Integer[8]	Indicates the changing amount of the fifth account. 0 means not unchanged. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
96	CurrentAcctAmount5	Integer[8]	Indicates the current amount of the fifth account. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
97	AccountType6	Integer[4]	Indicates the account type of the sixth account. 0 means not used. Not null The default value is **0**.
98	FeeType6	Integer[4]	Indicates the fee type of the sixth account. 0 means not used. Not null The default value is **0**.

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SN	Field Name	Field Type	Description
99	ChargeAmount6	Integer[8]	Indicates the changing amount of the sixth account. 0 means not unchanged. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
100	CurrentAcctAmount6	Integer[8]	Indicates the current amount of the sixth account. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
101	AccountType7	Integer[4]	Indicates the account type of the seventh account. 0 means not used. Not null The default value is **0**.
102	FeeType7	Integer[4]	Indicates the fee type of the seventh account. 0 means not used. Not null The default value is **0**.
103	ChargeAmount7	Integer[8]	Indicates the changing amount of the seventh account. 0 means not unchanged. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
104	CurrentAcctAmount7	Integer[8]	Indicates the current amount of the seventh account. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
105	AccountType8	Integer[4]	Indicates the account type of the eighth account. 0 means not used. Not null The default value is **0**.
106	FeeType8	Integer[4]	Indicates the fee type of the eighth account. 0 means not used. Not null The default value is **0**.

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SN	Field Name	Field Type	Description
107	ChargeAmount8	Integer[8]	Indicates the changing amount of the eighth account. 0 means not unchanged. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
108	CurrentAcctAmount8	Integer[8]	Indicates the current amount of the eighth account. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
109	AccountType9	Integer[4]	Indicates the account type of the ninth account. 0 means not used. Not null The default value is **0**.
110	FeeType9	Integer[4]	Indicates the fee type of the ninth account. 0 means not used. Not null The default value is **0**.
111	ChargeAmount9	Integer[8]	Indicates the changing amount of the ninth account. 0 means not unchanged. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
112	CurrentAcctAmount9	Integer[8]	Indicates the current amount of the ninth account. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
113	AccountType10	Integer[4]	Indicates the account type of the tenth account. 0 means not used. Not null The default value is **0**.
114	FeeType10	Integer[4]	Indicates the fee type of the tenth account. 0 means not used. Not null The default value is **0**.

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SN	Field Name	Field Type	Description
115	ChargeAmount10	Integer[8]	Indicates the changing amount of the tenth account. 0 means not unchanged. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
116	CurrentAcctAmount10	Integer[8]	Indicates the current amount of the tenth account. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
117	BonusValidity1	Char[14]	Indicates the validity period of the first rewarded resource. The validity period does not change when fees are deducted from accounts. In this case, the validity periods of consumption resources are not recorded. Format: YYYYMMDDhhmmss
118	BonusValidity2	Char[14]	Indicates the validity period of the second rewarded resource. The validity period does not change when fees are deducted from accounts. In this case, the validity periods of consumption resources are not recorded. Format: YYYYMMDDhhmmss
119	BonusValidity3	Char[14]	Indicates the validity period of the third rewarded resource. The validity period does not change when fees are deducted from accounts. In this case, the validity periods of consumption resources are not recorded. Format: YYYYMMDDhhmmss
120	BonusValidity4	Char[14]	Indicates the validity period of the fourth rewarded resource. The validity period does not change when fees are deducted from accounts. In this case, the validity periods of consumption resources are not recorded. Format: YYYYMMDDhhmmss
121	BonusValidity5	Char[14]	Indicates the validity period of the fifth rewarded resource. The validity period does not change when fees are deducted from accounts. In this case, the validity periods of consumption resources are not recorded. Format: YYYYMMDDhhmmss
122	BonusValidity6	Char[14]	Indicates the validity period of the sixth rewarded resource. The validity period does not change when fees are deducted from accounts. In this case, the validity periods of consumption

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SN	Field Name	Field Type	Description
			resources are not recorded. Format: YYYYMMDDhhmmss
123	BonusValidity7	Char[14]	Indicates the validity period of the seventh rewarded resource. The validity period does not change when fees are deducted from accounts. In this case, the validity periods of consumption resources are not recorded. Format: YYYYMMDDhhmmss
124	BonusValidity8	Char[14]	Indicates the validity period of the eighth rewarded resource. The validity period does not change when fees are deducted from accounts. In this case, the validity periods of consumption resources are not recorded. Format: YYYYMMDDhhmmss
125	BonusValidity9	Char[14]	Indicates the validity period of the ninth rewarded resource. The validity period does not change when fees are deducted from accounts. In this case, the validity periods of consumption resources are not recorded. Format: YYYYMMDDhhmmss
126	BonusValidity10	Char[14]	Indicates the validity period of the tenth rewarded resource. The validity period does not change when fees are deducted from accounts. In this case, the validity periods of consumption resources are not recorded. Format: YYYYMMDDhhmmss
127	CallingVPNTopGroupNumber	Char[15]	Indicates the top group number of the calling party. Value range: numerals within 0-9 Can be null
128	CallingVPNGroupNumber	Char[10]	Indicates the group number of the calling party. Value range: numerals within 0-9 Can be null
129	CallingVPNShortNumber	Char[10]	Indicates the short number of the calling party. Value range: numerals within 0-9 Can be null
130	CalledVPNTopGroupNumber	Char[15]	Indicates the top group number of the called party. Value range: numerals within 0-9 Can be null
131	CalledVPNGroupNumber	Char[10]	Indicates the group number of the called party. Value range: numerals within 0-9

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SN	Field Name	Field Type	Description
			Can be null
132	CalledVPNShortNumber	Char[10]	Indicates the short number of the called party. Value range: numerals within 0-9 Can be null
133	CallingNetworkType	Char[10]	Indicates the type of the network accessed by the calling party. The options are as follows: • 100: PSTN • 101: PBX • 102: PHS • 200: GSM • 201: GPRS • 202: EDGE • 300: CDMA • 301: CDMA 450 • 400: WCDMA • 500: TD-SCDMA • 600: IP/IMS • 601: WIFI • 602: WIMAX • Other to be extended Can be null
134	CalledNetworkType	Char[10]	Indicates the type of the network accessed by the called party. The options are as follows: • 100: PSTN • 101: PBX • 102: PHS • 200: GSM • 201: GPRS • 202: EDGE • 300: CDMA • 301: CDMA 450 • 400: WCDMA • 500: TD-SCDMA • 600: IP/IMS • 601: WIFI • 602: WIMAX • Other to be extended

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SN	Field Name	Field Type	Description
			Can be null
135	GroupCallType	Char[10]	Indicates the call type when the charged subscriber is a VPN user. The options are as follows: • 0: On-net call • 1: Off-net call • 2: cross-net call • 3: Common call • 4: Off-net number group call • 5: Attendant call • 6: Charge-free call • 7: Closed group call • 8: super-group call • 9: cell based charge • 10: AIP attendant call • 11: Off-net number group call at group member level • Other: to be extended Can be null
136	GroupPayFlag	Char[10]	Indicates the payment mode of the group users. The options are as follows: • 0: Personal payment • 1: Group payment • 2: Group and Personal payment Can be null
137	OnlineType	Integer[4]	Indicates online type. The options are as follows: • 0: Online • 1: Offline Can be null
138	Reserved	Char[10]	-
139	Reserved	Char[10]	-
140	Reserved	Char[10]	-
141	Reserved	Char[10]	-
142	AddtionalInfo	Char[256]	Reserved
143	StartTimeOfBill Cycle	Char[14]	Indicates the start time of bill cycle. The default value is **0**.

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SN	Field Name	Field Type	Description
144	SecondChargingPartyNumber	Char[64]	Indicates the number of the other charging party of the child-parent cards. In the example, the value of the field is the number of subscriber C. The number is same as calling party number in MO and called party number in MT. There are several special situations. Like Child-Parent card feature, the calling party number is the child, the charging party number is the parent's number. This field is null when the service is paid only by account. The number format are as follows: • **Area number + PSTN** The number format is the format of the fixed-line telephone number. • **MSISDN** The number format is the format of the mobile phone number. Value range: numerals within 0-9, *, #
145	SecondChargingPartyPayType	Char[1]	Indicates the payment type of the other charging party of the child-parent cards. For example, the payment type of the parent card. The options are as follows: • 0: Prepaid • 1: Postpaid **NOTE** If the value of the field is null, the system records the payment type of the parent card subscriber.
146	SecondChargingPartyBillCycleID	Char[20]	Indicates the bill cycle ID of the other charging party of the child-parent cards. For example, the bill cycle ID of the parent card. Format: YYYYMMTT The description of the format is as follows: • YYYY: indicates the year. • MM: indicates the month. • TT: indicates the type of a bill cycle.
147	ChargeOfItemsSecondAccounts	Integer[8]	Indicates the free call times used from others, such as the parent card. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
148	ChargeOfDurationSecondAccounts	Integer[8]	Indicates the total call time charge using reward resource from others, such as the parent card. Value range: -2^{63}-2^{63}-1

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SN	Field Name	Field Type	Description
			Not null The default value is **0**.
149	ChargeOfFundSecondAccounts	Integer[8]	Indicates the total cost amount charging from fund account of others, such as the parent card. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
150	ChargeFromSecondPrepaidAccount	Integer[8]	Indicates the charge amount from other prepaid account, such as the parent card. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
151	SecondPrepaidAccountBalance	Integer[8]	Indicates the balance of other prepaid account. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
152	ChargeFromSecondPostpaidAccount	Integer[8]	Indicates the charge amount from other postpaid account, such as the parent card. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
153	SecondPostpaidAccountBalance	Integer[8]	Indicates the balance of other postpaid account. Value range: -2^{63}-2^{63}-1 The default value is **0**.
154	SecondAccountID	Char[20]	Indicates the main account ID of the other charging party that the system deducts fees from. Value range: characters in [0-9, a-z, A-Z]
155	SecondAccountKey	Integer[8]	Indicates the main account key of the other charging party that the system deducts fees from. Value range: -2^{63}-2^{63}-1 The default value is **0**.
156	SecondAccountCurrencyCode	Integer[4]	Currency code of others, such as the parent card. Also, if this field missed or present be Zero, it means the system use local minimal currency as default, and no need to specify in CDR. The default value is **0**.
157	Reserved	Char[20]	-
158	SecondAccountType1	Integer[4]	Indicates the account type of the first account of others, such as the parent card.

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SN	Field Name	Field Type	Description
			0 means not used. Not null The default value is **0**.
159	SecondFeeType1	Integer[4]	Indicates the fee type of the first account of others, such as the parent card. 0 means not used. Not null The default value is **0**.
160	SecondChargeAmount1	Integer[8]	Indicates the changing amount of the first account of others, such as the parent card. 0 means not unchanged. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
161	SecondCurrentAcctAmount1	Integer[8]	Indicates the current amount of the first account of others, such as the parent card. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
162	SecondAccountType2	Integer[4]	Indicates the account type of the second account of others, such as the parent card. 0 means not used. Not null The default value is **0**.
163	SecondFeeType2	Integer[4]	Indicates the fee type of the second account of others, such as the parent card. 0 means not used. Not null The default value is **0**.
164	SecondChargeAmount2	Integer[8]	Indicates the changing amount of the second account of others, such as the parent card. 0 means not unchanged. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
165	SecondCurrentAcctAmount2	Integer[8]	Indicates the current amount of the second account of others, such as the parent card. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.

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SN	Field Name	Field Type	Description
166	SecondAccountType3	Integer[4]	Indicates the account type of the third account of others, such as the parent card. 0 means not used. Not null The default value is **0**.
167	SecondFeeType3	Integer[4]	Indicates the fee type of the third account of others, such as the parent card. 0 means not used. Not null The default value is **0**.
168	SecondChargeAmount3	Integer[8]	Indicates the changing amount of the third account of others, such as the parent card. 0 means not unchanged. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
169	SecondCurrentAcctAmount3	Integer[8]	Indicates the current amount of the third account of others, such as the parent card. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
170	SecondAccountType4	Integer[4]	Indicates the account type of the fourth account of others, such as the parent card. 0 means not used. Not null The default value is **0**.
171	SecondFeeType4	Integer[4]	Indicates the fee type of the fourth account of others, such as the parent card. 0 means not used. Not null The default value is **0**.
172	SecondChargeAmount4	Integer[8]	Indicates the changing amount of the fourth account of others, such as the parent card. 0 means not unchanged. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
173	SecondCurrentAcctAmount4	Integer[8]	Indicates the current amount of the fourth account of others, such as the parent card. Value range: -2^{63}-2^{63}-1

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SN	Field Name	Field Type	Description
			Not null The default value is **0**.
174	SecondAccountType5	Integer[4]	Indicates the account type of the fifth account of others, such as the parent card. 0 means not used. Not null The default value is **0**.
175	SecondFeeType5	Integer[4]	Indicates the fee type of the fifth account of others, such as the parent card. 0 means not used. Not null The default value is **0**.
176	SecondChargeAmount5	Integer[8]	Indicates the changing amount of the fifth account of others, such as the parent card. 0 means not unchanged. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
177	SecondCurrentAcctAmount5	Integer[8]	Indicates the current amount of the fifth account of others, such as the parent card. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
178	SecondBonusValidity1	Char[14]	Indicates the validity period of the first rewarded resource of others, such as the parent card. The validity period does not change when fees are deducted from accounts. In this case, the validity periods of consumption resources are not recorded. Format: YYYYMMDDhhmmss
179	SecondBonusValidity2	Char[14]	Indicates the validity period of the second rewarded resource of others, such as the parent card. The validity period does not change when fees are deducted from accounts. In this case, the validity periods of consumption resources are not recorded. Format: YYYYMMDDhhmmss
180	SecondBonusValidity3	Char[14]	Indicates the validity period of the third rewarded resource of others, such as the parent card. The validity period does not change when fees are deducted from accounts. In this case, the validity periods of consumption resources are not recorded.

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SN	Field Name	Field Type	Description
			Format: YYYYMMDDhhmmss
181	SecondBonusValidity4	Char[14]	Indicates the validity period of the fourth rewarded resource of others, such as the parent card. The validity period does not change when fees are deducted from accounts. In this case, the validity periods of consumption resources are not recorded. Format: YYYYMMDDhhmmss
182	SecondBonusValidity5	Char[14]	Indicates the validity period of the fifth rewarded resource of others, such as the parent card. The validity period does not change when fees are deducted from accounts. In this case, the validity periods of consumption resources are not recorded. Format: YYYYMMDDhhmmss
183	CallingClusterNumber	Integer[4]	Indicates the calling cluster number. The default value is **-1**.
184	CalledClusterNumber	Integer[4]	Indicates the called cluster number. The default value is **-1**.
185	OriginalCallingParty	Char[64]	Indicates the original calling number. The field is used in the forwarding process. For example, subscriber A calls subscriber B and the call is forwarded to subscriber C. In this case, subscriber B is the calling party, subscriber A is the original calling party, and the number of subscriber A is the original calling number. Value range: numerals within 0-9, *, #
186	PulaPulaProperty	Integer[4]	Reserved
187	BumerangeProperty	Integer[4]	Reserved
188	OppositePartyNumber	Char[64]	Indicates the number of the opposite party. In calling process, the number is the number of the called party. In called process, the number is the number of the calling party. The number format are as follows: • **Area number + PSTN** The number format is the format of the fixed-line telephone number. • **MSISDN** The number format is the format of the mobile phone number.

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SN	Field Name	Field Type	Description
			If it is a international number, the number format are as follows: • **Country code + Area number + PSTN** The number format is the format of the fixed-line phone number. • **Country code + MSISDN** The number format is the format of the mobile phone number. Value range: numerals within 0-9, *, # Not null The default value is **null**.
189	RegionalZoneID	Integer[4]	Reserved
190	DestinationName	Char[25]	Reserved
191	RateName	Char[25]	Reserved
192	DiameterSessionID	Char[128]	Indicates a diameter session ID. The format is as follows: <DiameterIdentity>;<high 32 bits>;<low 32 bits>[;<optional value>] • <DiameterIdentity>: the same as the field in the Origin-Host AVP. • <high 32 bits>: indicates the current time of the system. • <low 32 bits>: indicates the cycle count-up. The initial value is **0**. When you restart the system, set the value to **0**. • <optional value>: indicates implementation specific but may include a modem's device Id, a layer 2 address, timestamp, etc. Separate the preceding fields with the character ";". For example: SCP001.huawei.com;1876543210;523
193	StartcountryGroup	Integer[4]	Indicates the start charging country group, that is, the country group on the start location in toll call charging. Not null The default value is **0**.
194	EndcountryGroup	Integer[4]	Indicates the end charging country group, that is, the country group on the end location in toll call charging. Not null

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SN	Field Name	Field Type	Description
			The default value is **0**.
195	usergrade	Integer[4]	Reserved
196	IsTestNumber	Integer[4]	Reserved
197	MainProductStatus	Integer[4]	Reserved
198	SubscriberIDType	Char[1]	Reserved
199	OriginatingIOI	Char[32]	Reserved
200	TerminatingIOI	Char[32]	Reserved
201	IMSChargingIdentifier	Char[64]	Reserved
202	Pulse	Integer[4]	Reserved
203	StartTimOfOCS	Char[14]	Reserved
204	DiscountOfLastEffProduct	Integer[4]	Indicates the discount percentage for the last rated product that takes effect. For a product that does not provide discounts, the value is **100**. Value range: 0-100 **NOTE** The value 0 is not included. Not null The default value is **100**.
205	subscriberkey	Integer[4]	Indicates the key of a subscriber. The default value is **0**.
206	accumulationchange	Char[256]	Reserved
207	UsageServiceType	Integer[4]	Indicates the type of a real-time service. The default value is **1100**.
208	SourceSerialNo	Integer[8]	Reserved
209	SourceSubSequence	Integer[4]	Reserved
210	AcctBalChgList	Char[768]	Reserved
211	SubscriberType	Integer[4]	Reserved
212	OriginCdrFilename	Char[64]	Reserved
213	CurCDRLineNo	Integer[4]	Reserved
214	DDSDiscountO	Integer[4]	Reserved

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SN	Field Name	Field Type	Description
	ff		
215	DDSLocationNumber	Char[64]	Indicates to record the information of **LocationNumber** in DDS feature. Value range: degitals within [0,9],*,# **NOTE** The field is recorded in calling flow, and is null in called flow.
216	DTDiscount	String	Indicates dynamic discount. Can be null.
217	DTCellId	String	Indicates the Cell ID considered for DT discount. In case of default DT discount, record it as '000000000000000'. Can be null.

3.1.3 sms CDR

Description

The sms CDRs are the CDRs for the SMS. The CDRs record the following information in SMS charging requests: the service scenario information reported by external NEs and the information generated by the charging modules, such as rating and fee deduction information.

Generation Scenario

When OCS subscriber A sends an SM to subscriber B, and the system successfully deducts the fee from the account of subscriber A, an sms CDR is generated.

NOTE
For the description of child-parent cards, see section 3.1.2 rec CDR .

CDR File Name Format

smsYYYYMMDD_CBPNODE_Seqno.unl

Example of an sms CDR file name: sms20071112_101_000151.unl

Format

Table 3-4 describes each field of sms CDR file.

Table 3-4 Format of sms CDR

SN	Field Name	Field Type	Description
1	SerialNo	Integer[8]	Indicates the serial number. It is generated by the system and the serial

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SN	Field Name	Field Type	Description
			number is exclusive. Not null The default value is **0**.
2	SubSequence	Integer[4]	Indicates the sequence number when the bill is split for mid-night and end of the bill cycle. The default value is **0**.
3	TimeStamp	Char[14]	Indicates the timestamp of the bill item generation. Format: YYYYMMDDhhmmss Not null
4	ServiceKey	Integer[4]	Indicates the service key of the intelligent service to which the subscriber subscribes. The default value is **1**.
5	CallingPartyNumber	Char[64]	Indicates the calling number. The number format are as follows: • **Country code + Area number + PSTN** The number format is the format of the fixed-line phone number. • **Country code + MSISDN** The number format is the format of the mobile phone number. **NOTE** For E164 number, using international format. For SPID, keep the original format. Value range: numerals within 0-9, *, #
6	CalledPartyNumber	Char[64]	Indicates the called number. The number format are as follows: • **Country code + Area number + PSTN** The number format is the format of the fixed-line phone number. • **Country code + MSISDN** The number format is the format of the mobile phone number. **NOTE** For E164 number, using international format. For SPID, keep the original format. Value range: numerals within 0-9, *, #
7	CallingPartyIMSI	Char[64]	Indicates the IMSI of calling party. • When the SCP transfers an IMSI, the

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SN	Field Name	Field Type	Description
			value of this field is the IMSI transferred • When the SCP does not transfer an IMSI, the value of this field is the value of the corresponding field in the attribute table of the charged subscriber. • The value is **Null** when the relevant value cannot be obtained. Value range: numerals within 0-9, *, #
8	CalledPartyIMSI	Char[64]	Indicates the IMSI of called party. • When the SCP transfers an IMSI, the value of this field is the IMSI transferred • When the SCP does not transfer an IMSI, the value of this field is the value of the corresponding field in the attribute table of the charged subscriber. • The value is **Null** when the relevant value cannot be obtained. Value range: numerals within 0-9, *, #
9	DialedNumber	Char[64]	Indicates the original number that calling dialed in calling process. It is valid in MO process. Value range: numerals within 0-9, *, #
10	OriginalCalledParty	Char[64]	Indicates the original called number in a forwarding process. Special number reroute. International number form in call forwarding, actual call number in special number reroute, filling in original called number when dialed through an international gateway office. Value range: numerals within 0-9, *, #
11	ServiceFlow	Char[1]	Indicates the call process indicator. The options are as follows: • 1: MO process • 2: MT process
12	CallForwardIndicator	Char[1]	Indicates the SMS forwarding flag. In MT: A-B-C, In MT process of B, it needs to know that whether B party transfers the SMS to C. Usually, the MT of B party is set free and all the cost will be charged in MF of B party in this

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SN	Field Name	Field Type	Description
			situation. In MF: this field is related to the parameter of "redirect reason" in IDP. • 0: No forwarding • 1: SMS forwarding conditional • 2: SMS forwarding unconditional The default value is **0**.
13	CallingRoamInfo	Char[20]	Indicates the information used to analyze the roaming state of calling party. It can be the location number, MSC address, VLR number, and so on. The actual value is related with the service configuration. Value range: numerals within 0-9, *, #
14	CallingCellID	Char[20]	Indicates the cell ID of the calling party, which is directly filled in according to the IDP report. Value range: numerals within 0-9, *, #
15	CalledRoamInfo	Char[20]	Indicates the information used to analyze the roaming state of called party. It can be the location number, MSC address, VLR number, and so on. The actual value is related with the service configuration. Value range: numerals within 0-9, *, #
16	CalledCellID	Char[20]	Indicates the cell ID of the called party, which is handled just as that for **CallingCellID**. Value range: numerals within 0-9, *, #
17	TimeStampOfSSP	Char[14]	Indicates the time stamp report from SSP in IDP. If missing in charging interface, this field will be filled with the local time stamp of OCS. Format: YYYYMMDDhhmmss
18	TimeZoneOfSSP	Integer[4]	Indicates the time zone report from SSP in IDP. If missing in charging interface, this field will be filled with the time zone of the location of OCS, 15 min. per sector. Value range: -48-48 **NOTE** The value -48 is not included.

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SN	Field Name	Field Type	Description
			The default value is **0**.
19	BearerCapability	Char[32]	Reserved
20	call ID	Char[32]	Reserved
21	ChargingTime	Char[14]	Indicates the time stamp of the charging beginning. Format: YYYYMMDDhhmmss Not null
22	package	Char[32]	Reserved
23	ChargeAmount	Integer[8]	Indicates the deduction amount that is requested from an external network element, such as MDSP. The default value is **0**. **NOTE** NE = Network element
24	SendResult	Integer[4]	Indicates the results of short message sending: • 0: Successfully • Other: The value is based on the value transferred from the SCP. Value range: -2^{31}-2^{31}-1 The default value is **0**.
25	SMSID	Char[24]	Indicates the ID of a short message.
26	IMEI	Char[24]	Indicates the IE containing the IMEISV of the ME in use by the served subscriber.
27	SMLength	Integer[4]	Indicates the length of short message. The default value is **0**.
28	Reserved	Char[20]	-
29	APN	Char[32]	Reserved
30	RefundIndicator	Char[20]	Indicates the deduction or refund request. The options are as follows: • 0: Deduct • 1: Refund
31	BrandID	Integer[4]	Indicates the brand ID of subscribers. Not null The default value is **0**.
32	SubCOSID	Integer[4]	Indicates the main product ID of subscribers. Not null

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SN	Field Name	Field Type	Description
			The default value is **0**.
33	ChargingPartyNumber	Char[64]	Indicates the telephone number of the charging party. The number is same as calling party number in MO and called party number in MT. There are several special situations. Like Child-Parent card feature, the calling party number is the child, the charging party number is the parent's number. This field is null when the service is paid only by account. The number format are as follows: • **Area number + PSTN** The number format is the format of the fixed-line telephone number. • **MSISDN** The number format is the format of the mobile phone number. Value range: numerals within 0-9, *, #
34	ChargePartyIndicator	Char[1]	Indicates the charging party. The options are as follows: • 1: The calling party is charged • 2: The called party is charged • 3: Charge of third party • 4: Charge of parent & child card • 5: Charge of calling party & VPN
35	PayType	Char[1]	Indicates the pay type of subscribers. The options are as follows: • 0: Prepaid • 1: Postpaid
36	BillCycleID	Char[20]	Indicates the bill cycle ID. Format: YYYYMMTT • YYYY: indicates the year. • MM: indicates the month. • TT: indicates the type of a bill cycle. Value range: numerals within 0-9
37	ChargingType	Char[1]	Indicates the charging type. For MO/MF process: • 0: V-H • 1: H-H

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SN	Field Name	Field Type	Description
			• 2: V-V • 3: H-V For MT process: • 0: For free • 1: Roaming fee Not null
38	OnNetINdicator	Char[1]	Indicates the short message types. The options are as follows: • 0: on-net • 1: off-net • 2: unknown (when network type cannot be obtained in situations such as international roaming)
39	RoamState	Char[1]	Indicates the roaming state of charging party. The options are as follows: • 0: No roaming • 1: Roaming within charging area • 2: Roaming out of charging area • 3: International Roaming
40	ResultCode	Integer[4]	Indicates the internal result code. Value range: -2^{31}-2^{31}-1 The default value is **0**.
41	SameBrandPre	Integer[4]	Reserved
42	CallingHomeCountryCode	Integer[4]	Indicates the home country code information of calling party. >=0, -1, 0 or 9999 means unknown. The default value is **0**.
43	CallingHomeAreaNumber	Integer[4]	Indicates the home area number of the calling party. >=0, -1, 0 or 9999 means unknown. The default value is **0**.
44	CallingHomeNetworkCode	Integer[4]	Indicates the home network code of the calling party. >=0, -1, 0 or 9999 means unknown. The default value is **0**.
45	CallingRoamCountryCode	Integer[4]	Indicates the roam country code of the calling party. >=0, -1, 0 or 9999 means unknown.

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SN	Field Name	Field Type	Description
			The default value is **0**.
46	CallingRoamAreaNumber	Integer[4]	Indicates the roam area number of the calling party. >=0, -1, 0 or 9999 means unknown. The default value is **0**.
47	CallingRoamNetworkCode	Integer[4]	Indicates the roam network code of the calling party. >=0, -1, 0 or 9999 means unknown. The default value is **0**.
48	CalledHomeCountryCode	Integer[4]	Indicates the home country code of the called party. >=0, -1, 0 or 9999 means unknown. The default value is **0**.
49	CalledHomeAreaNumber	Integer[4]	Indicates the home area number of the called party. >=0, -1, 0 or 9999 means unknown. The default value is **0**.
50	CalledHomeNetworkCode	Integer[4]	Indicates the home network code of the called party. >=0, -1, 0 or 9999 means unknown. The default value is **0**.
51	CalledRoamCountryCode	Integer[4]	Indicates the roam country code of the called party. >=0, -1, 0 or 9999 means unknown. The default value is **0**.
52	CalledRoamAreaNumber	Integer[4]	Indicates the roam area number of the called party. >=0, -1, 0 or 9999 means unknown. The default value is **0**.
53	CalledRoamNetworkCode	Integer[4]	Indicates the roam network code of the called party. >=0, -1, 0 or 9999 means unknown. The default value is **0**.
54	ProductID	Char[20]	Indicates the last effect rating product.
55	ServiceType	Char[32]	Indicates the service included in the product. They are predefined in OCS. The options are as follows: • 1st digit: Normal tariff • 2nd digit: Familiarity number service

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SN	Field Name	Field Type	Description
			• 3rd digit: Data service • 4th digit: IP/international carrier call • 5th digit: pre-IP call • 6th digit: International Call Forward • 7th digit: Call from Home Zone • 8th digit: UCB service • 9th digit: P_FPH service • 10th digit: Remote Access • 11th digit: Call from Special Zone • Other: reserved The value range of each digit indicates whether the service is activated. • 0: Not activated • 1: Activated
56	SpecialNumberIndicator	Integer[4]	Types of special numbers that send short messages, which are used to identify call-free numbers, for example, A2P or P2A. The options are as follows: • 0: Normal SMS/P2P • 1: Free SMS • 2: BarringNumber The default value is **0**.
57	HomeZoneID	Integer[4]	Indicates the home zone ID. When a subscriber enjoys the home zone discount, the value of this field is the home zone ID of the area where the subscriber currently locates. The default value is **0**.
58	NPFlag	Char[1]	Indicates the flag of number portable. The options are as follows: • 0: Not NP • 1: NP out The default value is **0**.
59	NPPrefix	Char[6]	Indicates the number portability prefix of called party in the MO process or number portability prefix of the calling party in the MT process. Value range: numerals within 0-9, *, #
60	CallingGroupNo	Integer[4]	Indicates the group number of the calling party when enjoying CUG service. It is the

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SN	Field Name	Field Type	Description
			group number related to the rating. For VPN, it is the nearest level groupNo. The default value is **0**.
61	CalledGroupNo	Integer[4]	Indicates the group number of the called party when enjoying CUG or VPN service. It is the group number related to the rating. The default value is **0**.
62	UserState	Char[10]	Indicates the life cycle state and the management state of a subscriber. The **UserState** parameter is a numeric string of seven digits, in the format CCMMMMM. The first and second digits C indicate the life cycle state of a subscriber. The first digit C indicates the state of a prepaid subscriber, and the second digit C indicates the state of a postpaid subscriber. The options are as follows: • 0: Idle • 1: Active • 2: Suspended • 3: Disable • 4: Pool • 5: Pool without activation The next five digits MMMMM indicate the management state of a subscriber. The options are as follows: Third digit: Whether the subscriber claims missing. • 0: no • 1: yes Fourth digit: Whether the subscriber suspends the service. • 0: non-suspend • 1: suspend • 2: suspend by operator (B1W) • 3: suspend by operator (B2W) Fifth digit: Whether the subscriber applies for suspension with the number being reserved. • 0: no • 1: yes Sixth digit: Whether the subscriber is

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SN	Field Name	Field Type	Description
			blocked. • 0: no • 1: yes Seventh digit: Whether the subscriber is blacklisted. • 0: no • 1: yes
63	SubscriberID	Char[20]	Indicates the subscriber ID. Value range: characters in [0-9, a-z, A-Z]
64	OpposeNetworkType	Integer[4]	Indicates the network type of oppose side. The options are as follows: • 1: PSTN • 2: Mobile • 3: Native Unknown • 4: International Unknown • Other: to be extended The default value is **0**.
65	SMSType	Integer[4]	Indicates the type of a short message call. The options are as follows: • 0: local • 1: intra-charging area toll • 2: inter-charging area toll • 3: international toll The default value is **0**.
66	ChargeOfItemsAccounts	Integer[8]	Indicates the free short message items used. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
67	Reserved	Char[20]	-
68	ChargeOfFundAccounts	Integer[8]	Indicates the total cost amount charging from fund account. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
69	ChargeFromPrepaid	Integer[8]	Indicates the charge amount from prepaid account. Value range: -2^{63}-2^{63}-1 Not null

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SN	Field Name	Field Type	Description
			The default value is **0**.
70	PrepaidBalance	Integer[8]	Indicates the balance of prepaid account. Not null The default value is **0**.
71	ChargeFromPostpaid	Integer[8]	Indicates the charge amount from postpaid account. Not null The default value is **0**.
72	PostPaidBalance	Integer[8]	Indicates the balance of postpaid available limitation. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
73	AccountID	Char[20]	Indicates the ID of the main account. Value range: characters in [0-9, a-z, A-Z]
74	AccountKey	Integer[8]	Indicates the key of the main account. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
75	CurrencyCode	Integer[4]	Currency code. Also, if this field missed or present be Zero, it means the system use local minimal currency as default, and no need to specify in CDR. The default value is **0**.
76	Reserved	Char[20]	-
77	AccountType1	Integer[4]	Indicates the account type of the first account. 0 means not used. Not null The default value is **0**.
78	FeeType1	Integer[4]	Indicates the fee type of the first account. 0 means not used. Not null The default value is **0**.
79	ChargeAmount1	Integer[8]	Indicates the changing amount of the first account. 0 means unchanged. Value range: -2^{63}-2^{63}-1 Not null

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SN	Field Name	Field Type	Description
			The default value is **0**.
80	CurrentAcctAmount1	Integer[8]	Indicates the current amount of the first account. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
81	AccountType2	Integer[4]	Indicates the account type of the second account. 0 means not used. Not null The default value is **0**.
82	FeeType2	Integer[4]	Indicates the fee type of the second account. 0 means not used. Not null The default value is **0**.
83	ChargeAmount2	Integer[8]	Indicates the changing amount of the second account. 0 means unchanged. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
84	CurrentAcctAmount2	Integer[8]	Indicates the current amount of the second account. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
85	AccountType3	Integer[4]	Indicates the account type of the third account. 0 means not used. Not null The default value is **0**.
86	FeeType3	Integer[4]	Indicates the fee type of the third account. 0 means not used. Not null The default value is **0**.
87	ChargeAmount3	Integer[8]	Indicates the changing amount of the third account. 0 means unchanged.

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SN	Field Name	Field Type	Description
			Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
88	CurrentAcctAmount3	Integer[8]	Indicates the current amount of the third account. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
89	AccountType4	Integer[4]	Indicates the account type of the fourth account. 0 means not used. Not null The default value is **0**.
90	FeeType4	Integer[4]	Indicates the fee type of the fourth account. 0 means not used. Not null The default value is **0**.
91	ChargeAmount4	Integer[8]	Indicates the changing amount of the fourth account. 0 means unchanged. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
92	CurrentAcctAmount4	Integer[8]	Indicates the current amount of the fourth account. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
93	AccountType5	Integer[4]	Indicates the account type of the fifth account. 0 means not used. Not null The default value is **0**.
94	FeeType5	Integer[4]	Indicates the fee type of the fifth account. 0 means not used. Not null The default value is **0**.
95	ChargeAmount5	Integer[8]	Indicates the changing amount of the fifth

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SN	Field Name	Field Type	Description
			account. 0 means unchanged. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
96	CurrentAcctAmount5	Integer[8]	Indicates the current amount of the fifth account. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
97	AccountType6	Integer[4]	Indicates the account type of the sixth account. 0 means not used. Not null The default value is **0**.
98	FeeType6	Integer[4]	Indicates the fee type of the sixth account. 0 means not used. Not null The default value is **0**.
99	ChargeAmount6	Integer[8]	Indicates the changing amount of the sixth account. 0 means unchanged. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
100	CurrentAcctAmount6	Integer[8]	Indicates the current amount of the sixth account. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
101	AccountType7	Integer[4]	Indicates the account type of the seventh account. 0 means not used. Not null The default value is **0**.
102	FeeType7	Integer[4]	Indicates the fee type of the seventh account. 0 means not used. Not null

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SN	Field Name	Field Type	Description
			The default value is **0**.
103	ChargeAmount7	Integer[8]	Indicates the changing amount of the seventh account. 0 means unchanged. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
104	CurrentAcctAmount7	Integer[8]	Indicates the current amount of the seventh account. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
105	AccountType8	Integer[4]	Indicates the account type of the eighth account. 0 means not used. Not null The default value is **0**.
106	FeeType8	Integer[4]	Indicates the fee type of the eighth account. 0 means not used. Not null The default value is **0**.
107	ChargeAmount8	Integer[8]	Indicates the changing amount of the eighth account. 0 means unchanged. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
108	CurrentAcctAmount8	Integer[8]	Indicates the current amount of the eighth account. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
109	AccountType9	Integer[4]	Indicates the account type of the ninth account. 0 means not used. Not null
110	FeeType9	Integer[4]	Indicates the fee type of the ninth account. 0 means not used.

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SN	Field Name	Field Type	Description
			Not null The default value is **0**.
111	ChargeAmount9	Integer[8]	Indicates the changing amount of the ninth account. 0 means unchanged. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
112	CurrentAcctAmount9	Integer[8]	Indicates the current amount of the ninth account. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
113	AccountType10	Integer[4]	Indicates the account type of the tenth account. 0 means not used. Not null The default value is **0**.
114	FeeType10	Integer[4]	Indicates the fee type of the tenth account. 0 means not used. Not null The default value is **0**.
115	ChargeAmount10	Integer[8]	Indicates the changing amount of the tenth account. 0 means unchanged. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
116	CurrentAcctAmount10	Integer[8]	Indicates the current amount of the tenth account. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
117	BonusValidity1	Char[14]	Indicates the validity period of the first rewarded resource. The validity period does not change when fees are deducted from accounts. In this case, the validity periods of consumption resources are not recorded. Format: YYYYMMDDhhmmss

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SN	Field Name	Field Type	Description
118	BonusValidity2	Char[14]	Indicates the validity period of the second rewarded resource. The validity period does not change when fees are deducted from accounts. In this case, the validity periods of consumption resources are not recorded. Format: YYYYMMDDhhmmss
119	BonusValidity3	Char[14]	Indicates the validity period of the third rewarded resource. The validity period does not change when fees are deducted from accounts. In this case, the validity periods of consumption resources are not recorded. Format: YYYYMMDDhhmmss
120	BonusValidity4	Char[14]	Indicates the validity period of the fourth rewarded resource. The validity period does not change when fees are deducted from accounts. In this case, the validity periods of consumption resources are not recorded. Format: YYYYMMDDhhmmss
121	BonusValidity5	Char[14]	Indicates the validity period of the fifth rewarded resource. The validity period does not change when fees are deducted from accounts. In this case, the validity periods of consumption resources are not recorded. Format: YYYYMMDDhhmmss
122	BonusValidity6	Char[14]	Indicates the validity period of the sixth rewarded resource. The validity period does not change when fees are deducted from accounts. In this case, the validity periods of consumption resources are not recorded. Format: YYYYMMDDhhmmss
123	BonusValidity7	Char[14]	Indicates the validity period of the seventh rewarded resource. The validity period does not change when fees are deducted from accounts. In this case, the validity periods of consumption resources are not recorded. Format: YYYYMMDDhhmmss
124	BonusValidity8	Char[14]	Indicates the validity period of the eighth rewarded resource. The validity period does not change when fees are deducted from accounts. In this case, the validity periods of consumption resources are not

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SN	Field Name	Field Type	Description
			recorded. Format: YYYYMMDDhhmmss
125	BonusValidity9	Char[14]	Indicates the validity period of the ninth rewarded resource. The validity period does not change when fees are deducted from accounts. In this case, the validity periods of consumption resources are not recorded. Format: YYYYMMDDhhmmss
126	BonusValidity10	Char[14]	Indicates the validity period of the tenth rewarded resource. The validity period does not change when fees are deducted from accounts. In this case, the validity periods of consumption resources are not recorded. Format: YYYYMMDDhhmmss
127	Reserved	Char[15]	Indicates the top group number of the calling party. Value range: numerals within 0-9 Can be null
128	CallingVPNGroupNumber	Char[10]	Indicates the group number of the calling party. Value range: numerals within 0-9 Can be null
129	CallingVPNShortNumber	Char[10]	Indicates the short number of the calling party. Value range: numerals within 0-9 Can be null
130	CalledVPNTopGroupNumber	Char[15]	Indicates the top group number of the called party. Value range: numerals within 0-9 Can be null
131	CalledVPNGroupNumber	Char[10]	Indicates the group number of the called party. Value range: numerals within 0-9 Can be null
132	CalledVPNShortNumber	Char[10]	Indicates the short number of the called party Value range: numerals within 0-9 Can be null
133	CallingNetworkT	Char[10]	Indicates the type of the network accessed

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SN	Field Name	Field Type	Description
	ype		by the calling party. The options are as follows: • 100: PSTN • 101: PBX • 102: PHS • 200: GSM • 201: GPRS • 202: EDGE • 300: CDMA • 301: CDMA 450 • 400: WCDMA • 500: TD-SCDMA • 600: IP/IMS • 601: WIFI • 602: WIMAX • Other to be extended Can be null
134	CalledNetworkType	Char[10]	Indicates the type of the network accessed by the called party. The options are as follows: • 100: PSTN • 101: PBX • 102: PHS • 200: GSM • 201: GPRS • 202: EDGE • 300: CDMA • 301: CDMA 450 • 400: WCDMA • 500: TD-SCDMA • 600: IP/IMS • 601: WIFI • 602: WIMAX • Other to be extended Can be null
135	GroupCallType	Char[10]	Reserved
136	GroupPayFlag	Char[10]	Indicates the payment mode of the group users. The options are as follows:

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SN	Field Name	Field Type	Description
			• 0: Personal payment • 1: Group payment • 2: Group and Personal payment Can be null
137	OnlineType	Integer[4]	Indicates the charge type when the charged subscriber is a VPN user. The options are as follows: • 0: Online • 1: Offline Can be null
138	session id	Char[20]	Reserved
139	Service Name	Char[32]	Reserved
140	Consumed Resources	Integer[4]	Reserved
141	Original Amount	Integer[4]	Reserved
142	AddtionalInfo	Char[256]	Reserved
143	Reserved	Char[20]	-
144	StartTimeOfBillCycle	Char[14]	Indicates the start time of bill cycle. The default value is **0**.
145	SecondChargingPartyNumber	Char[64]	Indicates the number of the other charging party. In the example, the value of the field is the number of subscriber C. This field is null when the service is paid only by account. There are several special situations. Like Child-Parent card feature, the calling party number is the child, the charging party number is the parent's number. The number format is as follows: • **Area number + PSTN** The number format is the format of the fixed-line telephone number. • **MSISDN** The number format is the format of the mobile phone number. Value range: numerals within 0-9, *, #
146	SecondChargingPartyPayType	Char[1]	Indicates the payment type of the other charging party. For example, the payment type of the parent card. The options are as follows:

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SN	Field Name	Field Type	Description
			• 0: Prepaid • 1: Postpaid **NOTE** If the value of the field is null, the system records the payment type of the parent card subscriber.
147	SecondChargingPartyBillCycleID	Char[20]	Indicates the bill cycle ID of the other charging party. For example, the bill cycle ID of the parent card. Format: YYYYMMTT The description of the format is as follows: • YYYY: indicates the year. • MM: indicates the month. • TT: indicates the type of a bill cycle.
148	ChargeOfItemsSecondAccounts	Integer[8]	Indicates the free call times used from others, such as the parent card. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
149	ChargeOfDurationSecondAccounts	Integer[8]	Indicates the total call time charge using reward resource from others, such as the parent card. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
150	ChargeOfFundSecondAccounts	Integer[8]	Indicates the total cost amount charging from fund account of others, such as the parent card. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
151	ChargeFromSecondPrepaidAccount	Integer[8]	Indicates the charge amount from other prepaid account, such as the parent card. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
152	SecondPrepaidAccountBalance	Integer[8]	Indicates the balance of other prepaid account. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.

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SN	Field Name	Field Type	Description
153	ChargeFromSecondPostpaidAccount	Integer[8]	Indicates the charge amount from other postpaid account, such as the parent card. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
154	SecondPostpaidAccountBalance	Integer[8]	Indicates the balance of other postpaid account. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
155	SecondAccountID	Char[20]	Indicates the main account ID of the other charging party that the system deducts fees from. Value range: characters in [0-9, a-z, A-Z]
156	SecondAccountKey	Integer[8]	Indicates the main account key of the other charging party that the system deducts fees from. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
157	SecondAccountCurrencyCode	Integer[4]	Currency code of others, such as the parent card. Also, if this field missed or present be Zero, it means the system use local minimal currency as default, and no need to specify in CDR. The default value is **0**.
158	Reserved	Char[20]	-
159	SecondAccountType1	Integer[4]	Indicates the account type of the first account of others, such as the parent card. 0 means not used. Not null The default value is **0**.
160	SecondFeeType1	Integer[4]	Indicates the fee type of the first account of others, such as the parent card. 0 means not used. Not null The default value is **0**.
161	SecondChargeAmount1	Integer[8]	Indicates the changing amount of the first account of others, such as the parent card. 0 means unchanged.

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SN	Field Name	Field Type	Description
			Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
162	SecondCurrentActAmount1	Integer[8]	Indicates the current amount of the first account of others, such as the parent card. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
163	SecondAccountType2	Integer[4]	Indicates the account type of the second account of others, such as the parent card. 0 means not used. Not null The default value is **0**.
164	SecondFeeType2	Integer[4]	Indicates the fee type of the second account of others, such as the parent card. 0 means not used. Not null The default value is **0**.
165	SecondChargeAmount2	Integer[8]	Indicates the changing amount of the second account of others, such as the parent card. 0 means unchanged. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
166	SecondCurrentActAmount2	Integer[8]	Indicates the current amount of the second account of others, such as the parent card. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
167	SecondAccountType3	Integer[4]	Indicates the account type of the third account of others, such as the parent card. 0 means not used. Not null The default value is **0**.
168	SecondFeeType3	Integer[4]	Indicates the fee type of the third account of others, such as the parent card. 0 means not used. Not null The default value is **0**.

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SN	Field Name	Field Type	Description
169	SecondChargeAmount3	Integer[8]	Indicates the changing amount of the third account of others, such as the parent card. 0 means unchanged. Value range: -2^{63}-2^{63}-1 Not null
170	SecondCurrentActAmount3	Integer[8]	Indicates the current amount of the third account of others, such as the parent card. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
171	SecondAccountType4	Integer[4]	Indicates the account type of the fourth account of others, such as the parent card. 0 means not used. Not null The default value is **0**.
172	SecondFeeType4	Integer[4]	Indicates the fee type of the fourth account of others, such as the parent card. 0 means not used. Not null The default value is **0**.
173	SecondChargeAmount4	Integer[8]	Indicates the changing amount of the fourth account of others, such as the parent card. 0 means unchanged. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
174	SecondCurrentActAmount4	Integer[8]	Indicates the current amount of the fourth account of others, such as the parent card. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
175	SecondAccountType5	Integer[4]	Indicates the account type of the fifth account of others, such as the parent card. 0 means not used. Not null The default value is **0**.
176	SecondFeeType5	Integer[4]	Indicates the fee type of the fifth account of others, such as the parent card. 0 means not used.

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SN	Field Name	Field Type	Description
			Not null The default value is **0**.
177	SecondChargeAmount5	Integer[8]	Indicates the changing amount of the fifth account of others, such as the parent card. 0 means unchanged. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
178	SecondCurrentActAmount5	Integer[8]	Indicates the current amount of the fifth account of others, such as the parent card. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
179	SecondBonusValidity1	Char[14]	Indicates the validity period of the first rewarded resource of others, such as the parent card. The validity period does not change when fees are deducted from accounts. In this case, the validity periods of consumption resources are not recorded. Format: YYYYMMDDhhmmss
180	SecondBonusValidity2	Char[14]	Indicates the validity period of the second rewarded resource of others, such as the parent card. The validity period does not change when fees are deducted from accounts. In this case, the validity periods of consumption resources are not recorded. Format: YYYYMMDDhhmmss
181	SecondBonusValidity3	Char[14]	Indicates the validity period of the third rewarded resource of others, such as the parent card. The validity period does not change when fees are deducted from accounts. In this case, the validity periods of consumption resources are not recorded. Format: YYYYMMDDhhmmss
182	SecondBonusValidity4	Char[14]	Indicates the validity period of the fourth rewarded resource of others, such as the parent card. The validity period does not change when fees are deducted from accounts. In this case, the validity periods of consumption resources are not recorded. Format: YYYYMMDDhhmmss

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SN	Field Name	Field Type	Description
183	SecondBonusValidity5	Char[14]	Indicates the validity period of the fifth rewarded resource of others, such as the parent card. The validity period does not change when fees are deducted from accounts. In this case, the validity periods of consumption resources are not recorded. Format: YYYYMMDDhhmmss
184	PulaPulaProperty	Integer[4]	Reserved
185	BumerangeProperty	Integer[4]	Reserved
186	OppositePartyNumber	Char[64]	Indicates the number of the opposite party. In calling process, the number is the number of the called party. In called process, the number is the number of the calling party. The number format is as follows: • **Area number + PSTN** The number format is the format of the fixed-line telephone number. • **MSISDN** The number format is the format of the mobile phone number. If it is a international number, the number format are as follows: • Country code + MSISDN • Country code + Area number + PSTN Value range: numerals within 0-9, *, # Not null
187	RegionalZoneID	Integer[4]	Reserved
188	RateName	Char[25]	Reserved
189	DiameterSessionID	Char[128]	Indicates a diameter session ID. The format is as follows: <DiameterIdentity>;<high 32 bits>;<low 32 bits>[;<optional value>] • <DiameterIdentity>: the same as the field in the Origin-Host AVP. • <high 32 bits>: indicates the current time of the system. • <low 32 bits>: indicates the cycle count-up. The initial value is **0**. When you restart the system, set the value to

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SN	Field Name	Field Type	Description
			0. • <optional value>: indicates implementation specific but may include a modem is device Id, a layer 2 address, timestamp, etc. Separate the preceding fields with the character ";". For example: SCP001.huawei.com;1876543210;523
190	IsTestNumber	Integer[4]	Reserved
191	MainProductStatus	Integer[4]	Reserved
192	DiscountOfLastEffProduct	Integer[4]	Indicates the discount percentage for the last rated product that takes effect. For a product that does not provide discounts, the value is **100**. Value range: 0-100 **NOTE** The value **0** is not included. Not null The default value is **100**.
193	subscriberkey	Integer[4]	Indicates the key of a subscriber. The default value is **0**.
194	accumulationchange	Char[256]	Reserved
195	UsageServiceType	Integer[4]	Indicates the type of a real-time service. The default value is **1300**.
196	SourceSerialNo	Integer[8]	Reserved
197	SourceSubSequence	Integer[4]	Reserved
198	AcctBalChgList	Char[768]	Reserved
199	OriginCdrFilename	Char[64]	Reserved
200	CurCDRLineNo	Integer[4]	Reserved
201	DTDiscount	String	Indicates dynamic discount. Can be null.
202	DTCellId	String	Indicates the Cell ID considered for DT discount. In case of default DT discount, record it as ‘000000000000000’.

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SN	Field Name	Field Type	Description
			Can be null.

3.1.4 mms CDR

Description

The mms CDRs are the CDRs for the MMS. The CDRs record the following information in MMS charging requests: the service scenario information reported by external NEs and the information generated by the charging modules, such as rating and fee deduction information.

Generation Scenario

When OCS subscriber A sends a multimedia message to subscriber B, and the system successfully deducts the fee from the account of subscriber A, an mms CDR is generated.

NOTE
For the description of child-parent cards, see section 3.1.2 rec CDR .

CDR File Name Format

mmsYYYYMMDD_CBPNODE_Seqno.unl

Example of an mms CDR file name: mms20071012_101_000151.unl

Format

Table 3-5 describes each field of mms CDR file.

Table 3-5 Format of mms CDR

SN	Field Name	Field Type	Description
1	SerialNo	Integer[8]	Indicates the serial number. It is generated by the system and the serial number is exclusive. Not null The default value is **0**.
2	SubSequence	Integer[4]	Indicates the sequence number when the bill is split for mid-night and end of the bill cycle. The default value is **0**.
3	TimeStamp	Char[14]	Indicates the time stamp of the bill item generation. Format: YYYYMMDDhhmmss Not null
4	ServiceKey	Integer[4]	Indicates the service key of the intelligent

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SN	Field Name	Field Type	Description
			service to which the subscriber subscribes. The default value is **1**.
5	CallingPartyNumber	Char[64]	Indicates the calling number. The number format are as follows: • **Country code + Area number + PSTN** The number format is the format of the fixed-line phone number. • **Country code + MSISDN** The number format is the format of the mobile phone number. Value range: numerals within 0-9, *, #
6	CalledPartyNumber	Char[64]	Indicates the called number. The number format are as follows: • **Country code + Area number + PSTN** The number format is the format of the fixed-line phone number. • **Country code + MSISDN** The number format is the format of the mobile phone number. Value range: numerals within 0-9, *, #
7	CallingPartyIMSI	Char[64]	Indicates the IMSI of calling party. • When the SCP transfers an IMSI, the value of this field is the IMSI transferred • When the SCP does not transfer an IMSI, the value of this field is the value of the corresponding field in the attribute table of the charged subscriber. • The value is **Null** when the relevant value cannot be obtained. Value range: numerals within 0-9, *, #
8	CalledPartyIMSI	Char[64]	Indicates the IMSI of called party. • When the SCP transfers an IMSI, the value of this field is the IMSI transferred • When the SCP does not transfer an IMSI, the value of this field is the value of the corresponding field in the attribute table of the charged subscriber. • The value is **Null** when the relevant value cannot be obtained. Value range: numerals within 0-9, *, #
9	DialedNumber	Char[64]	Indicates the original number that the MMS is sent to. It is valid in MO flow.

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SN	Field Name	Field Type	Description
			Value range: numerals within 0-9, *, #
10	OriginalCalledParty	Char[64]	Indicates the original Receiver number in a MMS forwarding flow. Special number reroute. International number form in MMS forwarding, actual call number in special number reroute, filling in original called number when dialed through an international gateway office. It is valid in MF flow. Value range: numerals within 0-9, *, #
11	ServiceFlow	Char[1]	Indicates the call process indicator. The options are as follows: • 1: MO process • 2: MT process • 3: MF process (reserved) • 4: Read Reply • 5: Delivery Report • 6: Replied Report
12	CallForwardIndicator	Char[1]	Indicates the MMS forwarding flag. • In MT: A-B-C, In MT process of B, it needs to know that whether B party transfers the MMS to C. Usually the MT of B party is set free and all the cost will be charged in MF of B party in this situation. • In MF: This field should be defined in the interface between MMSC and SCP. The forwarding options are as follows: • 0: No forwarding • 1: MMS forwarding conditional • 2: MMS forwarding unconditional
13	CallingRoamInfo	Char[20]	Indicates the information used to analyze the roaming state of calling party. It can be the location number, MSC address, VLR number, and so on. The actual value is related with the service configuration.
14	CallingCellID	Char[20]	Indicates the cell ID of the calling party, which is directly filled in according to the IDP report. Value range: numerals within 0-9, *, #
15	CalledRoamInfo	Char[20]	Indicates the information used to analyze the roaming state of called party. It can be the location number, MSC address,

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SN	Field Name	Field Type	Description
			VLR number, and so on. The actual value is related with the service configuration. Value range: numerals within 0-9, *, # The wildcard value is **-1**.
16	CalledCellID	Char[20]	Indicates the cell ID of the called party, which is handled just as that for **CallingCellID**. Value range: numerals within 0-9, *, #
17	TimeStampOfSSP	Char[14]	Indicates the time stamp report from SSP in IDP. If missing in charging interface, this field will be filled with the local time stamp of OCS. Format: YYYYMMDDhhmmss
18	TimeZoneOfSSP	Integer[4]	Indicates the time zone report from SSP in IDP. If missing in charging interface, this field will be filled with the time zone of the location of OCS, 15 min. per sector. Value range: -48-48 **NOTE** The value -48 is not included. The default value is **0**.
19	BearerCapability	Char[32]	Reserved
20	BearerCapability2	Char[32]	Reserved
21	ChargingTime	Char[14]	Indicates the time stamp of the charging beginning. Format: YYYYMMDDhhmmss
22	Reserved	Integer[4]	-
23	ChargeAmount	Integer[8]	Indicates the charge amount from exterior. The default value is **0**.
24	SendResult	Integer[4]	Results of short message sending: • 0: Successfully • Other: The value is based on the value transferred from the SCP. Value range: -2^{31}-2^{31}-1 The default value is **0**.
25	MMSID	Char[24]	Indicates the ID of an MMS. The ID of each MMS is unique.

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SN	Field Name	Field Type	Description
26	IMEI	Char[24]	Indicates the IE containing the IMEISV of the ME in use by the served subscriber.
27	MMSLength	Integer[4]	Indicates the length of MMS. The default value is **0**.
28	call ID	Char[32]	Reserved
29	DiameterSessionID	Char[128]	Indicates a diameter session ID. The format is as follows: <DiameterIdentity>;<high 32 bits>;<low 32 bits>[;<optional value>] • <DiameterIdentity>: the same as the field in the Origin-Host AVP. • <high 32 bits>: indicates the current time of the system. • <low 32 bits>: indicates the cycle count-up. The initial value is 0. When you restart the system, set the value to 0. • <optional value>: indicates implementation specific but may include a modem is device Id, a layer 2 address, timestamp, etc. Separate the preceding fields with the character ";". For example: SCP001.huawei.com;1876543210;523
30	RefundIndicator	Char[20]	Indicates the deduction or refund request. The options are as follows: • 0: Deduct • 1: Refund
31	BrandID	Integer[4]	Indicates the brand ID of subscribers. The default value is **0**.
32	SubCOSID	Integer[4]	Indicates the main product ID of subscribers. The default value is **0**.
33	ChargingPartyNumber	Char[64]	Indicates the telephone number of the charging party. The number is same as calling party number in MO and called party number in MT. There are several special situations. Like Child-Parent card feature, the calling party number is the child, the charging party number is the parent's number. This field is null when the service is paid only by account.

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SN	Field Name	Field Type	Description
			The number format are as follows: • **Area number + PSTN** The number format is the format of the fixed-line telephone number. • **MSISDN** The number format is the format of the mobile phone number. Value range: numerals within 0-9, *, #
34	ChargePartyIndicator	Char[1]	Indicates the charging party. The options are as follows: • 1: The calling party is charged • 2: The called party is charged • 3: Charge of third party • 4: Charge of parent & child card • 5: Charge of calling party & VPN
35	PayType	Char[1]	Indicates the pay type of subscribers. The options are as follows: • 0: Prepaid • 1: Postpaid
36	BillCycleID	Char[20]	Indicates the bill cycle ID. Format: YYYYMMTT • YYYY: indicates the year. • MM: indicates the month. • TT: indicates the type of a bill cycle. Value range: numerals within 0-9
37	ChargingType	Char[1]	Indicates the charging type. For MO/MF process: • 0: V-H • 1: H-H • 2: V-V • 3: H-V For MT process: • 0: For free • 1: Roaming fee
38	OnNetIndicator	Char[1]	Indicates the MM types. The options are as follows: • 0: on-net • 1: off-net • 2: International

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SN	Field Name	Field Type	Description
39	RoamState	Char[1]	Indicates the roaming state of charging party. The options are as follows: • 0: No roaming • 1: Roaming within charging area • 2: Roaming out of charging area • 3: International Roaming
40	ResultCode	Integer[4]	Indicates the internal result code. Value range: -2^{31}-2^{31}-1 The default value is **0**.
41	SameBrandPre	Integer[4]	Reserved
42	CallingHomeCountryCode	Integer[4]	Indicates the home country code information of calling party. >=0, -1, 0 or 9999 means unknown. The default value is **0**.
43	CallingHomeAreaNumber	Integer[4]	Indicates the home area number of the calling party. >=0, -1, 0 or 9999 means unknown. The default value is **0**.
44	CallingHomeNetworkCode	Integer[4]	Indicates the home network code of the calling party. >=0, -1, 0 or 9999 means unknown. The default value is **0**.
45	CallingRoamCountryCode	Integer[4]	Indicates the roam country code of the calling party. >=0, -1, 0 or 9999 means unknown. The default value is **0**.
46	CallingRoamAreaNumber	Integer[4]	Indicates the roam area number of the calling party. >=0, -1, 0 or 9999 means unknown. The default value is **0**.
47	CallingRoamNetworkCode	Integer[4]	Indicates the roam network code of the calling party. >=0, -1, 0 or 9999 means unknown. The default value is **0**.
48	CalledHomeCountryCode	Integer[4]	Indicates the home country code of the called party. >=0, -1, 0 or 9999 means unknown. The default value is **0**.

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SN	Field Name	Field Type	Description
49	CalledHomeAreaNumber	Integer[4]	Indicates the home area number of the called party. >=0, -1, 0 or 9999 means unknown. The default value is **0**.
50	CalledHomeNetworkCode	Integer[4]	Indicates the home network code of the called party. >=0, -1, 0 or 9999 means unknown. The default value is **0**.
51	CalledRoamCountryCode	Integer[4]	Indicates the roam country code of the called party. >=0, -1, 0 or 9999 means unknown. The default value is **0**.
52	CalledRoamAreaNumber	Integer[4]	Indicates the roam area number of the called party. >=0, -1, 0 or 9999 means unknown. The default value is **0**.
53	CalledRoamNetworkCode	Integer[4]	Indicates the roam network code of the called party. >=0, -1, 0 or 9999 means unknown. The default value is **0**.
54	ProductID	Char[20]	Indicates the last effect rating product.
55	ServiceType	Char[20]	Indicates the service included in the product. They are predefined in OCS. The options are as follows: • 1st digit: Normal Voice tariff • 2nd digit: Familiarity number service • 3rd digit: Data service • 4th digit: IP/international carrier call • 5th digit: pre-IP call • 6th digit: International Call Forward • 7th digit: Call from Home Zone • 8th digit: UCB service • 9th digit: P_FPH service • 10th digit: Remote Access • 11th digit: Call from Special Zone • Other: reserved The value range of each digit indicates whether the service is activated. • 0: Not activated • 1: Activated

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SN	Field Name	Field Type	Description
			Not null
56	SpecialNumberIndicator	Integer[4]	Types of special numbers that send short messages, which are used to identify call-free numbers, for example, A2P or P2A. The options are as follows: • 0: Normal MMS/P2P • 1: BarringMMS • 2: Free MMS • 3: CommonMMSHotLineNumber >=0, -1, 0 or 9999 means unknown. The default value is **0**.
57	HomeZoneID	Integer[4]	Indicates the home zone ID. When a subscriber enjoys the home zone discount, the value of this field is the home zone ID of the area where the subscriber currently locates. >=0, -1, 0 or 9999 means unknown. The default value is **0**.
58	NPFlag	Char[1]	Indicates the flag of number portable. The options are as follows: • 0: Not NP • 1: NP out
59	NPPrefix	Char[6]	Indicates the number portability prefix of called party in the MO process or number portability prefix of the calling party in the MT process. Value range: numerals within 0-9, *, #
60	CallingGroupNo	Integer[4]	Indicates the group number of the calling party when enjoying CUG service. It is the group number related to the rating. For VPN, it is the nearest level groupNo.
61	CalledGroupNo	Integer[4]	Indicates the group number of the called party when enjoying CUG or VPN service. It is the group number related to the rating. The default value is **0**.
62	UserState	Char[10]	Indicates the life cycle state and the management state of a subscriber. The **UserState** parameter is a numeric string of seven digits, in the format CCMMMMM. The first and second digits C indicate the life cycle state of a subscriber. The first digit C indicates the state of a prepaid subscriber,

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SN	Field Name	Field Type	Description
			and the second digit C indicates the state of a postpaid subscriber. The options are as follows: • 0: Idle • 1: Active • 2: Suspended • 3: Disable • 4: Pool • 5: Pool without activation The next five digits MMMMM indicate the management state of a subscriber. The options are as follows: Third digit: Whether the subscriber claims missing. • 0: no • 1: yes Fourth digit: Whether the subscriber suspends the service. • 0: non-suspend • 1: suspend • 2: suspend by operator (B1W) • 3: suspend by operator (B2W) Fifth digit: Whether the subscriber applies for suspension with the number being reserved. • 0: no • 1: yes Sixth digit: Whether the subscriber is blocked. • 0: no • 1: yes Seventh digit: Whether the subscriber is blacklisted. • 0: no • 1: yes The default value is **0**.
63	SubscriberID	Char[20]	Indicates the subscriber ID. Value range: characters in [0-9, a-z, A-Z]
64	OpposeNetworkType	Integer[4]	Indicates the network type of oppose side. The options are as follows: • 1: PSTN • 2: Mobile

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SN	Field Name	Field Type	Description
			• 3: Native Unknown • 4: International Unknown • Other: to be extended The default value is **0**. The wildcard value is **-1**.
65	Reserved	Char[20]	-
66	ChargeOfItemsAccounts	Integer[8]	Indicates the free call times used. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
67	ChargeOfDurationAccounts	Integer[8]	Indicates the total call time charge using reword resource. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
68	ChargeOfFundAccounts	Integer[8]	Indicates the total cost amount charging from fund account. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
69	ChargeFromPrepaid	Integer[8]	Indicates the charge amount from prepaid account. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
70	PrepaidBalance	Integer[8]	Indicates the balance of prepaid account. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
71	ChargeFromPostpaid	Integer[8]	Indicates the charge amount from postpaid account. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
72	PostPaidBalance	Integer[8]	Indicates the balance of postpaid available limitation. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.

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SN	Field Name	Field Type	Description
73	AccountID	Char[20]	Indicates the ID of the main account. Value range: characters in [0-9, a-z, A-Z]
74	AccountKey	Integer[8]	Indicates the key of the main account. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
75	CurrencyCode	Integer[4]	Currency code. Also, if this field missed or present be Zero, it means the system use local minimal currency as default, and no need to specify in CDR. The default value is **0**.
76	package	Char[32]	Reserved
77	AccountType1	Integer[4]	Indicates the account type of the first account. 0 means not used. Not null The default value is **0**.
78	FeeType1	Integer[4]	Indicates the fee type of the first account. 0 means not used. Not null The default value is **0**.
79	ChargeAmount1	Integer[8]	Indicates the changing amount of the first account. 0 means not unchanged. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
80	CurrentAcctAmount1	Integer[8]	Indicates the current amount of the first account. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
81	AccountType2	Integer[4]	Indicates the account type of the second account. 0 means not used. Not null The default value is **0**.
82	FeeType2	Integer[4]	Indicates the fee type of the second account. 0 means not used.

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SN	Field Name	Field Type	Description
			Not null The default value is **0**.
83	ChargeAmount2	Integer[8]	Indicates the changing amount of the second account. 0 means not unchanged. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
84	CurrentAcctAmount2	Integer[8]	Indicates the current amount of the second account. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
85	AccountType3	Integer[4]	Indicates the account type of the third account. 0 means not used. Not null The default value is **0**.
86	FeeType3	Integer[4]	Indicates the fee type of the third account. 0 means not used. Not null The default value is **0**.
87	ChargeAmount3	Integer[8]	Indicates the changing amount of the third account. 0 means not unchanged. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
88	CurrentAcctAmount3	Integer[8]	Indicates the current amount of the third account. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
89	AccountType4	Integer[4]	Indicates the account type of the fourth account. 0 means not used. Not null The default value is **0**.
90	FeeType4	Integer[4]	Indicates the fee type of the fourth account.

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SN	Field Name	Field Type	Description
			0 means not used. Not null The default value is **0**.
91	ChargeAmount4	Integer[8]	Indicates the changing amount of the fourth account. 0 means not unchanged. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
92	CurrentAcctAmount4	Integer[8]	Indicates the current amount of the fourth account. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
93	AccountType5	Integer[4]	Indicates the account type of the fifth account. 0 means not used. Not null The default value is **0**.
94	FeeType5	Integer[4]	Indicates the fee type of the fifth account. 0 means not used. Not null The default value is **0**.
95	ChargeAmount5	Integer[8]	Indicates the changing amount of the fifth account. 0 means not unchanged. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
96	CurrentAcctAmount5	Integer[8]	Indicates the current amount of the fifth account. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
97	AccountType6	Integer[4]	Indicates the account type of the sixth account. 0 means not used. Not null The default value is **0**.

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SN	Field Name	Field Type	Description
98	FeeType6	Integer[4]	Indicates the fee type of the sixth account. 0 means not used. Not null The default value is **0**.
99	ChargeAmount6	Integer[8]	Indicates the changing amount of the sixth account. 0 means not unchanged. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
100	CurrentAcctAmount6	Integer[8]	Indicates the current amount of the sixth account. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
101	AccountType7	Integer[4]	Indicates the account type of the seventh account. 0 means not used. Not null The default value is **0**.
102	FeeType7	Integer[4]	Indicates the fee type of the seventh account. 0 means not used. Not null The default value is **0**.
103	ChargeAmount7	Integer[8]	Indicates the changing amount of the seventh account. 0 means not unchanged. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
104	CurrentAcctAmount7	Integer[8]	Indicates the current amount of the seventh account. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
105	AccountType8	Integer[4]	Indicates the account type of the eighth account. 0 means not used. Not null

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SN	Field Name	Field Type	Description
			The default value is **0**.
106	FeeType8	Integer[4]	Indicates the fee type of the eighth account. 0 means not used. Not null The default value is **0**.
107	ChargeAmount8	Integer[8]	Indicates the changing amount of the eighth account. 0 means not unchanged. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
108	CurrentAcctAmount8	Integer[8]	Indicates the current amount of the eighth account. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
109	AccountType9	Integer[4]	Indicates the account type of the ninth account. 0 means not used. Not null The default value is **0**.
110	FeeType9	Integer[4]	Indicates the fee type of the ninth account. 0 means not used. Not null The default value is **0**.
111	ChargeAmount9	Integer[8]	Indicates the changing amount of the ninth account. 0 means not unchanged. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
112	CurrentAcctAmount9	Integer[8]	Indicates the current amount of the ninth account. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
113	AccountType10	Integer[4]	Indicates the account type of the tenth account. 0 means not used.

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SN	Field Name	Field Type	Description
			Not null The default value is **0**.
114	FeeType10	Integer[4]	Indicates the fee type of the tenth account. 0 means not used. Not null The default value is **0**.
115	ChargeAmount10	Integer[8]	Indicates the changing amount of the tenth account. 0 means not unchanged. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
116	CurrentAcctAmount10	Integer[8]	Indicates the current amount of the tenth account. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
117	BonusValidity1	Char[14]	Indicates the validity period of the first rewarded resource. The validity period does not change when fees are deducted from accounts. In this case, the validity periods of consumption resources are not recorded. Format: YYYYMMDDhhmmss
118	BonusValidity2	Char[14]	Indicates the validity period of the second rewarded resource. The validity period does not change when fees are deducted from accounts. In this case, the validity periods of consumption resources are not recorded. Format: YYYYMMDDhhmmss
119	BonusValidity3	Char[14]	Indicates the validity period of the third rewarded resource. The validity period does not change when fees are deducted from accounts. In this case, the validity periods of consumption resources are not recorded. Format: YYYYMMDDhhmmss
120	BonusValidity4	Char[14]	Indicates the validity period of the fourth rewarded resource. The validity period does not change when fees are deducted from accounts. In this case, the validity periods of consumption resources are not recorded. Format: YYYYMMDDhhmmss
121	BonusValidity5	Char[14]	Indicates the validity period of the fifth

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SN	Field Name	Field Type	Description
			rewarded resource. The validity period does not change when fees are deducted from accounts. In this case, the validity periods of consumption resources are not recorded. Format: YYYYMMDDhhmmss
122	BonusValidity6	Char[14]	Indicates the validity period of the sixth rewarded resource. The validity period does not change when fees are deducted from accounts. In this case, the validity periods of consumption resources are not recorded. Format: YYYYMMDDhhmmss
123	BonusValidity7	Char[14]	Indicates the validity period of the seventh rewarded resource. The validity period does not change when fees are deducted from accounts. In this case, the validity periods of consumption resources are not recorded. Format: YYYYMMDDhhmmss
124	BonusValidity8	Char[14]	Indicates the validity period of the eighth rewarded resource. The validity period does not change when fees are deducted from accounts. In this case, the validity periods of consumption resources are not recorded. Format: YYYYMMDDhhmmss
125	BonusValidity9	Char[14]	Indicates the validity period of the ninth rewarded resource. The validity period does not change when fees are deducted from accounts. In this case, the validity periods of consumption resources are not recorded. Format: YYYYMMDDhhmmss
126	BonusValidity10	Char[14]	Indicates the validity period of the tenth rewarded resource. The validity period does not change when fees are deducted from accounts. In this case, the validity periods of consumption resources are not recorded. Format: YYYYMMDDhhmmss
127	CallingVPNTopGroupNumber	Char[15]	Indicates the top group number of the calling party. Value range: numerals within 0-9 Can be null
128	CallingVPNGroupNumber	Char[10]	Indicates the group number of the calling party. Value range: numerals within 0-9 Can be null
129	CallingVPNShortN	Char[10]	Indicates the short number of the calling

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SN	Field Name	Field Type	Description
	umber		party. Value range: numerals within 0-9 Can be null
130	CalledVPNTopGroupNumber	Char[15]	Indicates the top group number of the called party. Value range: numerals within 0-9 Can be null
131	CalledVPNGroupNumber	Char[10]	Indicates the group number of the called party. Value range: numerals within 0-9 Can be null
132	CalledVPNShortNumber	Char[10]	Indicates the short number of the called party Value range: numerals within 0-9 Can be null
133	CallingNetworkType	Char[10]	Indicates the type of the network accessed by the calling party. The options are as follows: • 100: PSTN • 101: PBX • 102: PHS • 200: GSM • 201: GPRS • 202: EDGE • 300: CDMA • 301: CDMA 450 • 400: WCDMA • 500: TD-SCDMA • 600: IP/IMS • 601: WIFI • 602: WIMAX • Other to be extended Can be null
134	CalledNetworkType	Char[10]	Indicates the type of the network accessed by the called party. The options are as follows: • 100: PSTN • 101: PBX • 102: PHS • 200: GSM

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SN	Field Name	Field Type	Description
			• 201: GPRS • 202: EDGE • 300: CDMA • 301: CDMA 450 • 400: WCDMA • 500: TD-SCDMA • 600: IP/IMS • 601: WIFI • 602: WIMAX • Other to be extended Can be null
135	GroupCallType	Char[10]	Reserved
136	GroupPayFlag	Char[10]	Indicates the payment mode of the group users. The options are as follows: • 0: Personal payment • 1: Group payment • 2: Group and Personal pay Can be null
137	OnlineType	Integer[4]	Indicates the charge type when the charged subscriber is a VPN user. The options are as follows: • 0: Online • 1: Offline Can be null
138	session id	Char[20]	Reserved
139	Service Name	Char[32]	Reserved
140	Consumed Resources	Integer[4]	Reserved
141	Original Amount	Integer[4]	Reserved
142	AddtionalInfo	Char[256]	Reserved
143	OriginAddressTyp e	Char[2]	The type of source address: • 0: e-mail address • 1: MSISDN • 2: IPv4 Address • 3: IPv6 Address • 4: Numeric Shortcode • 5: Alphanumeric Shortcode

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SN	Field Name	Field Type	Description
			• 6: Other The default value is **-1**.
144	DestAddressType	Char[2]	The type of purpose address: • 0: e-mail address • 1: MSISDN • 2: IPv4 Address • 3: IPv6 Address • 4: Numeric Shortcode • 5: Alphanumeric Shortcode • 6: Other The default value is **-1**.
145	Message-Type	Integer[4]	Indicates the message type. The options are as follows: • 1: m-send-req • 2: m-send-conf • 3: m-notification-ind • 4: m-notifyresp-ind • 5: m-retrieve-conf • 6: m-acknowledge-ind • 7: m-delivery-ind • 8: m-read-rec-ind • 9: m-read-orig-ind • 10: m-forward-req • 11: m-forward-conf • 12: m-mbox-store-conf • 13: m-mbox-view-conf • 14: m-mbox-upl • 101: MOAT • 102: AOMT • 103: MOMT • 104: MOET • 105: EOMT • 106: DlvRpt • 107: RdRpt • 108: Reply • 109: P2PMT • 110: FOMT • 111: FOAT • 112: FOET

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SN	Field Name	Field Type	Description
			• 113: MOFT • 114~: reserved The default value is **-1**.
146	VASP-Id	Char[20]	Indicates the ID of a service provider (SP).
147	VAS-Id	Char[20]	Indicates the service ID of an SP.
148	Service-Id	Char[20]	Indicates the service type ID.
149	Msg-Seq	Char[20]	Indicates the MSG-SEQ from EMPP interface of MMS.
150	StartTimeOfBillCycle	Char[14]	Indicates the start time of bill cycle.
151	SecondChargingPartyNumber	Char[64]	Indicates the number of the other charging party of the child-parent cards. In the example, the value of the field is the number of subscriber C. The number is same as calling party number in MO and called party number in MT. There are several special situations. Like Child-Parent card feature, the calling party number is the child, the charging party number is the parent's number. This field is null when the service is paid only by account. The number format are as follows: • **Area number + PSTN** The number format is the format of the fixed-line telephone number. • **MSISDN** The number format is the format of the mobile phone number. Value range: numerals within 0-9, *, #
152	SecondChargingPartyPayType	Char[1]	Indicates the payment type of the other charging party of the child-parent cards. For example, the payment type of the parent card. The options are as follows: • 0: Prepaid • 1: Postpaid **NOTE** If the value of the field is null, the system records the payment type of the parent card subscriber.
153	SecondChargingPartyBillCycleID	Char[20]	Indicates the bill cycle ID of the other charging party of the child-parent cards. For example, the bill cycle ID of the parent card.

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SN	Field Name	Field Type	Description
			Format: YYYYMMTT The description of the format is as follows: • YYYY: indicates the year. • MM: indicates the month. • TT: indicates the type of a bill cycle.
154	ChargeOfItemsSecondAccounts	Integer[8]	Indicates the free call times used from others, such as the parent card. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
155	ChargeOfDurationSecondAccounts	Integer[8]	Indicates the total call time charge using reword resource from others, such as the parent card. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
156	ChargeOfFundSecondAccounts	Integer[8]	Indicates the total cost amount charging from fund account of others, such as the parent card. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
157	ChargeFromSecondPrepaidAccount	Integer[8]	Indicates the charge amount from other prepaid account. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
158	SecondPrepaidAccountBalance	Integer[8]	Indicates the balance of other prepaid account. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
159	ChargeFromSecondPostpaidAccount	Integer[8]	Indicates the charge amount from other postpaid account. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
160	SecondPostpaidAccountBalance	Integer[8]	Indicates the balance of other postpaid account. Value range: -2^{63}-2^{63}-1

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SN	Field Name	Field Type	Description
			Not null The default value is **0**.
161	SecondAccountID	Char[20]	Indicates the main account ID of the other charging party that the system deducts fees from. Value range: characters in [0-9, a-z, A-Z] Not null
162	SecondAccountKey	Integer[8]	Indicates the main account key of the other charging party that the system deducts fees from. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
163	SecondAccountCurrencyCode	Integer[4]	Indicates currency code of others, such as the parent card. Also, if this field missed or present be Zero, it means the system use local minimal currency as default, and no need to specify in CDR. Not null The default value is **0**.
164	APN	Char[32]	Reserved
165	SecondAccountType1	Integer[4]	Indicates the account type of the first account of others, such as the parent card. 0 means not used. Not null The default value is **0**.
166	SecondFeeType1	Integer[4]	Indicates the fee type of the first account of others, such as the parent card. 0 means not used. Not null The default value is **0**.
167	SecondChargeAmount1	Integer[8]	Indicates the changing amount of the first account of others, such as the parent card. 0 means not unchanged. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
168	SecondCurrentAcctAmount1	Integer[8]	Indicates the current amount of the first account of others, such as the parent card. Value range: -2^{63}-2^{63}-1

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SN	Field Name	Field Type	Description
			Not null The default value is **0**.
169	SecondAccountType2	Integer[4]	Indicates the account type of the second account of others, such as the parent card. 0 means not used. Not null The default value is **0**.
170	SecondFeeType2	Integer[4]	Indicates the fee type of the second account of others, such as the parent card. 0 means not used. Not null The default value is **0**.
171	SecondChargeAmount2	Integer[8]	Indicates the changing amount of the second account of others, such as the parent card. 0 means not unchanged. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
172	SecondCurrentAcctAmount2	Integer[8]	Indicates the current amount of the second account of others, such as the parent card. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
173	SecondAccountType3	Integer[4]	Indicates the account type of the third account of others, such as the parent card. 0 means not used. Not null The default value is **0**.
174	SecondFeeType3	Integer[4]	Indicates the fee type of the third account of others, such as the parent card. 0 means not used. Not null The default value is **0**.
175	SecondChargeAmount3	Integer[8]	Indicates the changing amount of the third account of others, such as the parent card. 0 means not unchanged. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.

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SN	Field Name	Field Type	Description
176	SecondCurrentAcctAmount3	Integer[8]	Indicates the current amount of the third account of others, such as the parent card. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
177	SecondAccountType4	Integer[4]	Indicates the account type of the fourth account of others, such as the parent card. 0 means not used. Not null The default value is **0**.
178	SecondFeeType4	Integer[4]	Indicates the fee type of the fourth account of others, such as the parent card. 0 means not used. Not null The default value is **0**.
179	SecondChargeAmount4	Integer[8]	Indicates the changing amount of the fourth account of others, such as the parent card. 0 means not unchanged. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
180	SecondCurrentAcctAmount4	Integer[8]	Indicates the current amount of the fourth account of others, such as the parent card. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
181	SecondAccountType5	Integer[4]	Indicates the account type of the fifth account of others, such as the parent card. 0 means not used. Not null The default value is **0**.
182	SecondFeeType5	Integer[4]	Indicates the fee type of the fifth account of others, such as the parent card. 0 means not used. Not null The default value is **0**.
183	SecondChargeAmount5	Integer[8]	Indicates the changing amount of the fifth account of others, such as the parent card. 0 means not unchanged. Value range: -2^{63}-2^{63}-1

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SN	Field Name	Field Type	Description
			Not null The default value is **0**.
184	SecondCurrentAcctAmount5	Integer[8]	Indicates the current amount of the fifth account of others, such as the parent card. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
185	SecondBonusValidity1	Char[14]	Indicates the validity period of the first rewarded resource of others, such as the parent card. The validity period does not change when fees are deducted from accounts. In this case, the validity periods of consumption resources are not recorded. Format: YYYYMMDDhhmmss
186	SecondBonusValidity2	Char[14]	Indicates the validity period of the second rewarded resource of others, such as the parent card. The validity period does not change when fees are deducted from accounts. In this case, the validity periods of consumption resources are not recorded. Format: YYYYMMDDhhmmss
187	SecondBonusValidity3	Char[14]	Indicates the validity period of the third rewarded resource of others, such as the parent card. The validity period does not change when fees are deducted from accounts. In this case, the validity periods of consumption resources are not recorded. Format: YYYYMMDDhhmmss
188	SecondBonusValidity4	Char[14]	Indicates the validity period of the fourth rewarded resource of others, such as the parent card. The validity period does not change when fees are deducted from accounts. In this case, the validity periods of consumption resources are not recorded. Format: YYYYMMDDhhmmss
189	SecondBonusValidity5	Char[14]	Indicates the validity period of the fifth rewarded resource of others, such as the parent card. The validity period does not change when fees are deducted from accounts. In this case, the validity periods of consumption resources are not recorded. Format: YYYYMMDDhhmmss
190	PulaPulaProperty	Integer[4]	Reserved
191	BumerangePropert	Integer[4]	Reserved

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SN	Field Name	Field Type	Description
	y		
192	OppositePartyNumber	Char[64]	Indicates the number of the opposite party. In calling process, the number is the number of the called party. In called process, the number is the number of the calling party. The number format are as follows: • **Area number + PSTN** The number format is the format of the fixed-line telephone number. • **MSISDN** The number format is the format of the mobile phone number. Value range: numerals within 0-9, *, #
193	ContentType	Char[8]	Reserved
194	RateName	Char[25]	Reserved
195	RatingGroup	Integer[4]	Reserved
196	IsTestNumber	Integer[4]	Reserved
197	MainProductStatus	Integer[4]	Reserved
198	DiscountOfLastEffProduct	Integer[4]	Indicates the discount percentage for the last rated product that takes effect. For a product that does not provide discounts, the value is **100**. Value range: 0-100 **NOTE** The value **0** is not included. Not null The default value is **100**.
199	subscriberkey	Integer[4]	Indicates the key of a subscriber. Not null The default value is **0**.
200	accumulationchange	Char[256]	Indicates the change value of an accumulative item. The value format is as follows: Accumulation ID 1, accumulation switch time 1, change value 1; accumulation ID 2, accumulation switch time 2, change value 2... The value contains only numerals. The account ID and balance change is separated by a comma (,), and information of different

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SN	Field Name	Field Type	Description
			accounts is separated by a semicolon (;). The information about up to 10 accounts can be included. The excess account information will be discarded.
201	UsageServiceType	Integer[4]	Indicates the type of a real-time service. The default value is **1350**. • **1350**: Indicates the special mms • **1351**: Indicates the normal mms
202	SourceSerialNo	Integer[8]	Reserved
203	SourceSubSequence	Integer[4]	Reserved
204	AcctBalChgList	Char[768]	Reserved
205	OriginCdrFilename	Char[64]	Reserved
206	CurCDRLineNo	Integer[4]	Reserved
207	DDSDiscountOff	Integer[4]	Indicates the number of the line where the original CDR is located. The default value is **0**.
208	Tax	Integer[8]	Reserved
209	Tax2	Integer[8]	Reserved
210	Tax3	Integer[8]	Reserved
211	Tax4	Integer[8]	Reserved

3.1.5 data CDR

Description

A data CDR records the basic traffic of the data service.

Generation Scenario

An OCS subscriber downloads a ring tone through the GPRS. The ring tone costs the subscriber 2 Ush, a data CDR is generated.

CDR File Name Format

dataYYYYMMDD_CBPNODE_Seqno.unl

Example of a data CDR file name: data20071112_101_000151.unl

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Format

Table 3-6 describes each field of data CDR file.

Table 3-6 Format of data CDR

SN	Field Name	Field Type	Description
1	SerialNo	Integer[8]	Indicates the serial number. It is generated by the system and the serial number is exclusive. Not null The default value is **0**.
2	SubSequence	Integer[4]	Indicates the sequence number when the bill is split. For example, special requirement of splitting on mid-night and end of the bill cycle. The default value is **0**.
3	TimeStamp	Char[14]	Indicates the timestamp of the bill item generation. Format: YYYYMMDDhhmmss Not null
4	ServiceKey	Integer[4]	Indicates the service key of the intelligent service to which the subscriber subscribes to. The default value is **1**. Not null
5	CallingPartyNumber	Char[64]	Indicates the calling number. The number format are as follows: • **Country code + Area number + PSTN** The number format is the format of the fixed-line phone number. • **Country code + MSISDN** The number format is the format of the mobile phone number. Value range: numerals within 0-9, *, # Not null
6	APN	Char[22]	Indicates the access point.
7	URL	Char[22]	Indicates the accessed URL.
8	CallingPartyIMSI	Char[64]	Indicates the IMSI of calling party. • When the SCP transfers an IMSI, the value of this field is the IMSI transferred • When the SCP does not transfer an

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SN	Field Name	Field Type	Description
			IMSI, the value of this field is the value of the corresponding field in the attribute table of the charged subscriber. • The value is **Null** when the relevant value cannot be obtained. Value range: numerals within 0-9, *, #
9	AccessNetworkAddress	Char[22]	Indicates the address of access network. For GPRS, it refers to the SGSN. For WLAN, it refers to the Hotspot ID.
10	GGSNAddress	Char[22]	Indicates the address of GGSN.
11	CallingRoamInfo	Char[20]	Indicates the information used to analyze the roaming state of calling party. It can be the location number, MSC address, VLR number, and so on. The actual value is related with the service configuration. Value range: numerals within 0-9, *, #
12	CallingCellID	Char[20]	Indicates the cell ID of the calling party, which is directly filled in according to the IDP report. Value range: numerals within 0-9, *, #
13	TimeStampOfSGSN	Char[14]	Indicates the start time of the service reported by the SGSN or external network elements. Format: YYYYMMDDhhmmss
14	TimeZoneOfSGSN	Integer[4]	Indicates the time zone in which the SGSN or external network elements locate. Value range: -48-48 **NOTE** The value -48 is not included. The default value is **0**.
15	BearerCapability	Char[32]	Reserved
16	ChargingTime	Char[14]	Indicates the time stamp of the charging beginning. Format: YYYYMMDDhhmmss
17	TotalFlux	Integer[8]	Indicates the total traffic. Unit: byte Not null The default value is **0**.
18	UpFlux	Integer[8]	Indicates the upstream traffic.

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SN	Field Name	Field Type	Description
			Unit: byte Not null The default value is **0**.
19	DownFlux	Integer[8]	Indicates the downstream traffic. Unit: byte Not null The default value is **0**.
20	ElapseDuration	Integer[4]	Indicates the duration of using GPRS service. Unit: second Not null The default value is **0**.
21	TerminationReason	Integer[4]	Indicates the reason why the call cannot be connected. The options are as follows: • 1: hang up • 2: insufficient balance • 3: over maximum duration • 99: other reasons The default value is **0**.
22	IMEI	Char[24]	Indicates the IE containing the IMEISV of the ME in use by the served subscriber.
23	TransitionID	Char[32]	Indicates the sequence number of a transaction.
24	ServiceID	Char[20]	Indicates the service ID of the SP. The default value is **-1**.
25	SPID	Char[20]	Indicates the SP ID or CP ID. The default value is **-1**.
26	CategoryID	Char[20]	The number of the category. The default value is **-1**.
27	ContentID	Char[20]	The number of the content.
28	QoS	Char[30]	Service quality.
29	OriginalNetworkType	Char[20]	Indicates the type of the original network. The options are as follows: • 1: PSTN • 2: Mobile • 3: Native Unknown • 4: International Unknown

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SN	Field Name	Field Type	Description
			• Other to be extended
30	DiameterSessionID	Char[128]	Indicates a diameter session ID. The format is as follows: <DiameterIdentity>;<high 32 bits>;<low 32 bits>[;<optional value>] • <DiameterIdentity>: the same as the field in the Origin-Host AVP. • <high 32 bits>: indicates the current time of the system. • <low 32 bits>: indicates the cycle count-up. The initial value is **0**. When you restart the system, set the value to **0**. • <optional value>: indicates implementation specific but may include a modem's device Id, a layer 2 address, timestamp, etc. Separate the preceding fields with the character ";". For example: SCP001.huawei.com;1876543210;523
31	BrandID	Integer[4]	Indicates the brand ID of subscribers. Not null The default value is **0**.
32	SubCOSID	Integer[4]	Indicates the main product ID of subscribers. Not null The default value is **0**.
33	ChargingPartyNumber	Char[64]	Indicates the telephone number of the charging party. The number is same as calling party number in MO and called party number in MT. There are several special situations. Like Child-Parent card feature, the calling party number is the child, the charging party number is the parent's number. This field is null when the service is paid only by account. The number format are as follows: • **Area number + PSTN** The number format is the format of the fixed-line telephone number. • **MSISDN**

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SN	Field Name	Field Type	Description
			The number format is the format of the mobile phone number. Value range: numerals within 0-9, *, #
34	PayType	Char[1]	Indicates the pay type of subscribers. The options are as follows: • 0: Prepaid • 1: Postpaid
35	BillCycleID	Char[20]	Indicates the bill cycle ID. Format: YYYYMMTT • YYYY: indicates the year. • MM: indicates the month. • TT: indicates the type of a bill cycle. Value range: numerals within 0-9
36	ChargingType	Char[1]	Indicates the charging type. The current version supports the following types: • 0: based on duration • 1: based on traffic
37	ResultCode	Integer[4]	Indicates the internal result code. Value range: -2^{31}-2^{31}-1 The default value is **0**.
38	RoamState	Char[1]	Indicates the roaming state of charging party. The options are as follows: • 0: No roaming • 1: Roaming within charging area • 2: Roaming out of charging area • 3: International Roaming The default value is **0**.
39	CallingHomeCountryCode	Integer[4]	Indicates the home country code information of calling party. >=0, -1, 0 or 9999 means unknown. The default value is **0**.
40	CallingHomeAreaNumber	Integer[4]	Indicates the home area number of the calling party. >=0, -1, 0 or 9999 means unknown. The default value is **0**.
41	CallingHomeNetworkCode	Integer[4]	Indicates the home network code of the calling party.

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SN	Field Name	Field Type	Description
			>=0, -1, 0 or 9999 means unknown. The default value is **0**.
42	CallingRoamCountryCode	Integer[4]	Indicates the roam country code of the calling party. >=0, -1, 0 or 9999 means unknown. The default value is **0**.
43	CallingRoamAreaNumber	Integer[4]	Indicates the roam area number of the calling party. >=0, -1, 0 or 9999 means unknown. The default value is **0**.
44	CallingRoamNetworkCode	Integer[4]	Indicates the roam network code of the calling party. >=0, -1, 0 or 9999 means unknown. The default value is **0**.
45	Operation Type	Char[20]	Reserved
46	msg id	Char[20]	Reserved
47	session id	Char[20]	Reserved
48	Service Name	Char[32]	Reserved
49	Consumed Resources	Integer[4]	Reserved
50	Original Amount	Integer[4]	Reserved
51	ProductID	Char[20]	Indicates the last effect rating product.
52	ServiceType	Char[20]	Reserved
53	Reserved	Char[20]	-
54	HomeZoneID	Integer[4]	Reserved
55	UserState	Char[10]	Indicates the life cycle state and the management state of a subscriber. The **UserState** parameter is a numeric string of seven digits, in the format CCMMMMM. The first and second digits C indicate the life cycle state of a subscriber. The first digit C indicates the state of a prepaid subscriber, and the second digit C indicates the state of a postpaid subscriber. The options are as follows: • 0: Idle • 1: Active • 2: Suspended

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SN	Field Name	Field Type	Description
			• 3: Disable • 4: Pool • 5: Pool without activation The next five digits MMMMM indicate the management state of a subscriber. The options are as follows: Third digit: Whether the subscriber claims missing. • 0: no • 1: yes Fourth digit: Whether the subscriber suspends the service. • 0: non-suspend • 1: suspend • 2: suspend by operator (B1W) • 3: suspend by operator (B2W) Fifth digit: Whether the subscriber applies for suspension with the number being reserved. • 0: no • 1: yes Sixth digit: Whether the subscriber is blocked. • 0: no • 1: yes Seventh digit: Whether the subscriber is blacklisted. • 0: no • 1: yes
56	SubscriberID	Char[20]	Indicates the subscriber ID. Value range: characters in [0-9, a-z, A-Z]
57	ChargeDuration	Integer[4]	Indicates the charge duration. Unit: second Not null The default value is **0**.
58	TotalChargeFlux	Integer[8]	Indicates total charging traffic. Unit: byte Not null The default value is **0**.
59	PDN Connection Id	Char[20]	Includes the charging ID of the first EPS bearer activated within the PDN

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SN	Field Name	Field Type	Description
			connection. This field is reserved for future use.
60	Serving-Node-Type	Char[20]	Present for the PDP context and EPS bearer. Identifies the type of serving node. • 0: SGSN • 1: PMIPSGW • 2: GTPSGW • 3: ePDG • 4: hSGW • 5: MME This field is reserved for future use
61	ChargeOfFluxAccounts	Integer[8]	Indicates the free flux used. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
62	ChargeOfDuratiion Accounts	Integer[8]	Indicates the total call time charge using reword resource. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
63	ChargeOfFundAccounts	Integer[8]	Indicates the total cost amount charging from fund account. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
64	ChargeFromPrepaid	Integer[8]	Indicates the charge amount from prepaid account. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
65	PrepaidBalance	Integer[8]	Indicates the balance of prepaid account. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
66	ChargeFromPostpaid	Integer[8]	Indicates the charge amount from postpaid account. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.

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SN	Field Name	Field Type	Description
67	PostPaidBalance	Integer[8]	Indicates the balance of postpaid available limitation. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
68	AccountID	Char[20]	Indicates the ID of the main account. Value range: characters in [0-9, a-z, A-Z] Not null The default value is **0**.
69	AccountKey	Integer[8]	Indicates the key of the main account. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
70	CurrencyCode	Integer[4]	Currency code. Also, if this field missed or present be Zero, it means the system use local minimal currency as default, and no need to specify in CDR. The default value is **0**.
71	Reserved	Char[20]	-
72	AccountType1	Integer[4]	Indicates the account type of the first account. 0 means not used. Not null The default value is **0**.
73	FeeType1	Integer[4]	Indicates the fee type of the first account. 0 means not used. Not null The default value is **0**.
74	ChargeAmount1	Integer[8]	Indicates the changing amount of the first account. 0 means not unchanged. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
75	CurrentAcctAmount1	Integer[8]	Indicates the current amount of the first account. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.

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SN	Field Name	Field Type	Description
76	AccountType2	Integer[4]	Indicates the account type of the second account. 0 means not used. Not null The default value is **0**.
77	FeeType2	Integer[4]	Indicates the fee type of the second account. 0 means not used. Not null The default value is **0**.
78	ChargeAmount2	Integer[8]	Indicates the changing amount of the second account. 0 means not unchanged. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
79	CurrentAcctAmount2	Integer[8]	Indicates the current amount of the second account. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
80	AccountType3	Integer[4]	Indicates the account type of the third account. 0 means not used. Not null The default value is **0**.
81	FeeType3	Integer[4]	Indicates the fee type of the third account. 0 means not used. Not null The default value is **0**.
82	ChargeAmount3	Integer[8]	Indicates the changing amount of the third account. 0 means not unchanged. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
83	CurrentAcctAmount3	Integer[8]	Indicates the current amount of the third account. Value range: -2^{63}-2^{63}-1

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SN	Field Name	Field Type	Description
			Not null The default value is **0**.
84	AccountType4	Integer[4]	Indicates the account type of the fourth account. 0 means not used. Not null The default value is **0**.
85	FeeType4	Integer[4]	Indicates the fee type of the fourth account. 0 means not used. Not null The default value is **0**.
86	ChargeAmount4	Integer[8]	Indicates the changing amount of the fourth account. 0 means not unchanged. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
87	CurrentAcctAmount4	Integer[8]	Indicates the current amount of the fourth account. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
88	AccountType5	Integer[4]	Indicates the account type of the fifth account. 0 means not used. Not null The default value is **0**.
89	FeeType5	Integer[4]	Indicates the fee type of the fifth account. 0 means not used. Not null The default value is **0**.
90	ChargeAmount5	Integer[8]	Indicates the changing amount of the fifth account. 0 means not unchanged. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
91	CurrentAcctAmount5	Integer[8]	Indicates the current amount of the fifth account.

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SN	Field Name	Field Type	Description
			Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
92	AccountType6	Integer[4]	Indicates the account type of the sixth account. 0 means not used. Not null The default value is **0**.
93	FeeType6	Integer[4]	Indicates the fee type of the sixth account. 0 means not used. Not null The default value is **0**.
94	ChargeAmount6	Integer[8]	Indicates the changing amount of the sixth account. 0 means not unchanged. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
95	CurrentAcctAmount6	Integer[8]	Indicates the current amount of the sixth account. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
96	AccountType7	Integer[4]	Indicates the account type of the seventh account. 0 means not used. Not null The default value is **0**.
97	FeeType7	Integer[4]	Indicates the fee type of the seventh account. 0 means not used. Not null The default value is **0**.
98	ChargeAmount7	Integer[8]	Indicates the changing amount of the seventh account. 0 means not unchanged. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.

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SN	Field Name	Field Type	Description
99	CurrentAcctAmount7	Integer[8]	Indicates the current amount of the seventh account. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
100	AccountType8	Integer[4]	Indicates the account type of the eighth account. 0 means not used. Not null The default value is **0**.
101	FeeType8	Integer[4]	Indicates the fee type of the eighth account. 0 means not used. Not null The default value is **0**.
102	ChargeAmount8	Integer[8]	Indicates the changing amount of the eighth account. 0 means not unchanged. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
103	CurrentAcctAmount8	Integer[8]	Indicates the current amount of the eighth account. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
104	AccountType9	Integer[4]	Indicates the account type of the ninth account. 0 means not used. Not null The default value is **0**.
105	FeeType9	Integer[4]	Indicates the fee type of the ninth account. 0 means not used. Not null The default value is **0**.
106	ChargeAmount9	Integer[8]	Indicates the changing amount of the ninth account. 0 means not unchanged. Value range: -2^{63}-2^{63}-1 Not null

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SN	Field Name	Field Type	Description
			The default value is **0**.
107	CurrentAcctAmount9	Integer[8]	Indicates the current amount of the ninth account. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
108	AccountType10	Integer[4]	Indicates the account type of the tenth account. 0 means not used. Not null The default value is **0**.
109	FeeType10	Integer[4]	Indicates the fee type of the tenth account. 0 means not used. Not null The default value is **0**.
110	ChargeAmount10	Integer[8]	Indicates the changing amount of the tenth account. 0 means not unchanged. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
111	CurrentAcctAmount10	Integer[8]	Indicates the current amount of the tenth account. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
112	BonusValidity1	Char[14]	Indicates the validity period of the first rewarded resource. The validity period does not change when fees are deducted from accounts. In this case, the validity periods of consumption resources are not recorded. Format: YYYYMMDDhhmmss
113	BonusValidity2	Char[14]	Indicates the validity period of the second rewarded resource. The validity period does not change when fees are deducted from accounts. In this case, the validity periods of consumption resources are not recorded. Format: YYYYMMDDhhmmss
114	BonusValidity3	Char[14]	Indicates the validity period of the third

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SN	Field Name	Field Type	Description
			rewarded resource. The validity period does not change when fees are deducted from accounts. In this case, the validity periods of consumption resources are not recorded. Format: YYYYMMDDhhmmss
115	BonusValidity4	Char[14]	Indicates the validity period of the fourth rewarded resource. The validity period does not change when fees are deducted from accounts. In this case, the validity periods of consumption resources are not recorded. Format: YYYYMMDDhhmmss
116	BonusValidity5	Char[14]	Indicates the validity period of the fifth rewarded resource. The validity period does not change when fees are deducted from accounts. In this case, the validity periods of consumption resources are not recorded. Format: YYYYMMDDhhmmss
117	BonusValidity6	Char[14]	Indicates the validity period of the sixth rewarded resource. The validity period does not change when fees are deducted from accounts. In this case, the validity periods of consumption resources are not recorded. Format: YYYYMMDDhhmmss
118	BonusValidity7	Char[14]	Indicates the validity period of the seventh rewarded resource. The validity period does not change when fees are deducted from accounts. In this case, the validity periods of consumption resources are not recorded. Format: YYYYMMDDhhmmss
119	BonusValidity8	Char[14]	Indicates the validity period of the eighth rewarded resource. The validity period does not change when fees are deducted from accounts. In this case, the validity periods of consumption resources are not recorded. Format: YYYYMMDDhhmmss
120	BonusValidity9	Char[14]	Indicates the validity period of the ninth rewarded resource. The validity period does not change when fees are deducted from accounts. In this case, the validity periods of consumption resources are not recorded.

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SN	Field Name	Field Type	Description
			Format: YYYYMMDDhhmmss
121	BonusValidity10	Char[14]	Indicates the validity period of the tenth rewarded resource. The validity period does not change when fees are deducted from accounts. In this case, the validity periods of consumption resources are not recorded. Format: YYYYMMDDhhmmss
122	OnlineType	Integer[4]	Indicates online type. The options are as follows: • 0: online • 1: offline The default value is **0**.
123	Reserved	Char[22]	-
124	Reserved	Char[23]	-
125	Reserved	Char[24]	-
126	AddtionalInfo	Char[256]	Reserved
127	BearerType	Integer[4]	Indicates the type of bearer. The default value is **0**.
128	StartTime	Char[14]	PDN session start time. Indicates the value passed from AVP field [Start-Time].
129	StopTime	Char[14]	PDN session stop time.Indicates the value passed from AVP field [Stop-Time].
130	BearerProtocolType	Integer[4]	Indicates the type of the bearer protocol. When the interconnection between the OCS system and the AAA system is enabled, this field indicates the access network type. The values of the field can be extended later. The options are as follows: • 01: WAP/IP • 10: HTTP/TCP/IP • 20: STMP • 30: RTSP • 40: FTP • 100: ADSL/LAN • 101: Dial-Up/ISDN • 102: WiFi • 200: GPRS/UMTS • 300: CDMA

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SN	Field Name	Field Type	Description
			• 400: WiMAX • 500: TD-SCDMA • other: to be extended The default value is **0**.
131	ChargingID	Char[20]	Indicates the charge ID, the system can charge the subscriber according to this ID.
132	ServiceLevel	Integer[4]	Indicates the level of the data service. The default value is **0**.
133	StartTimeOfBillCycle	Char[14]	Indicates the start time of bill cycle.
134	ChargePartyIndicator	Char[1]	Indicates the charging party. The options are as follows: • 1: The calling party is charged • 2: The called party is charged • 3: Charge of third party • 4: Charge of parent & child card • 5: Charge of calling party & VPN The default value is **1**.
135	SecondChargingPartyNumber	Char[64]	Indicates the number of the other charging party of the child-parent cards. In the example, the value of the field is the number of subscriber C. The number is same as calling party number in MO and called party number in MT. There are several special situations. Like Child-Parent card feature, the calling party number is the child, the charging party number is the parent's number. This field is null when the service is paid only by account. The number format are as follows: • **Area number + PSTN** The number format is the format of the fixed-line telephone number. • **MSISDN** The number format is the format of the mobile phone number. Value range: numerals within 0-9, *, #
136	SecondChargingPartyPayType	Char[1]	Indicates the payment type of the other charging party of the child-parent cards. The options are as follows:

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SN	Field Name	Field Type	Description
			• 0: Prepaid • 1: Postpaid **NOTE** If the value of the field is null, the system records the payment type of the parent card subscriber.
137	SecondChargingPartyBillCycleID	Char[20]	Indicates the bill cycle ID of the other charging party of the child-parent cards. For example, the bill cycle ID of the parent card. Format: YYYYMMTT The description of the format is as follows: • YYYY: indicates the year. • MM: indicates the month. • TT: indicates the type of a bill cycle.
138	CallingVPNTopGroupNumber	Char[15]	Indicates the top group number of the calling party. Value range: numerals within 0-9 Can be null
139	CallingVPNGroupNumber	Char[10]	Indicates the group number of the calling party. Value range: numerals within 0-9 Can be null
140	CallingVPNShortNumber	Char[10]	Indicates the short number of the calling party. Value range: numerals within 0-9 Can be null
141	CalledVPNTopGroupNumber	Char[15]	Indicates the top group number of the called party. Value range: numerals within 0-9 Can be null
142	CalledVPNGroupNumber	Char[10]	Indicates the group number of the called party. Value range: numerals within 0-9 Can be null
143	CalledVPNShortNumber	Char[10]	Indicates the short number of the called party Value range: numerals within 0-9 Can be null
144	CallingNetworkType	Char[10]	Indicates the type of the network accessed by the calling party.

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SN	Field Name	Field Type	Description
			The options are as follows: • 100: PSTN • 101: PBX • 102: PHS • 200: GSM • 201: GPRS • 202: EDGE • 300: CDMA • 301: CDMA 450 • 400: WCDMA • 500: TD-SCDMA • 600: IP/IMS • 601: WIFI • 602: WIMAX • Other to be extended Can be null
145	CalledNetworkType	Char[10]	Indicates the type of the network accessed by the called party. The options are as follows: • 100: PSTN • 101: PBX • 102: PHS • 200: GSM • 201: GPRS • 202: EDGE • 300: CDMA • 301: CDMA 450 • 400: WCDMA • 500: TD-SCDMA • 600: IP/IMS • 601: WIFI • 602: WIMAX • Other to be extended Can be null
146	GroupCallType	Char[10]	Reserved
147	GroupPayFlag	Char[10]	Indicates the payment mode of the group users. The options are as follows: • 0: Personal payment

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SN	Field Name	Field Type	Description
			• 1: Group payment • 2: Group and Personal payment Can be null
148	Diagnostics	Char[12]	This field provides a more detailed release cause value from PCN node. This field is reserved for future use.
149	Reserved	Char[10]	-
150	Reserved	Char[10]	-
151	Reserved	Char[10]	-
152	Reserved	Char[10]	-
153	ChargeOfItemsSecondAccounts	Integer[4]	Indicates the free call times used from others, such as the parent card. Value range: -2^{31}-2^{31}-1 Not null The default value is **0**.
154	ChargeOfDurationSecondAccounts	Integer[4]	Indicates the total call time charge using reward resource from others, such as the parent card. Value range: -2^{31}-2^{31}-1 Not null The default value is **0**.
155	ChargeOfFundSecondAccounts	Integer[8]	Indicates the total cost amount charging from fund account of others, such as the parent card. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
156	ChargeFromSecondPrepaidAccount	Integer[8]	Indicates the charge amount from other prepaid account. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
157	SecondPrepaidAccountBalance	Integer[8]	Indicates the balance of other prepaid account. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
158	ChargeFromSecondPostpaidAccount	Integer[8]	Indicates the charge amount from other postpaid account.

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SN	Field Name	Field Type	Description
			Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
159	SecondPostpaidAccountBalance	Integer[8]	Indicates the balance of other postpaid account. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
160	SecondAccountID	Char[20]	Indicates the main account ID of the other charging party that the system deducts fees from.
161	SecondAccountKey	Integer[8]	Indicates the main account key of the other charging party that the system deducts fees from. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
162	SecondAccountCurrencyCode	Integer[4]	Currency code of others, such as the parent card. Also, if this field missed or present be Zero, it means the system use local minimal currency as default, and no need to specify in CDR. The default value is **0**.
163	Reserved	Char[20]	-
164	SecondAccountType1	Integer[4]	Indicates the account type of the first account of others, such as the parent card. 0 means not used. Not null The default value is **0**.
165	SecondFeeType1	Integer[4]	Indicates the fee type of the first account of others, such as the parent card. 0 means not used. Not null The default value is **0**.
166	SecondChargeAmount1	Integer[8]	Indicates the changing amount of the first account of others, such as the parent card. 0 means unchanged. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.

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SN	Field Name	Field Type	Description
167	SecondCurrentAcctAmount1	Integer[8]	Indicates the current amount of the first account of others, such as the parent card. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
168	SecondAccountType2	Integer[4]	Indicates the account type of the second account of others, such as the parent card. 0 means not used. Not null The default value is **0**.
169	SecondFeeType2	Integer[4]	Indicates the fee type of the second account of others, such as the parent card. 0 means not used. Not null The default value is **0**.
170	SecondChargeAmount2	Integer[8]	Indicates the changing amount of the second account of others, such as the parent card. 0 means unchanged. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
171	SecondCurrentAcctAmount2	Integer[8]	Indicates the current amount of the second account of others, such as the parent card. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
172	SecondAccountType3	Integer[4]	Indicates the account type of the third account of others, such as the parent card. 0 means not used. Not null The default value is **0**.
173	SecondFeeType3	Integer[4]	Indicates the fee type of the third account of others, such as the parent card. 0 means not used. Not null The default value is **0**.
174	SecondChargeAmount3	Integer[8]	Indicates the changing amount of the third account of others, such as the parent card. 0 means unchanged.

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SN	Field Name	Field Type	Description
			Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
175	SecondCurrentAcctAmount3	Integer[8]	Indicates the current amount of the third account of others, such as the parent card. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
176	SecondAccountType4	Integer[4]	Indicates the account type of the fourth account of others, such as the parent card. 0 means not used. Not null The default value is **0**.
177	SecondFeeType4	Integer[4]	Indicates the fee type of the fourth account of others, such as the parent card. 0 means not used. Not null The default value is **0**.
178	SecondChargeAmount4	Integer[8]	Indicates the changing amount of the fourth account of others, such as the parent card. 0 means unchanged. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
179	SecondCurrentAcctAmount4	Integer[8]	Indicates the current amount of the fourth account of others, such as the parent card. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
180	SecondAccountType5	Integer[4]	Indicates the account type of the fifth account of others, such as the parent card. 0 means not used. Not null The default value is **0**.
181	SecondFeeType5	Integer[4]	Indicates the fee type of the fifth account of others, such as the parent card. 0 means not used. Not null The default value is **0**.

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SN	Field Name	Field Type	Description
182	SecondChargeAmount5	Integer[8]	Indicates the changing amount of the fifth account of others, such as the parent card. 0 means unchanged. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
183	SecondCurrentAcctAmount5	Integer[8]	Indicates the current amount of the fifth account of others, such as the parent card. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
184	SecondBonusValidity1	Char[14]	Indicates the validity period of the first rewarded resource of others, such as the parent card. The validity period does not change when fees are deducted from accounts. In this case, the validity periods of consumption resources are not recorded. Format: YYYYMMDDhhmmss
185	SecondBonusValidity2	Char[14]	Indicates the validity period of the second rewarded resource of others, such as the parent card. The validity period does not change when fees are deducted from accounts. In this case, the validity periods of consumption resources are not recorded. Format: YYYYMMDDhhmmss
186	SecondBonusValidity3	Char[14]	Indicates the validity period of the third rewarded resource of others, such as the parent card. The validity period does not change when fees are deducted from accounts. In this case, the validity periods of consumption resources are not recorded. Format: YYYYMMDDhhmmss
187	SecondBonusValidity4	Char[14]	Indicates the validity period of the fourth rewarded resource of others, such as the parent card. The validity period does not change when fees are deducted from accounts. In this case, the validity periods of consumption resources are not recorded. Format: YYYYMMDDhhmmss
188	SecondBonusValidity5	Char[14]	Indicates the validity period of the fifth rewarded resource of others, such as the parent card. The validity period does not change when fees are deducted from accounts. In this case, the validity periods of consumption resources are not recorded.

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SN	Field Name	Field Type	Description
			Format: YYYYMMDDhhmmss
189	3GPPRATType	Integer[4]	Reserved
190	PulaPulaProperty	Integer[4]	Reserved
191	BumerangeProperty	Integer[4]	Reserved
192	RateName	Char[25]	Reserved
193	IsTestNumber	Integer[4]	Reserved
194	MainProductStatus	Integer[4]	Reserved
195	SubscriberIDType	Char[1]	Indicates ID Type of Subscriber. The options are as follows: • 0: END_USER_E164 • 2: END_USER_SIP_URI Not null The default value is **0**.
196	DiscountOfLastEff Product	Integer[4]	Indicates the discount percentage for the last rated product that takes effect. For a product that does not provide discounts, the value is **100**. Value range: 0-100 **NOTE** The value **0** is not included. Not null The default value is **100**.
197	subscriberkey	Integer[4]	Indicates the key of a subscriber. The default value is **0**.
198	accumulationchang e	Char[256]	Reserved
199	UsageServiceType	Integer[4]	Indicates the type of a real-time service. The default value is **1400**.
200	SourceSerialNo	Integer[8]	Reserved
201	SourceSubSequence	Integer[4]	Reserved
202	AcctBalChgList	Char[768]	Reserved
203	OriginCdrFilename	Char[64]	Indicates the name of a rated CDR. The default value is **0**.
204	CurCDRLineNo	Integer[4]	Indicates the number of the line where the original CDR is located. The default value is **0**.

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SN	Field Name	Field Type	Description
205	Tax	Integer[4]	Reserved
206	Tax2	Integer[4]	Reserved
207	Tax3	Integer[4]	Reserved
208	Tax4	Integer[4]	Reserved
209	DTDiscount	String	Indicates dynamic discount. Can be null.
210	DTCellId	String	Indicates the Cell ID considered for DT discount. In case of default DT discount, record it as '000000000000000'. Can be null.

3.1.6 vou CDR

Description

A vou CDR is generated when recharging or present successfully.

Generation Scenario

A vou CDR is generated in the following scenarios:

- When an OCS subscriber uses the call service for the first time or activates the MSISDN through the IVR recharge, the subscriber can obtain a certain bonus.
- When an OCS subscriber uses a recharge card or cash for recharge, the system provides a bonus.
- The system successfully recharges an account in advance.
- An OCS subscriber successfully recharges an account in advance through the IVR management process.

CDR File Name Format

vouYYYYMMDD_CBPNODE_Seqno.unl

Example of a vou CDR file name: vou20071112_101_000151.unl

Format

Table 3-7 describes each field of vou CDR file.

Table 3-7 Format of vou CDR

SN	Field Name	Field Type	Description
1	SerialNo	Integer[8]	Indicates the serial number.

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SN	Field Name	Field Type	Description
			It is generated by the system and the serial number is exclusive. Not null The default value is **0**.
2	TimeStamp	Char[14]	Indicates the timestamp of the bill item generation. Format: YYYYMMDDhhmmss Not null
3	BrandID	Integer[4]	Indicates the brand ID of subscribers. Not null The default value is **0**.
4	SubCOSID	Integer[4]	Indicates the main product ID of subscribers. Not null The default value is **0**.
5	ChargingPartyNumber	Char[64]	Indicates the telephone number of the charging party. The number is same as calling party number in MO and called party number in MT. There are several special situations. Like Child-Parent card feature, the calling party number is the child, the charging party number is the parent's number. This field is null when the service is paid only by account. The number format are as follows: • **Area number + PSTN** The number format is the format of the fixed-line telephone number. • **MSISDN** The number format is the format of the mobile phone number. Value range: numerals within 0-9, *, # **NOTE** It can be null in MO.
6	PayType	Char[1]	Indicates the pay type of subscribers. The options are as follows: • 0: Prepaid • 1: Postpaid
7	VoucherPINNumber	Char[32]	Record as null.
8	VoucherBatchNum	Char[12]	Indicates the batch number of a recharge

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SN	Field Name	Field Type	Description
	ber		voucher. **NOTE** It can not be null for voucher card recharge.
9	VoucherSequence	Char[20]	Indicates the serial number of a recharge voucher. **NOTE** It can not be null for voucher card recharge.
10	VoucherCosID	Integer[4]	Indicates the type of a recharge voucher. The default value is **0**.
11	VoucherSPID	Char[20]	• Indicates the ID of the recharge card distributor in the case of using a recharge card to recharge. • Indicates the ID of the e-top or e-voucher distributor in the case of cash recharge. • Indicates the bank ID or the bank code in the case of recharging through an ATM or a bank. The default value is **0**.
12	CardFaceValue	Integer[4]	Indicates the face value of a recharge card. Unit: minimum currency unit Not null The default value is **0**.
13	CardValueAdded	Integer[4]	Indicates the money for preference. Unit: minimum currency unit Not null The default value is **0**.
14	CardValidityAdded	Integer[4]	Indicates the validity period for preference. Not null The default value is **0**.
15	TradeTime	Char[19]	Indicates the recharge time. Format: YYYYMMDDhhmmss Not null
16	OperatedBy	Char[64]	Indicates the number of the operator.
17	ThirdPartyNumber	Char[64]	Indicates the number that recharges accounts. The calling number with an international number prefix. The number format are as follows: • **Country code + Area number + PSTN** The number format is the format of the fixed-line phone number.

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SN	Field Name	Field Type	Description
			• **Country code + MSISDN** The number format is the format of the mobile phone number. Value range: numerals within 0-9, *, # **NOTE** It can not null when recharge to others.
18	AccessMethod	Integer[4]	Indicates the access method. • 1: IVR • 2: USSD • 3: SMS • 4: SMAP • 5: WebService interface • 6: RTBP(for Ufone) Not null The default value is **0**.
19	TradeType	Char[12]	Indicates the recharge type. The format combined with several factors, like access method, whether is third-party recharge. The options are as follows: • 0: first activation • 2: Voucher recharge through IVR from self phone • 6: Voucher recharge through USSD from self phone • 7: Voucher recharge by operator/external system (manual recharge) • 8: First activation with recharge card • 9: First activation with cash • 10: Loan from USSD • 11: Loan from IVR • 999: EVC recharge • 1001: Cash recharge • 1002: ATM recharge • 1007: Loan • 2018: S-Voucher by third-party • 2019: E-Voucher by third-party Not null
20	RechargeAreaNumber	Integer[4]	Indicates the number of the area in which the request is initiated. The area number is recorded here if it can be obtained from the interface.

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SN	Field Name	Field Type	Description
			If it can't be obtained from the interface, the value of area number is 0, -1 or 9999. The default value is **0**.
21	TransitionID	Char[36]	Indicates the transaction ID.
22	VoucherEncryptNumber	Char[22]	Record as null.
23	CallingCellID	Char[20]	It will record CellID in case of IVR Recharge, USSD Recharge and Cash Recharge. Value range: numerals within 0-9, *, #
24	SubscriberType	Integer[4]	Indicates the subscriber type. The options are as follows: • 0: Normal OCS • 1: CST Master • 2: CST Client • 3: Corporate Master • 4: Corporate Slave • 5: Commitment Subscriber
25	Reserved	Char[25]	-
26	ProductID	Char[20]	Indicates the last effect rating product. For example, ordinary recharges, recharge rewards, and recharge packages.
27	ServiceType	Char[20]	Reserved
28	OldUserState	Char[10]	Indicates the life cycle state and the management state of a subscriber before recharging. The **OldUserState** parameter is a numeric string of seven digits, in the format CCMMMMM. The first and second digits C indicate the life cycle state of a subscriber. The first digit C indicates the state of a prepaid subscriber, and the second digit C indicates the state of a postpaid subscriber. The options are as follows: • 0: Idle • 1: Active • 2: Suspended • 3: Disable • 4: Pool • 5: Pool without activation

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SN	Field Name	Field Type	Description
			The next five digits MMMMM indicate the management state of a subscriber. The options are as follows: Third digit: Whether the subscriber claims missing. • 0: no • 1: yes Fourth digit: Whether the subscriber suspends the service. • 0: non-suspend • 1: suspend • 2: suspend by operator (B1W) • 3: suspend by operator (B2W) Fifth digit: Whether the subscriber applies for suspension with the number being reserved. • 0: no • 1: yes Sixth digit: Whether the subscriber is blocked. • 0: no • 1: yes Seventh digit: Whether the subscriber is blacklisted. • 0: no • 1: yes
29	CurrentUserSate	Char[10]	Indicates the life cycle state and the management state of a subscriber after recharging. The **CurrUserState** parameter is a numeric string of seven digits, in the format CCMMMMM. The first and second digits C indicate the life cycle state of a subscriber. The first digit C indicates the state of a prepaid subscriber, and the second digit C indicates the state of a postpaid subscriber. The options are as follows: • 0: Idle • 1: Active • 2: Suspended • 3: Disable • 4: Pool • 5: Pool without activation

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SN	Field Name	Field Type	Description
			The next five digits MMMMM indicate the management state of a subscriber. The options are as follows: Third digit: Whether the subscriber claims missing. • 0: no • 1: yes Fourth digit: Whether the subscriber suspends the service. • 0: non-suspend • 1: suspend • 2: suspend by operator (B1W) • 3: suspend by operator (B2W) Fifth digit: Whether the subscriber applies for suspension with the number being reserved. • 0: no • 1: yes Sixth digit: Whether the subscriber is blocked. • 0: no • 1: yes Seventh digit: Whether the subscriber is blacklisted. • 0: no • 1: yes
30	PreviousActivestop	Char[14]	Indicates the expiration of the Active before recharging. Format: YYYYMMDDhhmmss
31	NewActivestop	Char[14]	Indicates the expiration of the Active after recharging. Format: YYYYMMDDhhmmss
32	SuspendStop	Char[14]	Indicates the expiration of the Suspend after recharging. Format: YYYYMMDDhhmmss
33	DisableStop	Char[14]	Indicates the expiration of the Disable after recharging. Format: YYYYMMDDhhmmss
34	ValidityAdded	Integer[4]	Indicates the actually validity added after recharging. Not null The default value is **0**.

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SN	Field Name	Field Type	Description
35	RewardCycles	Integer[4]	Indicates the number of installment rebate periods. For the one-off reward, the value of this field is **1**. Not null The default value is **0**.
36	RebateSchemaID	Char[20]	Indicates the ID of the installment rebate plan.
37	BillCycleID	Char[20]	Indicates the bill cycle ID. Format: YYYYMMTT • YYYY: indicates the year. • MM: indicates the month. • TT: indicates the type of a bill cycle. Value range: numerals within 0-9
38	SubscriberID	Char[20]	Indicates the subscriber ID. Value range: characters in [0-9, a-z, A-Z]
39	Reserved	Char[20]	-
40	ResultCode	Integer[4]	Indicates the internal result code. Value range: -2^{31}-2^{31}-1 The default value is **0**.
41	AccountID	Char[20]	Indicates the ID of the main account. Value range: characters in [0-9, a-z, A-Z]
42	RechargeTax	Integer[4]	Indicates the recharge tax. The system deducts the recharge tax when a subscriber recharges an account. The default value is **0**.
43	RechargePenalty	Integer[4]	Indicates the penalty for the recharge. Sometimes, some kind of one-off charge for first activation or recharge will also be recorded in this field. The default value is **0**.
44	CurrencyCode	Integer[4]	Currency code. Also, if this field missed or present be Zero, it means the system use local minimal currency as default, and no need to specify in CDR. The default value is **0**.
45	Reserved	Char[20]	-
46	PrepaidBalanceBefore	Integer[8]	Indicates the balance of prepaid account before recharging.

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SN	Field Name	Field Type	Description
			Unit: minimum currency unit Not null The default value is **0**.
47	RechargeForPrepaid	Integer[8]	Indicates the recharge amount for prepaid account. Unit: minimum currency unit Not null The default value is **0**.
48	PrepaidBalance	Integer[8]	Indicates the balance of prepaid account after recharging. Unit: minimum currency unit Not null The default value is **0**.
49	PostPaidBalanceBefore	Integer[8]	Indicates the balance of postpaid account before recharging. Value range: -2^{63}-2^{63}-1 Unit: minimum currency unit Not null The default value is **0**.
50	RechargeForPostpaid	Integer[8]	Indicates the recharge amount for postpaid account. Unit: minimum currency unit Not null The default value is **0**.
51	PostPaidBalance	Integer[8]	Indicates the balance of postpaid account after recharging. Value range: -2^{63}-2^{63}-1 Unit: minimum currency unit Not null The default value is **0**.
52	AccountType1	Integer[4]	Indicates the account type of the first account. Not null The default value is **0**.
53	ChargeAmount1	Integer[8]	Indicates the reward amount of the first account. Unit: minimum currency unit Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.

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SN	Field Name	Field Type	Description
54	CurrentAcctAmount1	Integer[8]	Indicates the current changing amount of the first account. Unit: minimum currency unit Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
55	AccountType2	Integer[4]	Indicates the account type of the second account. Not null The default value is **0**.
56	ChargeAmount2	Integer[8]	Indicates the reward amount of the second account. Unit: minimum currency unit Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
57	CurrentAcctAmount2	Integer[8]	Indicates the current changing amount of the second account. Unit: minimum currency unit Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
58	AccountType3	Integer[4]	Indicates the account type of the third account. Not null The default value is **0**.
59	ChargeAmount3	Integer[8]	Indicates the reward amount of the third account. Unit: minimum currency unit Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
60	CurrentAcctAmount3	Integer[8]	Indicates the current changing amount of the third account. Unit: minimum currency unit Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
61	AccountType4	Integer[4]	Indicates the account type of the fourth account.

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SN	Field Name	Field Type	Description
			Not null The default value is **0**.
62	ChargeAmount4	Integer[8]	Indicates the reward amount of the fourth account. Unit: minimum currency unit Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
63	CurrentAcctAmount4	Integer[8]	Indicates the current changing amount of the fourth account. Unit: minimum currency unit Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
64	AccountType5	Integer[4]	Indicates the account type of the fifth account. Not null The default value is **0**.
65	ChargeAmount5	Integer[8]	Indicates the reward amount of the fifth account. Unit: minimum currency unit Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
66	CurrentAcctAmount5	Integer[8]	Indicates the current changing amount of the fifth account. Unit: minimum currency unit Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
67	AccountType6	Integer[4]	Indicates the account type of the sixth account. Not null The default value is **0**.
68	ChargeAmount6	Integer[8]	Indicates the reward amount of the sixth account. Unit: minimum currency unit Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.

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SN	Field Name	Field Type	Description
69	CurrentAcctAmount6	Integer[8]	Indicates the current changing amount of the sixth account. Unit: minimum currency unit Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
70	AccountType7	Integer[4]	Indicates the account type of the seventh account. Not null The default value is **0**.
71	ChargeAmount7	Integer[8]	Indicates the reward amount of the seventh account. Unit: minimum currency unit Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
72	CurrentAcctAmount7	Integer[8]	Indicates the current changing amount of the seventh account. Unit: minimum currency unit Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
73	AccountType8	Integer[4]	Indicates the account type of the eighth account. Not null The default value is **0**.
74	ChargeAmount8	Integer[8]	Indicates the reward amount of the eighth account. Unit: minimum currency unit Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
75	CurrentAcctAmount8	Integer[8]	Indicates the current changing amount of the eighth account. Unit: minimum currency unit Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
76	AccountType9	Integer[4]	Indicates the account type of the ninth account.

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SN	Field Name	Field Type	Description
			Not null The default value is **0**.
77	ChargeAmount9	Integer[8]	Indicates the reward amount of the ninth account. Unit: minimum currency unit Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
78	CurrentAcctAmount9	Integer[8]	Indicates the current changing amount of the ninth account. Unit: minimum currency unit Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
79	AccountType10	Integer[4]	Indicates the account type of the tenth account. Not null The default value is **0**.
80	ChargeAmount10	Integer[8]	Indicates the reward amount of the tenth account. Unit: minimum currency unit Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
81	CurrentAcctAmount10	Integer[8]	Indicates the current changing amount of the tenth account. Unit: minimum currency unit Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
82	BonusValidity1	Char[14]	Indicates the validity period of the first rewarded resource. The validity period does not change when fees are deducted from accounts. In this case, the validity periods of consumption resources are not recorded. Format: YYYYMMDDhhmmss
83	BonusValidity2	Char[14]	Indicates the validity period of the second rewarded resource. The validity period does not change when fees are deducted from accounts. In this case, the validity periods of consumption resources are not recorded.

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SN	Field Name	Field Type	Description
			Format: YYYYMMDDhhmmss
84	BonusValidity3	Char[14]	Indicates the validity period of the third rewarded resource. The validity period does not change when fees are deducted from accounts. In this case, the validity periods of consumption resources are not recorded. Format: YYYYMMDDhhmmss
85	BonusValidity4	Char[14]	Indicates the validity period of the fourth rewarded resource. The validity period does not change when fees are deducted from accounts. In this case, the validity periods of consumption resources are not recorded. Format: YYYYMMDDhhmmss
86	BonusValidity5	Char[14]	Indicates the validity period of the fifth rewarded resource. The validity period does not change when fees are deducted from accounts. In this case, the validity periods of consumption resources are not recorded. Format: YYYYMMDDhhmmss
87	BonusValidity6	Char[14]	Indicates the validity period of the sixth rewarded resource. The validity period does not change when fees are deducted from accounts. In this case, the validity periods of consumption resources are not recorded. Format: YYYYMMDDhhmmss
88	BonusValidity7	Char[14]	Indicates the validity period of the seventh rewarded resource. The validity period does not change when fees are deducted from accounts. In this case, the validity periods of consumption resources are not recorded. Format: YYYYMMDDhhmmss
89	BonusValidity8	Char[14]	Indicates the validity period of the eighth rewarded resource. The validity period does not change when fees are deducted from accounts. In this case, the validity periods of consumption resources are not recorded. Format: YYYYMMDDhhmmss
90	BonusValidity9	Char[14]	Indicates the validity period of the ninth rewarded resource. The validity period does not change when fees are deducted from accounts. In this case, the validity periods of consumption resources are not recorded. Format: YYYYMMDDhhmmss
91	BonusValidity10	Char[14]	Indicates the validity period of the tenth rewarded resource. The validity period does

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SN	Field Name	Field Type	Description
			not change when fees are deducted from accounts. In this case, the validity periods of consumption resources are not recorded. Format: YYYYMMDDhhmmss
92	PreviousSuspendStop	Char[14]	Indicates the expiration date of **Suspend** state before recharging. Format: YYYYMMDDhhmmss
93	PreviousDisableStop	Char[14]	Indicates the expiry date of **Disable** state before recharging. Format: YYYYMMDDhhmmss
94	AgentName	Char[64]	The name of the agent. This field indicates the information of provider of recharge channel.
95	AddtionalInfo	Char[256]	Reserved
96	StartTimeOfBillCycle	Char[14]	Indicates the start time of bill cycle.
97	LoanAmount	Integer[8]	Reserved
98	LoanPoundage	Integer[8]	Reserved
99	ChargedSubAcctType	Integer[4]	Reserved
100	ChargedSubAcctKey	Integer[8]	Reserved
101	ChargedSubAcctAmount	Integer[8]	Reserved
102	CurrSubAcctBalance	Integer[8]	Reserved
103	ChangeSubAcctValidity	Integer[4]	Reserved
104	CurrSubAcctExpireTime	Char[14]	Reserved
105	PrepaidProGroupID	Char[14]	Reserved
106	IsTestNumber	Integer[4]	Reserved
107	MainProductStatus	Integer[4]	Reserved
108	BankCode	Char[20]	Reserved
109	SubscriberIDType	Char[1]	Indicates ID Type of Subscriber. The options are as follows: 0: END_USER_E164

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SN	Field Name	Field Type	Description
			2: END_USER_SIP_URI Not null The default value is **0**.
110	DiameterSessionID	Char[128]	Indicates the session id that generated during the interaction between the SCP and the CBP. The field includes parts as follows: <DiameterIdentity>;<high 32 bits>;<low 32 bits>[;<optional value>]. • <DiameterIdentity>: same as Origin-Host; • <high 32 bits>: the current day • <low 32 bits>: increase by degrees circulatory, the start value is 0. Set the value is 0 when the system restarts. • <optional value>: reserved. Every fields are separated by a character ";". Example: SCP001.huawei.com;1876543210;523
111	subscriberkey	Integer[4]	Indicates the key of a subscriber. The default value is **0**.
112	accountkey	Integer[8]	Indicates the key of a account. The default value is **0**.
113	UsageServiceType	Integer[4]	Indicates the type of a real-time service. The default value is **1700**.
114	OriginCdrFilename	Char[64]	Reserved
115	CurCDRLineNo	Integer[4]	Reserved
116	Reason	Integer[4]	Indicates the recharge reason type. The default value is **-1**.
117	Comment	Char[256]	Indicates the recharge reason description.
118	LoanGrade	Integer[4]	Reserved
119	LoanAcctType	Integer[4]	Reserved
120	LoanAcctBalance	Integer[8]	Reserved
121	ETUGracePeriod	Char[14]	Reserved
122	Tax	Integer[8]	Reserved
123	Tax2	Integer[8]	Reserved
124	Tax3	Integer[8]	Reserved

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SN	Field Name	Field Type	Description
125	Tax4	Integer[8]	Reserved
126	FeeType1	Integer[4]	Indicates the fee type of the first account. Fee types record fee change information such as consumption fees and rewarded fees. 0 means not used. Not null The default value is **0**.
127	FeeType2	Integer[4]	Indicates the fee type of the second account. Fee types record fee change information such as consumption fees and rewarded fees. 0 means not used. Not null The default value is **0**.
128	FeeType3	Integer[4]	Indicates the fee type of the third account. Fee types record fee change information such as consumption fees and rewarded fees. 0 means not used. Not null The default value is **0**.
129	FeeType4	Integer[4]	Indicates the fee type of the fourth account. Fee types record fee change information such as consumption fees and rewarded fees. 0 means not used. Not null The default value is **0**.
130	FeeType5	Integer[4]	Indicates the fee type of the fifth account. Fee types record fee change information such as consumption fees and rewarded fees. 0 means not used. Not null The default value is **0**.
131	FeeType6	Integer[4]	Indicates the fee type of the sixth account. Fee types record fee change information such as consumption fees and rewarded fees. 0 means not used. Not null The default value is **0**.
132	FeeType7	Integer[4]	Indicates the fee type of the seventh account. Fee types record fee change information such

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SN	Field Name	Field Type	Description
			as consumption fees and rewarded fees. 0 means not used. Not null The default value is **0**.
133	FeeType8	Integer[4]	Indicates the fee type of the eighth account. Fee types record fee change information such as consumption fees and rewarded fees. 0 means not used. Not null The default value is **0**.
134	FeeType9	Integer[4]	Indicates the fee type of the ninth account. Fee types record fee change information such as consumption fees and rewarded fees. 0 means not used. Not null The default value is **0**.
135	FeeType10	Integer[4]	Indicates the fee type of the tenth account. Fee types record fee change information such as consumption fees and rewarded fees. 0 means not used. Not null The default value is **0**.

3.1.7 mon CDR

Description

An mon CDR records the information about the monthly fee and the bonus.

Generation Scenario

When an OCS subscriber subscribes to a package, and the system successfully deducts the monthly fee from the account of the subscriber or the system rebates resources to the subscriber periodically, an mon CDR is generated.

NOTE

For the description of child-parent cards, see section 3.1.2 rec CDR .

CDR File Name Format

monYYYYMMDD_CBPNODE_Seqno.unl

Example of an mon CDR file name: mon20071112_101_000151.unl

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Format

Table 3-8 describes each field of mon CDR file.

Table 3-8 Format of mon CDR

SN	Field Name	Field Type	Description
1	SerialNo	Integer[8]	Indicates the serial number. It is generated by the system and the serial number is exclusive. Not null The default value is **0**.
2	TimeStamp	Char[14]	Indicates the timestamp of the bill item generation. Format: YYYYMMDDhhmmss Not null
3	BrandID	Integer[4]	Indicates the brand ID of subscriber. Not null The default value is **0**.
4	SubCOSID	Integer[4]	Indicates the main product ID of subscribers. Not null The default value is **0**.
5	PayType	Char[1]	Indicates the pay type of subscribers. The options are as follows: • 0: Prepaid • 1: Postpaid Not null
6	ChargingPartyNumber	Char[64]	Indicates the charging party ID. It can be the MSISDN, customer ID. Value range: numerals within 0-9, *, # Not null
7	ChargingPartyType	Char[1]	Indicates the type of charging party. The options are as follows: • 0: MSISDN subscriber • 1: Customer ID • 2: Group No Not null
8	Reserved	Integer[4]	-
9	BillCycleID	Char[20]	Indicates the bill cycle ID. Format: YYYYMMTT • YYYY: indicates the year.

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SN	Field Name	Field Type	Description
			• MM: indicates the month. • TT: indicates the type of a bill cycle. Value range: numerals within 0-9
10	CycleBeginTime	Char[14]	Indicates the start time of a rental cycle. Format: YYYYMMDDhhmmss Not null The default value is **0**.
11	CycleEndTime	Char[14]	Indicates the end time of a rental cycle. The default value is **0**.
12	CycleType	Integer[4]	Indicates the cycle type. The value of this field is configurable according to different template parameters. The options are as follows: • 1: bill cycle Indicates that the cycle is the bill cycle. • 2: days Indicates that the cycle is days. • 3: weeks Indicates that the cycle is weeks. • 4: months Indicates that the cycle is months. • Other: reserved Not null The default value is **0**.
13	CycleLength	Integer[4]	Indicates the length of a bill cycle. The value of this field is configurable according to different template parameters. Not null The default value is **0**.
14	ElapseCycles	Integer[4]	Indicates the number of bill cycles. Not null The default value is **0**.
15	InnerCycleBeginTime	Char[14]	Indicates the start time of an inner bill cycle. Format: YYYYMMDDhhmmss Not null The default value is **0**.
16	InnerCycleEndTime	Char[14]	Indicates the end time of an inner bill cycle. Format: YYYYMMDDhhmmss

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SN	Field Name	Field Type	Description
			Not null The default value is **0**.
17	ProductID	Char[20]	Indicates the ID of a product or product package that corresponds to a rental type or package. Not null
18	ServiceType	Integer[4]	Indicates the service included in the product. For example, CID, RBT, FN, and CallScreen. They are predefined in the OCS, like: • 1: Voice • 2: SMS • 3: MMS • 4: GPRS • 6: RBT • 8: VoiceVideoPhoneService • 9: VoiceData • 10: VoiceFax • 11: Content • 20: FN • 21: HomeZone • 22: CallScreen • 24: VoIP • 26: CUG • 33: Package • 38: OneTimePackage • 41: MCN • 42: VMS • 45: DestNumber • 50: AOC • 73: DailyBucket Not null The default value is **0**.
19	UserState	Char[10]	Indicates the life cycle state and the management state of a subscriber. The **UserState** parameter is a numeric string of seven digits, in the format CCMMMMM. The first and second digits C indicate the life cycle state of a subscriber. The first digit C indicates the state of a prepaid subscriber, and the second digit C indicates the state of a postpaid subscriber. The options are as

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SN	Field Name	Field Type	Description
			follows: • 0: Idle • 1: Active • 2: Suspended • 3: Disable • 4: Pool • 5: Pool without activation The next five digits MMMMM indicate the management state of a subscriber. The options are as follows: Third digit: Whether the subscriber claims missing. • 0: no • 1: yes Fourth digit: Whether the subscriber suspends the service. • 0: non-suspend • 1: suspend • 2: suspend by operator (B1W) • 3: suspend by operator (B2W) Fifth digit: Whether the subscriber applies for suspension with the number being reserved. • 0: no • 1: yes Sixth digit: Whether the subscriber is blocked. • 0: no • 1: yes Seventh digit: Whether the subscriber is blacklisted. • 0: no • 1: yes Not null
20	SubscriberID	Char[20]	Indicates the subscriber ID. Value range: characters in [0-9, a-z, A-Z] Not null
21	OrderStatus	Char[20]	Indicates the subscription relation status. The options are as follows: • 0: normal • 1: suspend

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SN	Field Name	Field Type	Description
			• 2: not activated • 3: expired The default value is **0**.
22	TotalCycles	Integer[4]	Record the number of periods for installation reward. The default value is **0**.
23	ProductOrderKey	Char[20]	Indicates a product subscription key.
24	OperatorId	Char[20]	Indicates the Operator Id received from external system or SMAP during product subscription operation.
25	ResultCode	Integer[4]	Indicates the internal result code. Value range: -2^{31}-2^{31}-1 The default value is **0**.
26	ChargeOfItemsAccounts	Integer[8]	Indicates the free call times used. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
27	DeductResourceAccounts	Integer[4]	Indicates the total call time charge using reward resource. Not null The default value is **0**.
28	DeductFundAccounts	Integer[4]	Indicates the total cost amount charging from fund account. Not null The default value is **0**.
29	DeductFromPrepaid	Integer[8]	Indicates the charge amount from prepaid account. Not null The default value is **0**.
30	PrepaidBalance	Integer[8]	Indicates the balance of prepaid account. Not null The default value is **0**.
31	DeductFromPostpaid	Integer[8]	Indicates the charge amount from postpaid account. Not null The default value is **0**.
32	PostPaidBalance	Integer[8]	Indicates the balance of postpaid available limitation. **NOTE**

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SN	Field Name	Field Type	Description
			• If the rental of a postpaid subscriber is debited to the previous bill cycle, the rental CDR of the subscriber needs to record the balance of the previous bill cycle after fee deduction. • For postpaid subscribers, the value of this field does not include the fee that is deducted virtually. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
33	AccountID	Char[20]	Indicates the ID of the main account. Value range: characters in [0-9, a-z, A-Z]
34	AccountKey	Integer[8]	Indicates the key of the main account. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
35	CurrencyCode	Integer[4]	Currency code. Also, if this field missed or present be Zero, it means the system use local minimal currency as default, and no need to specify in CDR.
36	Reserved	Char[20]	-
37	AccountType1	Integer[4]	Indicates the account type of the first account. 0 means not used. The first group of account change information include the following field: • AccountType • ChargeAmount • CurrentAcctAmount **NOTE** In the recharge and rental deduction processes, this fee group records only reward-related information, but does not record consumption-, tax-, and rental-related information. The account change information structures from the second to the tenth groups are the same as that of the first group. The default value is **0**.
38	ChargeAmount1	Integer[8]	Indicates the reward amount of the first account. 0 means unchanged. Unit: minimum currency unit Value range: -2^{63}-2^{63}-1

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SN	Field Name	Field Type	Description
			The default value is **0**.
39	CurrentAcctAmount1	Integer[8]	Indicates the current amount of the first account. Unit: minimum currency unit Value range: -2^{63}-2^{63}-1 The default value is **0**.
40	AccountType2	Integer[4]	Indicates the account type of the second account. 0 means not used. The default value is **0**.
41	ChargeAmount2	Integer[8]	Indicates the reward amount of the second account. 0 means unchanged. Unit: minimum currency unit Value range: -2^{63}-2^{63}-1 The default value is **0**.
42	CurrentAcctAmount2	Integer[8]	Indicates the current amount of the second account. Unit: minimum currency unit Value range: -2^{63}-2^{63}-1 The default value is **0**.
43	AccountType3	Integer[4]	Indicates the account type of the third account. 0 means not used. The default value is **0**.
44	ChargeAmount3	Integer[8]	Indicates the reward amount of the third account. 0 means unchanged. Unit: minimum currency unit The default value is **0**. Value range: -2^{63}-2^{63}-1
45	CurrentAcctAmount3	Integer[8]	Indicates the current amount of the third account. Unit: minimum currency unit Value range: -2^{63}-2^{63}-1 The default value is **0**.
46	AccountType4	Integer[4]	Indicates the account type of the fourth account. 0 means not used. The default value is **0**.

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SN	Field Name	Field Type	Description
47	ChargeAmount4	Integer[8]	Indicates the reward amount of the fourth account. 0 means unchanged. Unit: minimum currency unit Value range: -2^{63}-2^{63}-1 The default value is **0**.
48	CurrentAcctAmount4	Integer[8]	Indicates the current amount of the fourth account. Unit: minimum currency unit The default value is **0**. Value range: -2^{63}-2^{63}-1
49	AccountType5	Integer[4]	Indicates the account type of the fifth account. 0 means not used.
50	ChargeAmount5	Integer[8]	Indicates the reward amount of the fifth account. 0 means unchanged. Unit: minimum currency unit Value range: -2^{63}-2^{63}-1 The default value is **0**.
51	CurrentAcctAmount5	Integer[8]	Indicates the current amount of the fifth account. Unit: minimum currency unit Value range: -2^{63}-2^{63}-1 The default value is **0**.
52	AccountType6	Integer[4]	Indicates the account type of the sixth account. 0 means not used. The default value is **0**.
53	ChargeAmount6	Integer[8]	Indicates the reward amount of the sixth account. 0 means unchanged. Unit: minimum currency unit Value range: -2^{63}-2^{63}-1 The default value is **0**.
54	CurrentAcctAmount6	Integer[8]	Indicates the current amount of the sixth account. Unit: minimum currency unit Value range: -2^{63}-2^{63}-1 The default value is **0**.

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SN	Field Name	Field Type	Description
55	AccountType7	Integer[4]	Indicates the account type of the seventh account. 0 means not used. The default value is **0**.
56	ChargeAmount7	Integer[8]	Indicates the reward amount of the seventh account. 0 means unchanged. Unit: minimum currency unit Value range: -2^{63}-2^{63}-1 The default value is **0**.
57	CurrentAcctAmount7	Integer[8]	Indicates the current amount of the seventh account. Unit: minimum currency unit Value range: -2^{63}-2^{63}-1 The default value is **0**.
58	AccountType8	Integer[4]	Indicates the account type of the eighth account. 0 means not used. The default value is **0**.
59	ChargeAmount8	Integer[8]	Indicates the reward amount of the eighth account. 0 means unchanged. Unit: minimum currency unit Value range: -2^{63}-2^{63}-1 The default value is **0**.
60	CurrentAcctAmount8	Integer[8]	Indicates the current amount of the eighth account. Unit: minimum currency unit Value range: -2^{63}-2^{63}-1 The default value is **0**.
61	AccountType9	Integer[4]	Indicates the account type of the ninth account. 0 means not used. The default value is **0**.
62	ChargeAmount9	Integer[8]	Indicates the reward amount of the ninth account. 0 means unchanged. Unit: minimum currency unit Value range: -2^{63}-2^{63}-1 The default value is **0**.

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SN	Field Name	Field Type	Description
63	CurrentAcctAmount9	Integer[8]	Indicates the current amount of the ninth account. Unit: minimum currency unit Value range: -2^{63}-2^{63}-1 The default value is **0**.
64	AccountType10	Integer[4]	Indicates the account type of the tenth account. 0 means not used. The default value is **0**.
65	ChargeAmount10	Integer[8]	Indicates the reward amount of the tenth account. 0 means unchanged. Unit: minimum currency unit Value range: -2^{63}-2^{63}-1 The default value is **0**.
66	CurrentAcctAmount10	Integer[8]	Indicates the current amount of the tenth account. Unit: minimum currency unit Value range: -2^{63}-2^{63}-1 The default value is **0**.
67	Reserved	Char[20]	-
68	BonusValidity1	Char[14]	Indicates the validity period of the first rewarded resource. The validity period does not change when fees are deducted from accounts. In this case, the validity periods of consumption resources are not recorded. Format: YYYYMMDDhhmmss
69	BonusValidity2	Char[14]	Indicates the validity period of the second rewarded resource. The validity period does not change when fees are deducted from accounts. In this case, the validity periods of consumption resources are not recorded. Format: YYYYMMDDhhmmss
70	BonusValidity3	Char[14]	Indicates the validity period of the third rewarded resource. The validity period does not change when fees are deducted from accounts. In this case, the validity periods of consumption resources are not recorded. Format: YYYYMMDDhhmmss
71	BonusValidity4	Char[14]	Indicates the validity period of the fourth rewarded resource. The validity period does not change when fees are deducted from

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SN	Field Name	Field Type	Description
			accounts. In this case, the validity periods of consumption resources are not recorded. Format: YYYYMMDDhhmmss
72	BonusValidity5	Char[14]	Indicates the validity period of the fifth rewarded resource. The validity period does not change when fees are deducted from accounts. In this case, the validity periods of consumption resources are not recorded. Format: YYYYMMDDhhmmss
73	BonusValidity6	Char[14]	Indicates the validity period of the sixth rewarded resource. The validity period does not change when fees are deducted from accounts. In this case, the validity periods of consumption resources are not recorded. Format: YYYYMMDDhhmmss
74	BonusValidity7	Char[14]	Indicates the validity period of the seventh rewarded resource. The validity period does not change when fees are deducted from accounts. In this case, the validity periods of consumption resources are not recorded. Format: YYYYMMDDhhmmss
75	BonusValidity8	Char[14]	Indicates the validity period of the eighth rewarded resource. The validity period does not change when fees are deducted from accounts. In this case, the validity periods of consumption resources are not recorded. Format: YYYYMMDDhhmmss
76	BonusValidity9	Char[14]	Indicates the validity period of the ninth rewarded resource. The validity period does not change when fees are deducted from accounts. In this case, the validity periods of consumption resources are not recorded. Format: YYYYMMDDhhmmss
77	BonusValidity10	Char[14]	Indicates the validity period of the tenth rewarded resource. The validity period does not change when fees are deducted from accounts. In this case, the validity periods of consumption resources are not recorded. Format: YYYYMMDDhhmmss
78	FreeRBTRentFlag	Integer[4]	Indicates whether the RBT monthly rental that is adopted is free of charge. The options are as follows: • 0: yes • 1: no

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SN	Field Name	Field Type	Description
			The default value is **0**.
79	Free_RBT_Monthlyfee	Integer[4]	Indicates the remaining number of events of charging no RBT monthly fee. For example, if the value of the field is 3, you can infer that a subscriber can enjoy charging no RBT monthly fee for three times. The default value is **0**.
80	Reserved	Char[24]	-
81	AddtionalInfo	Char[256]	Indicates the additional information about the service. Some additional information about the service is recorded. In most cases, the information is only useful for the specified carrier.
82	StartTimeOfBillCycle	Char[14]	Indicates the start time of bill cycle. The default value is **0**.
83	ChargePartyIndicator	Char[1]	Indicates the charging party. The options are as follows: • 1: The calling party is charged • 2: The called party is charged • 3: Charge of third party • 4: Charge of parent & child card • 5: Charge of calling party & VPN The default value is **1**.
84	SecondChargingPartyNumber	Char[64]	Indicates the number of the other charging party. In the example, the value of the field is the number of subscriber C. This field is null when the service is paid only by account. There are several special situations. Like Child-Parent card feature, the calling party number is the child, the charging party number is the parent's number. The number format is as follows: • **Area number + PSTN** The number format is the format of the fixed-line telephone number. • **MSISDN** The number format is the format of the mobile phone number. Value range: numerals within 0-9, *, #

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SN	Field Name	Field Type	Description
			Not null
85	SecondChargingPartyPayType	Char[1]	Indicates the payment type of the other charging party. The options are as follows: • 0: Prepaid • 1: Postpaid **NOTE** If the value of the field is null, the system records the payment type of the parent card subscriber.
86	SecondChargingPartyBillCycleID	Char[20]	Indicates the bill cycle ID of the other charging party. For example, the bill cycle ID of the parent card. Format: YYYYMMTT • YYYY: indicates the year. • MM: indicates the month. • TT: indicates the type of a bill cycle. Value range: numerals within 0-9
87	Reserved	Char[15]	Indicates the validity period of the sixth rewarded resource. The validity period does not change when fees are deducted from accounts. In this case, the validity periods of consumption resources are not recorded. Format: YYYYMMDDhhmmss
88	CallingVPNGroupNumber	Char[10]	Indicates the validity period of the seventh rewarded resource. The validity period does not change when fees are deducted from accounts. In this case, the validity periods of consumption resources are not recorded. Format: YYYYMMDDhhmmss
89	CallingVPNShortNumber	Char[10]	Indicates the validity period of the eighth rewarded resource. The validity period does not change when fees are deducted from accounts. In this case, the validity periods of consumption resources are not recorded. Format: YYYYMMDDhhmmss
90	CalledVPNTopGroupNumber	Char[15]	Indicates the validity period of the ninth rewarded resource. The validity period does not change when fees are deducted from accounts. In this case, the validity periods of consumption resources are not recorded. Format: YYYYMMDDhhmmss
91	CalledVPNGroupNumber	Char[10]	Indicates the validity period of the tenth rewarded resource. The validity period does not change when fees are deducted from

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SN	Field Name	Field Type	Description
			accounts. In this case, the validity periods of consumption resources are not recorded. Format: YYYYMMDDhhmmss
92	CalledVPNShortNumber	Char[10]	Indicates the expiration date of Suspend state before recharging. Format: YYYYMMDDhhmmss
93	CallingNetworkType	Char[10]	Indicates the expiry date of Disable state before recharging. Format: YYYYMMDDhhmmss
94	CalledNetworkType	Char[10]	The name of the agent. This field indicates the information of provider of recharge channel.
95	GroupCallType	Char[10]	Indicates additional service information. The additional service information is recorded in combination mode. In normal cases, the additional information is intended for specified carriers.
96	GroupPayFlag	Char[10]	Indicates the start time of bill cycle.
97	UsageServiceType	Integer[4]	Indicates the type of a real-time service. The default value is **1100**.
98	Reserved	Char[10]	-
99	Reserved	Char[10]	-
100	Reserved	Char[10]	-
101	Reserved	Char[10]	-
102	ChargeOfItemsSecondAccounts	Integer[8]	Indicates the free call times used from others, such as the parent card. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
103	ChargeOfDurationSecondAccounts	Integer[8]	Indicates the total call time charge using reward resource from others, such as the parent card. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
104	ChargeOfFundSecondAccounts	Integer[8]	Indicates the total cost amount charging from fund account of others, such as the parent card. Value range: -2^{63}-2^{63}-1

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SN	Field Name	Field Type	Description
			Not null The default value is **0**.
105	ChargeFromSecondPrepaidAccount	Integer[8]	Indicates the charge amount from other prepaid account. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
106	SecondPrepaidAccountBalance	Integer[8]	Indicates the balance of other prepaid account. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
107	ChargeFromSecondPostpaidAccount	Integer[8]	Indicates the charge amount from other postpaid account. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
108	SecondPostpaidAccountBalance	Integer[8]	Indicates the balance of other postpaid account. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
109	SecondAccountID	Char[20]	Indicates the main account ID of the other charging party that the system deducts fees from. Value range: characters in [0-9, a-z, A-Z]
110	SecondAccountKey	Integer[8]	Indicates the main account key of the other charging party that the system deducts fees from. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
111	SecondAccountCurrencyCode	Integer[4]	Currency code of others, such as the parent card. Also, if this field missed or present be Zero, it means the system use local minimal currency as default, and no need to specify in CDR. Not null The default value is **0**.
112	Reserved	Char[20]	-

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SN	Field Name	Field Type	Description
113	SecondAccountType1	Integer[4]	Indicates the account type of the first account of others, such as the parent card. 0 means not used. Not null The default value is **0**.
114	SecondFeeType1	Integer[4]	Indicates the fee type of the first account of others, such as the parent card. 0 means not used. Not null The default value is **0**.
115	SecondChargeAmount1	Integer[8]	Indicates the changing amount of the first account of others, such as the parent card. 0 means unchanged. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
116	SecondCurrentAcctAmount1	Integer[8]	Indicates the current amount of the first account of others, such as the parent card. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
117	SecondAccountType2	Integer[4]	Indicates the account type of the second account of others, such as the parent card. 0 means not used. Not null The default value is **0**.
118	SecondFeeType2	Integer[4]	Indicates the fee type of the second account of others, such as the parent card. 0 means not used. Not null The default value is **0**.
119	SecondChargeAmount2	Integer[8]	Indicates the changing amount of the second account of others, such as the parent card. 0 means unchanged. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
120	SecondCurrentAcctAmount2	Integer[8]	Indicates the current amount of the second account of others, such as the parent card. Value range: -2^{63}-2^{63}-1

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SN	Field Name	Field Type	Description
			Not null The default value is **0**.
121	SecondAccountType3	Integer[4]	Indicates the account type of the third account of others, such as the parent card. 0 means not used. Not null The default value is **0**.
122	SecondFeeType3	Integer[4]	Indicates the fee type of the third account of others, such as the parent card. 0 means not used. Not null The default value is **0**.
123	SecondChargeAmount3	Integer[8]	Indicates the changing amount of the third account of others, such as the parent card. 0 means unchanged. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
124	SecondCurrentAcctAmount3	Integer[8]	Indicates the current amount of the third account of others, such as the parent card. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
125	SecondAccountType4	Integer[4]	Indicates the account type of the fourth account of others, such as the parent card. 0 means not used. Not null The default value is **0**.
126	SecondFeeType4	Integer[4]	Indicates the fee type of the fourth account of others, such as the parent card. 0 means not used. Not null The default value is **0**.
127	SecondChargeAmount4	Integer[8]	Indicates the changing amount of the fourth account of others, such as the parent card. 0 means unchanged. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.

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SN	Field Name	Field Type	Description
128	SecondCurrentAcctAmount4	Integer[8]	Indicates the current amount of the fourth account of others, such as the parent card. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
129	SecondAccountType5	Integer[4]	Indicates the account type of the fifth account of others, such as the parent card. 0 means not used. Not null The default value is **0**.
130	SecondFeeType5	Integer[4]	Indicates the fee type of the fifth account of others, such as the parent card. 0 means not used. Not null The default value is **0**.
131	SecondChargeAmount5	Integer[8]	Indicates the changing amount of the fifth account of others, such as the parent card. 0 means unchanged. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
132	SecondCurrentAcctAmount5	Integer[8]	Indicates the current amount of the fifth account of others, such as the parent card. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
133	SecondBonusValidity1	Char[14]	Indicates the validity period of the first rewarded resource of others, such as the parent card. The validity period does not change when fees are deducted from accounts. In this case, the validity periods of consumption resources are not recorded. Format: YYYYMMDDhhmmss
134	SecondBonusValidity2	Char[14]	Indicates the validity period of the second rewarded resource of others, such as the parent card. The validity period does not change when fees are deducted from accounts. In this case, the validity periods of consumption resources are not recorded. Format: YYYYMMDDhhmmss
135	SecondBonusValidity3	Char[14]	Indicates the validity period of the third rewarded resource of others, such as the

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SN	Field Name	Field Type	Description
			parent card. The validity period does not change when fees are deducted from accounts. In this case, the validity periods of consumption resources are not recorded. Format: YYYYMMDDhhmmss
136	SecondBonusValidity4	Char[14]	Indicates the validity period of the fourth rewarded resource of others, such as the parent card. The validity period does not change when fees are deducted from accounts. In this case, the validity periods of consumption resources are not recorded. Format: YYYYMMDDhhmmss
137	SecondBonusValidity5	Char[14]	Indicates the validity period of the fifth rewarded resource of others, such as the parent card. The validity period does not change when fees are deducted from accounts. In this case, the validity periods of consumption resources are not recorded. Format: YYYYMMDDhhmmss
138	IsTestNumber	Integer[4]	Reserved
139	MainProductStatus	Integer[4]	Reserved
140	TransitionID	Char[32]	Indicates the ID of transition.
141	PreRentFlag	Integer[4]	Indicates the flag of pre-deduction of a rental. The default value is **0**.
142	PreRentNum	Integer[4]	Indicates a number whose rental is pre-deducted. The default value is **0**.
143	PreRentAmt	Integer[8]	Indicates an amount that is pre-deducted. The default value is **0**.
144	PreRentCycleOpen	Char[16]	Indicates the start time of the pre-deduction bill cycle. The default value is **19700101000000**.
145	PreRentCycleEnd	Char[16]	Indicates the end time of the pre-deduction bill cycle. The default value is **19700101000000**.
146	subscriberkey	Integer[4]	Indicates the key of a subscriber. Not null The default value is **0**.
147	ExpiredAirTime	Char[64]	Reserved

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SN	Field Name	Field Type	Description
148	OriginCdrFilenam e	Char[64]	Reserved
149	CurCDRLineNo	Integer[4]	Reserved
150	Tax	Integer[8]	Reserved
151	Tax2	Integer[8]	Reserved
152	Tax3	Integer[8]	Reserved
153	Tax4	Integer[8]	Reserved
154	SecondSubscriber Key	Integer[4]	Indicates the key value of a entrusted-payment subscriber. The default value is **0**.
155	SecondProductID	Integer[4]	Reserved
156	FeeType1	integer[4]	Indicates the fee type of the first account. Fee types record fee change information such as consumption fees and rewarded fees. 0 means not used. Not null The default value is **0**.
157	FeeType2	integer[4]	Indicates the fee type of the second account. Fee types record fee change information such as consumption fees and rewarded fees. 0 means not used. Not null The default value is **0**.
158	FeeType3	integer[4]	Indicates the fee type of the third account. Fee types record fee change information such as consumption fees and rewarded fees. 0 means not used. Not null The default value is **0**.
159	FeeType4	integer[4]	Indicates the fee type of the fourth account. Fee types record fee change information such as consumption fees and rewarded fees. 0 means not used. Not null The default value is **0**.
160	FeeType5	integer[4]	Indicates the fee type of the fifth account. Fee types record fee change information such as consumption fees and rewarded fees.

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SN	Field Name	Field Type	Description
			0 means not used. Not null The default value is **0**.
161	FeeType6	integer[4]	Indicates the fee type of the sixth account. Fee types record fee change information such as consumption fees and rewarded fees. 0 means not used. Not null The default value is **0**.
162	FeeType7	integer[4]	Indicates the fee type of the seventh account. Fee types record fee change information such as consumption fees and rewarded fees. 0 means not used. Not null The default value is **0**.
163	FeeType8	integer[4]	Indicates the fee type of the eighth account. Fee types record fee change information such as consumption fees and rewarded fees. 0 means not used. Not null The default value is **0**.
164	FeeType9	integer[4]	Indicates the fee type of the ninth account. Fee types record fee change information such as consumption fees and rewarded fees. 0 means not used. Not null The default value is **0**.
165	FeeType10	integer[4]	Indicates the fee type of the tenth account. Fee types record fee change information such as consumption fees and rewarded fees. 0 means not used. Not null The default value is **0**.

3.1.8 com CDR

Description

The com CDR indicates the general CDR.

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Generation Scenario

A com CDR is generated when a subscriber of the OCS service uses the third-party interface.

For example, when a subscriber recharges through the bank account, the system cannot identify the service type. In this case, the com CDR is generated.

NOTE
For the description of child-parent cards, see section 3.1.2 rec CDR .

CDR File Name Format

comYYYYMMDD_CBPNODE_Seqno.unl

Example of a com CDR file name: com20071112_101_000151.unl

Format

Table 3-9 describes each field of com CDR file.

Table 3-9 Format of com CDR

SN.	Field Name	Field Type	Description
1	SerialNo	Integer[8]	Indicates the serial number. It is generated by the system and the serial number is exclusive. Not null The default value is **0**.
2	SubSequence	Integer[4]	Indicates the sequence number when the bill is split for mid-night and end of the bill cycle. The default value is **0**.
3	TimeStamp	Char[14]	Indicates the timestamp of the bill item generation. Format: YYYYMMDDhhmmss Not null
4	ServiceKey	Integer[4]	Indicates the service key of the intelligent service to which the subscriber subscribes. The default value is **1**. Not null
5	ChargingPartyNumber	Char[64]	Indicates the telephone number of the charging party. The number is same as calling party number in MO and called party number in MT. There are several special situations. Like Child-Parent card feature, the calling party number is the child, the charging party number is the parent's number. This field is

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SN.	Field Name	Field Type	Description
			null when the service is paid only by account. The number format are as follows: • **Area number + PSTN** The number format is the format of the fixed-line telephone number. • **MSISDN** The number format is the format of the mobile phone number. Value range: numerals within 0-9, *, #
6	RequestAction	Integer[4]	Indicates the charging type, for example, fee deduction or refunding. The options are as follows: • 0: Deduct fees. • 1: Refund fees Not null The default value is **0**.
7	ChargeType	Char[10]	Indicates service-related information that is sent from external systems. Some fields are reserved for recording the original information (such as the RBT download charging information) sent from external systems. It is customized according to the special requirement. • 1001: RBT fee deduction interface • 1002: Content fee deduction interface • 1003: RTBP 2.0 interface • 1004: RTBP 3.0 interface
8	ServiceType	Integer[4]	Indicates the service-related information that is sent from external systems. • 1: Voice • 2: SMS • 3: MMS • 4: GPRS • 5: Recharge It is customized according to the special requirement. • When the value of **Charge type** is **1001: RBT fee deduction interface**, the values of **Service type** are as follows: – 1: Deducting the fees of downloading

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SN.	Field Name	Field Type	Description
			RBTs – 2: Deducting the fees of the music box • When the value of **Charge type** is **1002: content fee deduction interface**, the value of **Service type** is **1**, indicating the directly deducting content fees The default value is **0**.
9	BearedService	Char[20]	Indicates the beared service customized according to the special requirement.
10	SPID	Char[20]	Indicates the SP ID or CP ID.
11	CategoryID	Char[20]	Indicates the number of the category got from charge interface and field into CDR for customized CDR.
12	ContentID	Char[20]	Indicates the number of the content got from charge interface and field into CDR for customized CDR.
13	ServiceID	Char[20]	Indicates the service detail of the SP.
14	CalledPartyNumber	Char[64]	Indicates the called (receiver) number. Keep the original format.
15	ChargeAmount	Integer[8]	Indicates information about the deducted or refunded amount. The information is sent from external systems. The default value is **0**.
16	BrandID	Integer[4]	Indicates the brand ID of subscribers. Not null The default value is **0**.
17	SubCOSID	Integer[4]	Indicates the main product ID of subscribers. Not null The default value is **0**.
18	PayType	Char[1]	Indicates the pay type of subscribers. The options are as follows: • 0: Prepaid • 1: Postpaid
19	BillCycleID	Char[20]	Indicates the bill cycle ID. Format: YYYYMMTT • YYYY: indicates the year. • MM: indicates the month. • TT: indicates the type of a bill cycle. Value range: numerals within 0-9

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SN.	Field Name	Field Type	Description
20	ServiceBeginTime	Char[14]	Indicates the start time for charging, which is reported by the service. When the service network element does not report the start time, the value is the time when the OCS charges the service. Format: YYYYMMDDhhmmss Not null The default value is **null**.
21	ServiceEndTime	Char[14]	Indicates the end time for charging, which is reported by the service. The end time can be calculated according to the start time. If the end time is unavailable, the value of this field is equal to the value of the **ServiceBeginTime** field. Format: YYYYMMDDhhmmss Not null The default value is **null**.
22	UsageAmount	Integer[4]	Indicates service usage amount. Not null The default value is **0**.
23	ServiceUnit	Integer[4]	Indicates the measurement methods corresponding to the metric table. For example, times or second. If the system deducts fee or refunds fees, the unit is times. For the one-off charging, the value of **useageamount** is 1, and the value of **Serviceunit** is 3. Not null The default value is **0**.
24	ChargingTime	Char[14]	Indicates the time stamp of the charging beginning. Format: YYYYMMDDhhmmss Not null
25	TransitionID	Char[32]	Indicates the sequence number of a transaction.
26	UserState	Char[10]	Indicates the life cycle state and the management state of a subscriber. The **UserState** parameter is a numeric string of seven digits, in the format CCMMMMM. The first and second digits C indicate the life cycle state of a subscriber. The first digit C indicates the state of a prepaid subscriber, and

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SN.	Field Name	Field Type	Description
			the second digit C indicates the state of a postpaid subscriber. The options are as follows: • 0: Idle • 1: Active • 2: Suspended • 3: Disable • 4: Pool • 5: Pool without activation The next five digits MMMMM indicate the management state of a subscriber. The options are as follows: Third digit: Whether the subscriber claims missing. • 0: no • 1: yes Fourth digit: Whether the subscriber suspends the service. • 0: non-suspend • 1: suspend • 2: suspend by operator (B1W) • 3: suspend by operator (B2W) Fifth digit: Whether the subscriber applies for suspension with the number being reserved. • 0: no • 1: yes Sixth digit: Whether the subscriber is blocked. • 0: no • 1: yes Seventh digit: Whether the subscriber is blacklisted. • 0: no • 1: yes
27	SubscriberID	Char[20]	Indicates the subscriber ID. Value range: characters in [0-9, a-z, A-Z]
28	Reserved	Char[20]	-
29	Reserved	Char[20]	-
30	ResultCode	Integer[4]	Indicates the internal result code. Value range: -2^{31}-2^{31}-1

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SN.	Field Name	Field Type	Description
			The default value is **0**.
31	ChargeOfItemsAccounts	Integer[8]	Indicates the free times used. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
32	ChargeOfDurationAccounts	Integer[8]	Indicates the total call time charge using reword resource. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
33	ChargeOfFundAccounts	Integer[8]	Indicates the total cost amount charging from fund account. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
34	ChargeFromPrepaid	Integer[8]	Indicates the charge amount from prepaid account. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
35	PrepaidBalance	Integer[8]	Indicates the balance of prepaid account. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
36	ChargeFromPostpaid	Integer[8]	Indicates the charge amount from postpaid account. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
37	PostPaidBalance	Integer[8]	Indicates the balance of postpaid available limitation. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
38	AccountID	Char[20]	Indicates the ID of the main account. Value range: characters in [0-9, a-z, A-Z]
39	AccountKey	Integer[8]	Indicates the key of the main account. Value range: -2^{63}-2^{63}-1

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SN.	Field Name	Field Type	Description
			Not null The default value is **0**.
40	CurrencyCode	Integer[4]	Indicates currency code. Also, if this field missed or present be zero, it means the system use local minimal currency as default, and no need to specify in CDR. The default value is **0**.
41	Original Amount	Integer[4]	Reserved
42	AccountType1	Integer[4]	Indicates the account type of the first account. **0** means not used. Not null The default value is **0**.
43	FeeType1	Integer[4]	Indicates the fee type of the first account. **0** means not used. Not null The default value is **0**.
44	ChargeAmount1	Integer[8]	Indicates the changing amount of the first account. **0** means unchanged. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
45	CurrentAcctAmount1	Integer[8]	Indicates the current amount of the first account. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
46	AccountType2	Integer[4]	Indicates the account type of the second account. **0** means not used. Not null The default value is **0**.
47	FeeType2	Integer[4]	Indicates the fee type of the second account. **0** means not used. Not null The default value is **0**.
48	ChargeAmount2	Integer[8]	Indicates the changing amount of the second account.

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SN.	Field Name	Field Type	Description
			0 means unchanged. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
49	CurrentAcctAmount2	Integer[8]	Indicates the current amount of the second account. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
50	AccountType3	Integer[4]	Indicates the account type of the third account. **0** means not used. Not null The default value is **0**.
51	FeeType3	Integer[4]	Indicates the fee type of the third account. **0** means not used. Not null The default value is **0**.
52	ChargeAmount3	Integer[8]	Indicates the changing amount of the third account. **0** means unchanged. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
53	CurrentAcctAmount3	Integer[8]	Indicates the current amount of the third account. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
54	AccountType4	Integer[4]	Indicates the account type of the fourth account. **0** means not used. Not null The default value is **0**.
55	FeeType4	Integer[4]	Indicates the fee type of the fourth account. **0** means not used. Not null The default value is **0**.

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SN.	Field Name	Field Type	Description
56	ChargeAmount4	Integer[8]	Indicates the changing amount of the fourth account. **0** means unchanged. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
57	CurrentAcctAmount4	Integer[8]	Indicates the current amount of the fourth account. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
58	AccountType5	Integer[4]	Indicates the account type of the fifth account. **0** means not used. Not null The default value is **0**.
59	FeeType5	Integer[4]	Indicates the fee type of the fifth account. **0** means not used. Not null The default value is **0**.
60	ChargeAmount5	Integer[8]	Indicates the changing amount of the fifth account. **0** means unchanged. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
61	CurrentAcctAmount5	Integer[8]	Indicates the current amount of the fifth account. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
62	AccountType6	Integer[4]	Indicates the account type of the sixth account. **0** means not used. Not null The default value is **0**.
63	FeeType6	Integer[4]	Indicates the fee type of the sixth account. **0** means not used.

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SN.	Field Name	Field Type	Description
			Not null The default value is **0**.
64	ChargeAmount6	Integer[8]	Indicates the changing amount of the sixth account. **0** means unchanged. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
65	CurrentAcctAmount6	Integer[8]	Indicates the current amount of the sixth account. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
66	AccountType7	Integer[4]	Indicates the account type of the seventh account. **0** means not used. Not null The default value is **0**.
67	FeeType7	Integer[4]	Indicates the fee type of the seventh account. **0** means not used. Not null The default value is **0**.
68	ChargeAmount7	Integer[8]	Indicates the changing amount of the seventh account. **0** means unchanged. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
69	CurrentAcctAmount7	Integer[8]	Indicates the current amount of the seventh account. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
70	AccountType8	Integer[4]	Indicates the account type of the eighth account. **0** means not used. Not null The default value is **0**.

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SN.	Field Name	Field Type	Description
71	FeeType8	Integer[4]	Indicates the fee type of the eighth account. **0** means not used. Not null The default value is **0**.
72	ChargeAmount8	Integer[8]	Indicates the changing amount of the eighth account. **0** means unchanged. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
73	CurrentAcctAmount8	Integer[8]	Indicates the current amount of the eighth account. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
74	AccountType9	Integer[4]	Indicates the account type of the ninth account. **0** means not used. Not null The default value is **0**.
75	FeeType9	Integer[4]	Indicates the fee type of the ninth account. **0** means not used. Not null The default value is **0**.
76	ChargeAmount9	Integer[8]	Indicates the changing amount of the ninth account. **0** means unchanged. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
77	CurrentAcctAmount9	Integer[8]	Indicates the current amount of the ninth account. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
78	AccountType10	Integer[4]	Indicates the account type of the tenth account. **0** means not used.

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SN.	Field Name	Field Type	Description
			Not null The default value is **0**.
79	FeeType10	Integer[4]	Indicates the fee type of the tenth account. **0** means not used. Not null The default value is **0**.
80	ChargeAmount10	Integer[8]	Indicates the changing amount of the tenth account. **0** means unchanged. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
81	CurrentAcctAmount10	Integer[8]	Indicates the current amount of the tenth account. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
82	BonusValidity1	Char[14]	Indicates the validity period of the first rewarded resource. The validity period does not change when fees are deducted from accounts. In this case, the validity periods of consumption resources are not recorded. Format: YYYYMMDDhhmmss
83	BonusValidity2	Char[14]	Indicates the validity period of the second rewarded resource. The validity period does not change when fees are deducted from accounts. In this case, the validity periods of consumption resources are not recorded. Format: YYYYMMDDhhmmss
84	BonusValidity3	Char[14]	Indicates the validity period of the third rewarded resource. The validity period does not change when fees are deducted from accounts. In this case, the validity periods of consumption resources are not recorded. Format: YYYYMMDDhhmmss
85	BonusValidity4	Char[14]	Indicates the validity period of the fourth rewarded resource. The validity period does not change when fees are deducted from accounts. In this case, the validity periods of consumption resources are not recorded. Format: YYYYMMDDhhmmss

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SN.	Field Name	Field Type	Description
86	BonusValidity5	Char[14]	Indicates the validity period of the fifth rewarded resource. The validity period does not change when fees are deducted from accounts. In this case, the validity periods of consumption resources are not recorded. Format: YYYYMMDDhhmmss
87	BonusValidity6	Char[14]	Indicates the validity period of the sixth rewarded resource. The validity period does not change when fees are deducted from accounts. In this case, the validity periods of consumption resources are not recorded. Format: YYYYMMDDhhmmss
88	BonusValidity7	Char[14]	Indicates the validity period of the seventh rewarded resource. The validity period does not change when fees are deducted from accounts. In this case, the validity periods of consumption resources are not recorded. Format: YYYYMMDDhhmmss
89	BonusValidity8	Char[14]	Indicates the validity period of the eighth rewarded resource. The validity period does not change when fees are deducted from accounts. In this case, the validity periods of consumption resources are not recorded. Format: YYYYMMDDhhmmss
90	BonusValidity9	Char[14]	Indicates the validity period of the ninth rewarded resource. The validity period does not change when fees are deducted from accounts. In this case, the validity periods of consumption resources are not recorded. Format: YYYYMMDDhhmmss
91	BonusValidity10	Char[14]	Indicates the validity period of the tenth rewarded resource. The validity period does not change when fees are deducted from accounts. In this case, the validity periods of consumption resources are not recorded. Format: YYYYMMDDhhmmss
92	msg id	Char[20]	Reserved
93	session id	Char[20]	Reserved
94	Service Name	Char[32]	Reserved
95	Consumed Resources	Integer[4]	Reserved
96	AddtionalInfo	Char[512]	Reserved

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SN.	Field Name	Field Type	Description
97	ChargeMode	Integer[4]	Indicates the charge mode. The options are as follows: • 0: times Indicates charging by events. • 1: volume Indicates charging by volume. • 2: duration Indicates charging by duration. • 3: reserved The default value is **0**.
98	CDRProductID	Char[20]	Indicates the last effect rating product.
99	CDRChargeMode	Integer[4]	Indicates the charge mode. The default value is **0**.
100	CDRTimes	Integer[4]	Indicates the number of consumption times of a subscriber that is recorded in the CDR of charging by times. The default value is **0**.
101	CDRDuration	Integer[4]	Indicates the consumed duration of a subscriber that is recorded in the CDR of charging by duration. The default value is **0**.
102	CDRVolume	Integer[4]	Indicates the consumed volume of a subscriber that is recorded in the CDR of charging by volume. The default value is **0**.
103	CDRCDRType	Integer[4]	Indicates the CDR type. The default value is **0**.
104	CDRServiceType	Integer[4]	Indicates the service type referring to the following values: • 0: wap • 1: kjava • 2: sms • 3: mms • 4: Mail • 5: LBS • 6: StreamMedia • 7: IMPS • 8: UM • 9: WapPush

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SN.	Field Name	Field Type	Description
			• 11: CRBT • 12: General • 13: MC • 14: PIM • 15: VMB • 16: CMB • 17: PaymentGW • 18: Storage • 20: Service Package The default value is **0**.
105	CDRBeginTime	Integer[4]	Indicates the time that a subscriber starts to use a service, such as surfing Internet. The default value is **0**.
106	CDREndTime	Integer[4]	Indicates the time that a subscriber stops using a service, such as surfing Internet. The default value is **0**.
107	CDRPkgSpid	Char[20]	Indicates the SP ID or CP ID.
108	CDRPkgServiceid	Char[20]	Indicates the service ID of the SP.
109	CDRPkgProductid	Char[20]	Indicates the last effect rating product.
110	CDRSPName	Char[256]	Indicates the name of the SP or CP.
111	CDRServiceName	Char[256]	Indicates the name of the service.
112	StartTimeOfBillCycle	Char[14]	Indicates the start time of bill cycle. The default value is **0**.
113	ChargePartyIndicator	Char[1]	Indicates the charging party. The options are as follows: • 1: The calling party is charged • 2: The called party is charged • 3: Charge of third party • 4: Charge of parent & child card • 5: Charge of calling party & VPN
114	SecondChargingPartyNumber	Char[64]	Indicates the number of the other charging party. In the example, the value of the field is the number of subscriber C. This field is null when the service is paid only by account. There are several special situations. Like Child-Parent card feature, the calling party number is the child, the charging party number is the parent's number.

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SN.	Field Name	Field Type	Description
			The number format is as follows: • **Area number + PSTN** The number format is the format of the fixed-line telephone number. • **MSISDN** The number format is the format of the mobile phone number. Value range: numerals within 0-9, *, #
115	SecondChargingPartyPayType	Char[1]	Indicates the payment type of the other charging party. For example, the payment type of the parent card. The options are as follows: • 0: Prepaid • 1: Postpaid **NOTE** If the value of the field is null, the system records the payment type of the parent card subscriber.
116	SecondChargingPartyBillCycleID	Char[20]	Indicates the bill cycle ID of the other charging party. Format: YYYYMMTT The description of the format is as follows: • YYYY: indicates the year. • MM: indicates the month. • TT: indicates the type of a bill cycle.
117	CallingVPNTopGroupNumber	Char[15]	Indicates the top group number of the calling party. Value range: numerals within 0-9 Can be null
118	CallingVPNGroupNumber	Char[10]	Indicates the group number of the calling party. Value range: numerals within 0-9 Can be null
119	CallingVPNShortNumber	Char[10]	Indicates the short number of the calling party. Value range: numerals within 0-9 Can be null
120	CalledVPNTopGroupNumber	Char[15]	Indicates the top group number of the called party. Value range: numerals within 0-9 Can be null

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SN.	Field Name	Field Type	Description
121	CalledVPNGroupNumber	Char[10]	Indicates the group number of the called party. Value range: numerals within 0-9 Can be null
122	CalledVPNShortNumber	Char[10]	Indicates the short number of the called party Value range: numerals within 0-9 Can be null
123	CallingNetworkType	Char[10]	Indicates the type of the network accessed by the calling party. The options are as follows: • 100: PSTN • 101: PBX • 102: PHS • 200: GSM • 201: GPRS • 202: EDGE • 300: CDMA • 301: CDMA 450 • 400: WCDMA • 500: TD-SCDMA • 600: IP/IMS • 601: WIFI • 602: WIMAX • Other to be extended Can be null
124	CalledNetworkType	Char[10]	Indicates the type of the network accessed by the called party. Indicates the type of the network accessed by the called party. • 100: PSTN • 101: PBX • 102: PHS • 200: GSM • 201: GPRS • 202: EDGE • 300: CDMA • 301: CDMA 450 • 400: WCDMA • 500: TD-SCDMA

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SN.	Field Name	Field Type	Description
			• 600: IP/IMS • 601: WIFI • 602: WIMAX • Other to be extended Can be null
125	GroupCallType	Char[10]	Reserved
126	GroupPayFlag	Char[10]	Indicates the payment mode of the group users. The options are as follows: • 0: Personal payment • 1: Group payment • 2: Group and Personal payment Can be null
127	OnlineType	Integer[4]	Indicates online type. The options are as follows: • 0: Online • 1: Offline Can be null
128	call ID	Char[32]	Reserved
129	package	Char[32]	Reserved
130	APN	Char[22]	Reserved
131	Reserved	Char[10]	-
132	ChargeOfItemsSecondAccounts	Integer[8]	Indicates the free call times used from others, such as the parent card. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
133	ChargeOfDurationSecondAccounts	Integer[8]	Indicates the total call time charge using reward resource from others, such as the parent card. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
134	ChargeOfFundSecondAccounts	Integer[8]	Indicates the total cost amount charging from fund account of others, such as the parent card. Value range: -2^{63}-2^{63}-1 Not null

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SN.	Field Name	Field Type	Description
			The default value is **0**.
135	ChargeFromSecondPrepaidAccount	Integer[8]	Indicates the charge amount from other prepaid account. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
136	SecondPrepaidAccountBalance	Integer[8]	Indicates the balance of other prepaid account. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
137	ChargeFromSecondPostpaidAccount	Integer[8]	Indicates the charge amount from other postpaid account. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
138	SecondPostpaidAccountBalance	Integer[8]	Indicates the balance of other postpaid account. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
139	SecondAccountID	Char[20]	Indicates the main account ID of the other charging party that the system deducts fees from. Value range: characters in [0-9, a-z, A-Z]
140	SecondAccountKey	Integer[8]	Indicates the main account key of the other charging party that the system deducts fees from. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
141	SecondAccountCurrencyCode	Integer[4]	Indicates currency code of others, such as the parent card. Also, if this field missed or present be Zero, it means the system use local minimal currency as default, and no need to specify in CDR. The default value is **0**.
142	Reserved	Char[20]	-
143	SecondAccountTy	Integer[4]	Indicates the account type of the first account

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SN.	Field Name	Field Type	Description
	pe1		of others, such as the parent card. **0** means not used. Not null The default value is **0**.
144	SecondFeeType1	Integer[4]	Indicates the fee type of the first account of others, such as the parent card. **0** means not used. Not null The default value is **0**.
145	SecondChargeAmount1	Integer[8]	Indicates the changing amount of the first account of others, such as the parent card. **0** means unchanged. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
146	SecondCurrentAcctAmount1	Integer[8]	Indicates the current amount of the first account of others, such as the parent card. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
147	SecondAccountType2	Integer[4]	Indicates the account type of the second account of others, such as the parent card. **0** means not used. Not null The default value is **0**.
148	SecondFeeType2	Integer[4]	Indicates the fee type of the second account of others, such as the parent card. **0** means not used. Not null The default value is **0**.
149	SecondChargeAmount2	Integer[8]	Indicates the changing amount of the second account of others, such as the parent card. **0** means unchanged. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
150	SecondCurrentAcctAmount2	Integer[8]	Indicates the current amount of the second account of others, such as the parent card. Value range: -2^{63}-2^{63}-1

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SN.	Field Name	Field Type	Description
			Not null The default value is **0**.
151	SecondAccountType3	Integer[4]	Indicates the account type of the third account of others, such as the parent card. **0** means not used. Not null The default value is **0**.
152	SecondFeeType3	Integer[4]	Indicates the fee type of the third account of others, such as the parent card. **0** means not used. Not null The default value is **0**.
153	SecondChargeAmount3	Integer[8]	Indicates the changing amount of the third account of others, such as the parent card. **0** means unchanged. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
154	SecondCurrentAcctAmount3	Integer[8]	Indicates the current amount of the third account of others, such as the parent card. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
155	SecondAccountType4	Integer[4]	Indicates the account type of the fourth account of others, such as the parent card. **0** means not used. Not null The default value is **0**.
156	SecondFeeType4	Integer[4]	Indicates the fee type of the fourth account of others, such as the parent card. **0** means not used. Not null The default value is **0**.
157	SecondChargeAmount4	Integer[8]	Indicates the changing amount of the fourth account of others, such as the parent card. **0** means unchanged. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.

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SN.	Field Name	Field Type	Description
158	SecondCurrentAcctAmount4	Integer[8]	Indicates the current amount of the fourth account of others, such as the parent card. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
159	SecondAccountType5	Integer[4]	Indicates the account type of the fifth account of others, such as the parent card. **0** means not used. Not null The default value is **0**.
160	SecondFeeType5	Integer[4]	Indicates the fee type of the fifth account of others, such as the parent card. **0** means not used. Not null The default value is **0**.
161	SecondChargeAmount5	Integer[8]	Indicates the changing amount of the fifth account of others, such as the parent card. **0** means unchanged. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
162	SecondCurrentAcctAmount5	Integer[8]	Indicates the current amount of the fifth account of others, such as the parent card. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
163	SecondBonusValidity1	Char[14]	Indicates the validity period of the first rewarded resource of others, such as the parent card. The validity period does not change when fees are deducted from accounts. In this case, the validity periods of consumption resources are not recorded. Format: YYYYMMDDhhmmss
164	SecondBonusValidity2	Char[14]	Indicates the validity period of the second rewarded resource of others, such as the parent card. The validity period does not change when fees are deducted from accounts. In this case, the validity periods of consumption resources are not recorded. Format: YYYYMMDDhhmmss
165	SecondBonusVali	Char[14]	Indicates the validity period of the third

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SN.	Field Name	Field Type	Description
	dity3		rewarded resource of others, such as the parent card. The validity period does not change when fees are deducted from accounts. In this case, the validity periods of consumption resources are not recorded. Format: YYYYMMDDhhmmss
166	SecondBonusValidity4	Char[14]	Indicates the validity period of the fourth rewarded resource of others, such as the parent card. The validity period does not change when fees are deducted from accounts. In this case, the validity periods of consumption resources are not recorded. Format: YYYYMMDDhhmmss
167	SecondBonusValidity5	Char[14]	Indicates the validity period of the fifth rewarded resource of others, such as the parent card. The validity period does not change when fees are deducted from accounts. In this case, the validity periods of consumption resources are not recorded. Format: YYYYMMDDhhmmss
168	DiameterSessionID	Char[128]	Indicates a diameter session ID. The format is as follows: <DiameterIdentity>;<high 32 bits>;<low 32 bits>[;<optional value>] • <DiameterIdentity>: the same as the field in the Origin-Host AVP. • <high 32 bits>: indicates the current time of the system. • <low 32 bits>: indicates the cycle count-up. The initial value is **0**. When you restart the system, set the value to **0**. • <optional value>: indicates implementation specific but may include a modem's device Id, a layer 2 address, timestamp, etc. Separate the preceding fields with the character ";". For example: SCP001.huawei.com;1876543210;523
169	Merchant	Char[20]	Reserved
170	Service	Char[20]	Reserved
171	PulaPulaProperty	Integer[4]	Reserved
172	BumerangePropert	Integer[4]	Reserved

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SN.	Field Name	Field Type	Description
	y		
173	RateName	Char[25]	Reserved
174	IsTestNumber	Integer[4]	Reserved
175	MainProductStatus	Integer[4]	Reserved
176	subscriberkey	Integer[4]	Indicates the key of a subscriber. The default value is **0**.
177	accumulationchange	Char[256]	Reserved
178	UsageServiceType	Integer[4]	Indicates the type of a real-time service. The default value is **1500**.
179	ChargePartyIMSI	Char[64]	Indicates IMSI of the charged party.
180	SourceSerialNo	Integer[8]	Reserved
181	SourceSubSequence	Integer[4]	Reserved
182	AcctBalChgList	Char[768]	Reserved
183	OriginCdrFilename	Char[64]	Reserved
184	CurCDRLineNo	Integer[4]	Reserved
185	Tax	Integer[8]	Reserved
186	Tax2	Integer[8]	Reserved
187	Tax3	Integer[8]	Reserved
188	Tax4	Integer[8]	Reserved

3.1.9 mgr CDR

Description

An mgr CDR records the information about management charging.

Generation Scenario

An mgr CDR is generated in the following scenarios:

When a subscriber queries the balance of his/her account successfully.

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CDR File Name Format

mgrYYYYMMDD_CBPNODE_Seqno.unl

Example of an mgr CDR file name: mgr20071112_101_000151.unl

Format

Table 3-10 describes each field of mgr CDR file.

Table 3-10 Format of mgr CDR

SN.	Field Name	Field Type	Description
1	SerialNo	Integer[8]	Indicates the serial number. It is generated by the system and the serial number is exclusive. Not null The default value is **0**.
2	SubSequence	Integer[4]	Indicates the sequence number when the bill is split for mid-night and end of the bill cycle. The default value is **0**.
3	TimeStamp	Char[16]	Indicates the time stamp of the bill item generation. Format: YYYYMMDDhhmmss Not null
4	ServiceKey	Integer[4]	Reserved
5	ChargingParty Number	Char[64]	Indicates the telephone number of the charging party. The number is same as calling party number in MO and called party number in MT. There are several special situations. Like Child-Parent card feature, the calling party number is the child, the charging party number is the parent's number. This field is null when the service is paid only by account. The number format are as follows: • **Area number + PSTN** The number format is the format of the fixed-line telephone number. • **MSISDN** The number format is the format of the mobile phone number. Value range: numerals within 0-9, *, #
6	RequestAction	Integer[4]	Indicates the charging type, for example, fee deduction or refunding. The options are as follows:

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SN.	Field Name	Field Type	Description
			• 0: Deduct fees • 1: Refund fees Not null The default value is **0**. **NOTE** The parameter **RequestAction** is invalid in the Adjust Account flow.
7	OperatorID	Char[24]	Indicates the number of the operator.
8	OperationID	Integer[4]	Indicates the ID of an operation. For example, FN, transferring balances or adjusting balances. The value is configured according to the configurations on the BME. Please refer to Table 3-11. Not null The default value is **0**.
9	OperationType	Integer[4]	Indicates the operation type. The options are as follows: • 0: Cancel • 1: Enable • 4: Add • 5: Modify • 6: Delete • 7: Query • 22: Transfer-in • 23: Transfer-out • 999: Not specified **NOTE** When this value is used, the system does not specify the operation type. This operation type is used in special cases such as one-off fee deduction. The default value is **0**. Not null
10	OtherPartyNumber	Char[64]	Indicates the number of the third party. For URI, the format is the_name@site.com. For example, ABC_DE@wana.com For transfer service, it is the other party number. It is the transferor number in the bill of transferee, and it is the transferee number in the bill of transferor.
11	TransitionFee	Integer[8]	Indicates the operating charge. The operating charge, which is defined by external interfaces, can be rated by internal

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SN.	Field Name	Field Type	Description
			interfaces. Unit: minimum currency unit The default value is **0**.
12	TranferAmount	Integer[8]	Indicates the adjusted or transferred amount of the main capital account. Unit: minimum currency unit The default value is **0**.
13	SPID	Char[20]	Indicates the SP ID or CP ID.
14	ServiceID	Char[20]	Indicates the service detail of the SP.
15	ServiceType	Integer[4]	Indicates the service-related information that is sent from external systems. It is customized according to the special requirement. The default value is **0**.
16	SubscriberType	Integer[4]	Indicates the subscriber type. The options are as follows: • 0: Normal OCS • 1: CST Master • 2: CST Client • 3: Corporate Master • 4: Corporate Slave • 5: Commitment Subscriber
17	url	Char[32]	Reserved
18	contentid	Char[32]	Reserved
19	Reserved	Char[32]	Reserved
20	Reserved	Char[32]	Reserved
21	IsARAdjustAccount	Integer[4]	Indicates whether the adjustment is initiated by the AR. The value is follows: • 0: No • 1: Yes The default value is **0**. Not null
22	BrandID	Integer[4]	Indicates the brand ID of subscribers. Not null The default value is **0**.
23	SubCOSID	Integer[4]	Indicates the main product ID of subscribers. Not null The default value is **0**.

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SN.	Field Name	Field Type	Description
24	PayType	Char[4]	Indicates the pay type of subscribers. The options are as follows: • 0: Prepaid • 1: Postpaid Not null
25	BillCycleID	Char[20]	Indicates the bill cycle ID. Format: YYYYMMTT The description of the format is as follows: • YYYY: indicates the year. • MM: indicates the month. • TT: indicates the type of a bill cycle.
26	ServiceBeginTime	Char[16]	Indicates the start time for charging, which is reported by the service. When the service network element does not report the start time, the value is the time when the OCS charges the service. Format: YYYYMMDDhhmmss Not null
27	ServiceEndTime	Char[16]	Indicates the end time for charging, which is reported by the service. The end time can be calculated according to the start time. If the end time is unavailable, the value of this field is equal to the value of the **ServiceBeginTime** field. Format: YYYYMMDDhhmmss
28	ChargeAmount	Integer[8]	Indicates the amount of deducted fees or owing fees of the main capital account. The amount is a positive number. The value of the **ChargeAmount** field is equal to or greater than 0 all the time. The default value is **0**.
29	TotalOperation	Integer[4]	Indicates the service amount used. The value of **TotalOperation** equals the total values of **QueryOperation**, **AddOperation**, **DeleteOperation**, and **ModifyOperation**. The default value is **0**.
30	QueryOperation	Integer[4]	Indicates the service amount queried. In the one-off fee query or balance check process, the **QueryOperation** parameter is set to **1**. The default value is **0**.

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SN.	Field Name	Field Type	Description
31	AddOperation	Integer[4]	Indicates the service amount added. In the one-off fee adding operation, the **AddOperation** parameter is set to **1**. The default value is **0**.
32	DeleteOperation	Integer[4]	Indicates the service amount deleted. In the one-off fee deletion operation, the **AddOperation** parameter is set to **1**. The default value is **0**. Not null
33	ModifyOperation	Integer[4]	Indicates the service amount modified. Indicates the service amount modified. Presently, the **ModifyOperation** parameter does not need to be configured for any processes. Therefore, its value is fixed to **0**. Not null The default value is **0**.
34	ServiceUnit	Integer[4]	Indicates the measurement methods corresponding to the metric table. For example, times, or second. If the system deducts fee or refunds fees, the unit is times. For the one-off charging, the value of **useageamount** is 1, and the value of **Serviceunit** is 3. Not null The default value is **0**.
35	ChargingTime	Char[16]	Indicates the time stamp of the charging beginning. Format: YYYYMMDDhhmmss Not null
36	TransitionID	Char[32]	Indicates the sequence number of a transaction.
37	OldUserState	Char[12]	Indicates the life cycle state and the management state of a subscriber before recharging. The **OldUserState** parameter is a numeric string of seven digits, in the format CCMMMMM. The first and second digits C indicate the life cycle state of a subscriber. The first digit C indicates the state of a prepaid subscriber, and the second digit C indicates the state of a postpaid subscriber. The options are as follows: • 0: Idle • 1: Active

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SN.	Field Name	Field Type	Description
			• 2: Suspended • 3: Disable • 4: Pool • 5: Pool without activation The next five digits MMMMM indicate the management state of a subscriber. The options are as follows: Third digit: Whether the subscriber claims missing. • 0: no • 1: yes Fourth digit: Whether the subscriber suspends the service. • 0: non-suspend • 1: suspend • 2: suspend by operator (B1W) • 3: suspend by operator (B2W) Fifth digit: Whether the subscriber applies for suspension with the number being reserved. • 0: no • 1: yes Sixth digit: Whether the subscriber is blocked. • 0: no • 1: yes Seventh digit: Whether the subscriber is blacklisted. • 0: no • 1: yes Not null
38	SubscriberID	Char[32]	Indicates the subscriber ID. Value range: characters in [0-9, a-z, A-Z] Not null
39	ValidDayAdded	Integer[4]	When transfer money into validity, or extend validity, this field will record the extend validity. The default value is **0**.
40	CurrUserState	Char[10]	Indicates the life cycle state and the management state of a subscriber after recharging. The **CurrUserState** parameter is a numeric string of seven digits, in the format CCMMMMM.

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SN.	Field Name	Field Type	Description
			The first and second digits C indicate the life cycle state of a subscriber. The first digit C indicates the state of a prepaid subscriber, and the second digit C indicates the state of a postpaid subscriber. The options are as follows: • 0: Idle • 1: Active • 2: Suspended • 3: Disable • 4: Pool • 5: Pool without activation The next five digits MMMMM indicate the management state of a subscriber. The options are as follows: Third digit: Whether the subscriber claims missing. • 0: no • 1: yes Fourth digit: Whether the subscriber suspends the service. • 0: non-suspend • 1: suspend • 2: suspend by operator (B1W) • 3: suspend by operator (B2W) Fifth digit: Whether the subscriber applies for suspension with the number being reserved. • 0: no • 1: yes Sixth digit: Whether the subscriber is blocked. • 0: no • 1: yes Seventh digit: Whether the subscriber is blacklisted. • 0: no • 1: yes Not null
41	Reserved	Char[32]	-
42	Reserved	Char[32]	-
43	Reserved	Char[32]	-
44	Reserved	Char[32]	-
45	ResultCode	Integer[4]	Indicates the internal result code.

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SN.	Field Name	Field Type	Description
			Value range: -2^{31}-2^{31}-1 The default value is **0**.
46	ChargeOfItems Accounts	Integer[8]	Indicates the free call times used. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
47	ChargeOfDurat ionAccounts	Integer[8]	Indicates the total call time charge using reword resource. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
48	ChargeOfFund Accounts	Integer[8]	Indicates the total cost amount charging from fund account. Not null The default value is **0**.
49	ChargeFromPre paid	Integer[8]	Indicates the charge amount from prepaid account. Value range: -2^{63}-2^{63}-1 Not null The default value is **0**.
50	PrepaidBalance	Integer[8]	Indicates the charge amount from prepaid account. Not null The default value is **0**.
51	ChargeFromPo stpaid	Integer[8]	Indicates the balance of prepaid account. Not null The default value is **0**.
52	PostPaidBalanc e	Integer[8]	Indicates the charge amount from postpaid account. Not null The default value is **0**.
53	AccountID	Char[20]	Indicates the balance of postpaid account. Not null
54	CurrencyCode	Integer[4]	Currency code. Also, if this field missed or present be Zero, it means the system use local minimal currency as default, and no need to specify in CDR. The default value is **0**.
55	Reserved	Char[20]	-

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SN.	Field Name	Field Type	Description
56	AccountType1	Integer[4]	Indicates the account type of the first account. 0 means not used. Not null The default value is **0**.
57	FeeType1	Integer[4]	Indicates the fee type of the first account. 0 means not used. Not null The default value is **0**.
58	ChargeAmount1	Integer[8]	Indicates the changing amount of the first account. 0 means not unchanged. Not null The default value is **0**.
59	CurrentAcctAmount1	Integer[8]	Indicates the current amount of the first account. Not null The default value is **0**.
60	AccountType2	Integer[4]	Indicates the account type of the second account. 0 means not used. Not null The default value is **0**.
61	FeeType2	Integer[4]	Indicates the fee type of the second account. 0 means not used. Not null The default value is **0**.
62	ChargeAmount2	Integer[8]	Indicates the changing amount of the second account. 0 means not unchanged. Not null The default value is **0**.
63	CurrentAcctAmount2	Integer[8]	Indicates the current amount of the second account. Not null The default value is **0**.
64	AccountType3	Integer[4]	Indicates the account type of the third account. 0 means not used. Not null The default value is **0**.

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SN.	Field Name	Field Type	Description
65	FeeType3	Integer[4]	Indicates the fee type of the third account. 0 means not used. Not null The default value is **0**.
66	ChargeAmount3	Integer[8]	Indicates the changing amount of the third account. 0 means not unchanged. Not null The default value is **0**.
67	CurrentAcctAmount3	Integer[8]	Indicates the current amount of the third account. Not null The default value is **0**.
68	AccountType4	Integer[4]	Indicates the account type of the fourth account. 0 means not used. Not null The default value is **0**.
69	FeeType4	Integer[4]	Indicates the fee type of the fourth account. 0 means not used. Not null The default value is **0**.
70	ChargeAmount4	Integer[8]	Indicates the changing amount of the fourth account. 0 means not unchanged. Not null The default value is **0**.
71	CurrentAcctAmount4	Integer[8]	Indicates the current amount of the fourth account. Not null The default value is **0**.
72	AccountType5	Integer[4]	Indicates the account type of the fifth account. 0 means not used. Not null The default value is **0**.
73	FeeType5	Integer[4]	Indicates the fee type of the fifth account. 0 means not used. Not null The default value is **0**.

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SN.	Field Name	Field Type	Description
74	ChargeAmount5	Integer[8]	Indicates the changing amount of the fifth account. 0 means not unchanged. Not null The default value is **0**.
75	CurrentAcctAmount5	Integer[8]	Indicates the current amount of the fifth account. Not null The default value is **0**.
76	AccountType6	Integer[4]	Indicates the account type of the sixth account. 0 means not used. Not null The default value is **0**.
77	FeeType6	Integer[4]	Indicates the fee type of the sixth account. 0 means not used. Not null The default value is **0**.
78	ChargeAmount6	Integer[8]	Indicates the changing amount of the sixth account. 0 means not unchanged. Not null The default value is **0**.
79	CurrentAcctAmount6	Integer[8]	Indicates the current amount of the sixth account. Not null The default value is **0**.
80	AccountType7	Integer[4]	Indicates the account type of the seventh account. 0 means not used. Not null The default value is **0**.
81	FeeType7	Integer[4]	Indicates the fee type of the seventh account. 0 means not used. Not null The default value is **0**.
82	ChargeAmount7	Integer[8]	Indicates the changing amount of the seventh account. 0 means not unchanged. Not null

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SN.	Field Name	Field Type	Description
			The default value is **0**.
83	CurrentAcctAmount7	Integer[8]	Indicates the current amount of the seventh account. Not null The default value is **0**.
84	AccountType8	Integer[4]	Indicates the account type of the eighth account. 0 means not used. Not null The default value is **0**.
85	FeeType8	Integer[4]	Indicates the fee type of the eighth account. 0 means not used. Not null The default value is **0**.
86	ChargeAmount8	Integer[8]	Indicates the changing amount of the eighth account. 0 means not unchanged. Not null The default value is **0**.
87	CurrentAcctAmount8	Integer[8]	Indicates the current amount of the eighth account. Not null The default value is **0**.
88	AccountType9	Integer[4]	Indicates the account type of the ninth account. 0 means not used. Not null The default value is **0**.
89	FeeType9	Integer[4]	Indicates the fee type of the ninth account. 0 means not used. Not null The default value is **0**.
90	ChargeAmount9	Integer[8]	Indicates the changing amount of the ninth account. 0 means not unchanged. Not null The default value is **0**.
91	CurrentAcctAmount9	Integer[8]	Indicates the current amount of the ninth account. Not null

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SN.	Field Name	Field Type	Description
			The default value is **0**.
92	AccountType10	Integer[4]	Indicates the account type of the tenth account. 0 means not used. Not null The default value is **0**.
93	FeeType10	Integer[4]	Indicates the fee type of the tenth account. 0 means not used. Not null The default value is **0**.
94	ChargeAmount10	Integer[8]	Indicates the changing amount of the tenth account. 0 means not unchanged. Not null The default value is **0**.
95	CurrentAcctAmount10	Integer[8]	Indicates the current amount of the tenth account. Not null The default value is **0**.
96	PreviousActivestop	Char[14]	Indicates **Activestop** before operation. Format: YYYYMMDDhhmmss
97	NewActivestop	Char[14]	Indicates **Activestop** after operation. Format: YYYYMMDDhhmmss
98	PreviousSuspendStop	Char[14]	Indicates **SuspendStop** before operation. Format: YYYYMMDDhhmmss
99	NewSuspendStop	Char[14]	Indicates **SuspendStop** after operation. Format: YYYYMMDDhhmmss
100	PreviousDisableStop	Char[14]	Indicates **DisableStop** before operation. Format: YYYYMMDDhhmmss
101	NewDisableStop	Char[14]	Indicates **DisableStop** after operation. Format: YYYYMMDDhhmmss
102	StandardEventID	Integer[4]	Indicates the ID of the standard event.
103	EventRatingStartTime	Char[14]	Indicates the start time of the event rating.
104	EventChargedStartTime	Char[14]	Indicates the start time of the event charged.
105	CurChargerSub	Integer[4]	Indicates the subscriber object of the standard

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SN.	Field Name	Field Type	Description
	scriberObject		cur charger.
106	ChargedService Number	Char[64]	Indicates the service number of charged.
107	Reserved	Char[20]	-
108	Reserved	Char[20]	-
109	Reserved	Char[20]	-
110	AddtionalInfo	Char[512]	Reserved
111	StartTimeOfBil lCycle	Char[14]	Indicates the start time of bill cycle. The default value is **0**.
112	AccountKey1	Integer[8]	Indicates the key value of the first account. **NOTE** ● When changing the validity period of the account balance, you need to record this parameter. ● The changes of the values of the **AccountKey1**, **PreviousAcctExpireTime1**, **NewAcctExpireTime1**, **AccountType1**, **FeeType1**, **ChargeAmount1**, and **CurrentAcctAmount1** parameters indicate the change of a subaccount instance. The default value is **0**.
113	PreviousAcctE xpireTime1	Char[14]	Indicates the previous expire time of the first account. **NOTE** When changing the validity period of the account balance, you need to record this parameter.
114	NewAcctExpire Time1	Char[14]	Indicates the current expire time of the first account. **NOTE** When changing the validity period of the account balance, you need to record this parameter.
115	AccountKey2	Integer[8]	Indicates the key value of the second account. **NOTE** When changing the validity period of the account balance, you need to record this parameter. The default value is **0**.
116	PreviousAcctE xpireTime2	Char[14]	Indicates the previous expire time of the second account. **NOTE** When changing the validity period of the account balance, you need to record this parameter.
117	NewAcctExpire Time2	Char[14]	Indicates the current expire time of the second account. **NOTE**

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SN.	Field Name	Field Type	Description
			When changing the validity period of the account balance, you need to record this parameter.
118	AccountKey3	Integer[8]	Indicates the key value of the third account. **NOTE** When changing the validity period of the account balance, you need to record this parameter. The default value is **0**.
119	PreviousAcctExpireTime3	Char[14]	Indicates the previous expire time of the third account. **NOTE** When changing the validity period of the account balance, you need to record this parameter.
120	NewAcctExpireTime3	Char[14]	Indicates the current expire time of the third account. **NOTE** When changing the validity period of the account balance, you need to record this parameter.
121	AccountKey4	Integer[8]	Indicates the key value of the fourth account. **NOTE** When changing the validity period of the account balance, you need to record this parameter. The default value is **0**.
122	PreviousAcctExpireTime4	Char[14]	Indicates the previous expire time of the fourth account. **NOTE** When changing the validity period of the account balance, you need to record this parameter.
123	NewAcctExpireTime4	Char[14]	Indicates the current expire time of the fourth account. **NOTE** When changing the validity period of the account balance, you need to record this parameter.
124	AccountKey5	Integer[8]	Indicates the key value of the fifth account. **NOTE** When changing the validity period of the account balance, you need to record this parameter. The default value is **0**.
125	PreviousAcctExpireTime5	Char[14]	Indicates the previous expire time of the fifth account. **NOTE** When changing the validity period of the account balance, you need to record this parameter.
126	NewAcctExpireTime5	Char[14]	Indicates the current expire time of the fifth account.

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SN.	Field Name	Field Type	Description
			NOTE When changing the validity period of the account balance, you need to record this parameter.
127	AccountKey6	Integer[8]	Indicates the key value of the sixth account. **NOTE** When changing the validity period of the account balance, you need to record this parameter. The default value is **0**.
128	PreviousAcctExpireTime6	Char[14]	Indicates the previous expire time of the sixth account. **NOTE** When changing the validity period of the account balance, you need to record this parameter.
129	NewAcctExpireTime6	Char[14]	Indicates the current expire time of the sixth account. **NOTE** When changing the validity period of the account balance, you need to record this parameter.
130	AccountKey7	Integer[8]	Indicates the key value of the seventh account. **NOTE** When changing the validity period of the account balance, you need to record this parameter. The default value is **0**.
131	PreviousAcctExpireTime7	Char[14]	Indicates the previous expire time of the seventh account. **NOTE** When changing the validity period of the account balance, you need to record this parameter.
132	NewAcctExpireTime7	Char[14]	Indicates the current expire time of the seventh account. **NOTE** When changing the validity period of the account balance, you need to record this parameter.
133	AccountKey8	Integer[8]	Indicates the key value of the eighth account. **NOTE** When changing the validity period of the account balance, you need to record this parameter. The default value is **0**.
134	PreviousAcctExpireTime8	Char[14]	Indicates the previous expire time of the eighth account. **NOTE** When changing the validity period of the account balance, you need to record this parameter.
135	NewAcctExpire	Char[14]	Indicates the current expire time of the eighth

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SN.	Field Name	Field Type	Description
	Time8		account. **NOTE** When changing the validity period of the account balance, you need to record this parameter.
136	AccountKey9	Integer[8]	Indicates the key value of the ninth account. **NOTE** When changing the validity period of the account balance, you need to record this parameter. The default value is **0**.
137	PreviousAcctExpireTime9	Char[14]	Indicates the previous expire time of the ninth account. **NOTE** When changing the validity period of the account balance, you need to record this parameter.
138	NewAcctExpireTime9	Char[14]	Indicates the current expire time of the ninth account. **NOTE** When changing the validity period of the account balance, you need to record this parameter.
139	AccountKey10	Integer[8]	Indicates the key value of the tenth account. **NOTE** When changing the validity period of the account balance, you need to record this parameter. The default value is **0**.
140	PreviousAcctExpireTime10	Char[14]	Indicates the previous expire time of the tenth account. **NOTE** When changing the validity period of the account balance, you need to record this parameter.
141	NewAcctExpireTime10	Char[14]	Indicates the current expire time of the tenth account. **NOTE** When changing the validity period of the account balance, you need to record this parameter.
142	Reserved	Char[20]	Reserved
143	Merchant	Char[20]	Reserved
144	SERVICE	Char[20]	Reserved
145	LoanAmount	Integer[8]	Reserved
146	LoanPoundage	Integer[8]	Reserved
147	IsTestNumber	Integer[4]	Reserved
148	MainProductSt	Integer[4]	Reserved

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SN.	Field Name	Field Type	Description
	atus		
149	DiameterSessionID	Char[128]	Indicates the session id that generated during the interaction between the SCP and the CBP. The field includes parts as follows: <DiameterIdentity>;<high 32 bits>;<low 32 bits>[;<optional value>]. • <DiameterIdentity>: same as Origin-Host; • <high 32 bits>: the current day • <low 32 bits>: increase by degrees circulatory, the start value is 0. Set the value is 0 when the system restarts. • <optional value>: reserved. Every fields are separated by a character ";". Example: SCP001.huawei.com;1876543210;523
150	NewProductID	Integer[32]	Indicates the ID of the new product that the subscriber subscribes and unsubscribes. The default value is **0**. Value range: -2^{63}-2^{63}-1
151	ExpireTimeofNewProduct	Char[32]	Indicates the expire time of the new product that the subscriber subscribes. Format: YYYYMMDDhhmmss
152	subscriberkey	Integer[4]	Indicates the key of a subscriber. Not null The default value is **0**.
153	accountkey	Integer[8]	Indicates the key of a account. Not null The default value is **0**.
154	HistoryBillCycleFlag	Integer[4]	Indicates whether an adjustment is a history adjustment. The value is follows: • 0: No • 1: Yes The default value is **0**. Not null
155	UsageServiceType	Integer[4]	Indicates the type of a real-time service. The default value is **3901**.
156	OriginCdrFilename	Char[64]	Indicates the name of a rated CDR. The default value is **0**.
157	CurCDRLineN	Integer[4]	Indicates the number of the line where the

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SN.	Field Name	Field Type	Description
	o		original CDR is located. The default value is **0**.
158	ISQueryAccFree	Char[1]	Indicates whether the subscriber queries account balance is free. • 0: no • 1: yes The default value is **0**.
159	Tax	Integer[8]	Reserved
160	Tax2	Integer[8]	Reserved
161	Tax3	Integer[8]	Reserved
162	Tax4	Integer[8]	Reserved
163	ChargeOfItemsSecondAccounts	Integer[8]	Free call times be used from others, such as parent card, VPN. Not null
164	ChargeOfDurationSecondAccounts	Integer[8]	Indicates total call time charge using reword resource from others, such as parent card, VPN. Not null
165	ChargeOfFundSecondAccounts	Integer[8]	Indicates total cost amount charging from fund account. Not null
166	ChargeFromSecondAccount	Integer[8]	Indicates charge amount from other account, such as the parent card, VPN. Not null
167	SecondAccountBalance	Integer[8]	Reserved
168	SecondAccountID	Char[20]	Indicates the ID of the main capital account from which fees are deducted. The ID is transferred into the internal system by an external system when the account is created.
169	SecondAccountCurrencyCode	Integer[4]	Indicates the currency code, in minimum currency unit by default. The default value is **0**.
170	SecondAccountType1	Integer[4]	1st group of account change info. Including: • Account Type, 0 means not used. • Fee Type, 0 means not used. • Changed Amount, 0 means unchanged. • Current Amount. Not null

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SN.	Field Name	Field Type	Description
			The default value is **0**.
171	SecondFeeType1	Integer[4]	Fee Type, 0 means not used. Not null The default value is **0**.
172	SecondChargeAmount1	Integer[8]	Changed Amount, 0 means unchanged. Not null The default value is **0**.
173	SecondCurrentAcctAmount1	Integer[8]	Reserved
174	SecondAccountType2	Integer[4]	2nd group of account change info. The structure is same as the first group. Not null The default value is **0**.
175	SecondFeeType2	Integer[4]	Fee Type, 0 means not used. Not null The default value is **0**.
176	SecondChargeAmount2	Integer[8]	Changed Amount, 0 means unchanged. Not null The default value is **0**.
177	SecondCurrentAcctAmount2	Integer[8]	Reserved
178	SecondAccountType3	Integer[4]	3rd group of account change info. The structure is same as the first group. Not null The default value is **0**.
179	SecondFeeType3	Integer[4]	Fee Type, 0 means not used. Not null The default value is **0**.
180	SecondChargeAmount3	Integer[8]	Changed Amount, 0 means unchanged. Not null The default value is **0**.
181	SecondCurrentAcctAmount3	Integer[8]	Reserved
182	SecondAccountType4	Integer[4]	4th group of account change info. The structure is same as the first group. The default value is **0**.
183	SecondFeeType4	Integer[4]	Fee Type, 0 means not used. Not null

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SN.	Field Name	Field Type	Description
			The default value is **0**.
184	SecondChargeAmount4	Integer[8]	Changed Amount, 0 means unchanged. Not null The default value is **0**.
185	SecondCurrentAcctAmount4	Integer[8]	Reserved
188	SecondAccountType5	Integer[4]	5th group of account change info. The structure is same as the first group. The default value is **0**.
187	SecondFeeType5	Integer[4]	Fee Type, 0 means not used. Not null The default value is **0**.
188	SecondChargeAmount5	Integer[8]	Changed Amount, 0 means unchanged. Not null The default value is **0**.
189	SecondCurrentAcctAmount5	Integer[8]	Reserved
190	SecondBonusValidity1	Char[14]	Indicates the validity period of the first rewarded resource. Only the validity period of rewarded resources is recorded. The valid period of consumption resources is not recorded.
191	SecondBonusValidity2	Char[14]	Indicates the validity period of the second rewarded resource. Only the validity period of rewarded resources is recorded. The valid period of consumption resources is not recorded.
192	SecondBonusValidity3	Char[14]	Indicates the validity period of the third rewarded resource. Only the validity period of rewarded resources is recorded. The valid period of consumption resources is not recorded.
193	SecondBonusValidity4	Char[14]	Indicates the validity period of the fourth rewarded resource. Only the validity period of rewarded resources is recorded. The valid period of consumption resources is not recorded.
194	SecondBonusValidity5	Char[14]	Indicates the validity period of the fifth rewarded resource. Only the validity period of rewarded resources is recorded. The valid period of consumption resources is not recorded

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SN.	Field Name	Field Type	Description
195	SecondSubscriberKey	Integer[4]	Indicates the key value of a entrusted-payment subscriber. The default value is **0**.
196	SecondAccountKey	Integer[8]	Indicates the key value of a entrusted-payment account. The default value is **0**.
197	SecondProductID	Integer[4]	Reserved
198	SecondChargeFromPostpaid	Integer[8]	Indicates the amount that is paid by the postpaid account of a entrusted-payment subscriber. The default value is **0**.
199	SecondChargeFromPrepaid	Integer[8]	Indicates the amount that is paid by the prepaid account of a entrusted-payment subscriber. The default value is **0**.
200	SecondChargingPartyNumber	Char[32]	Indicates the number of the entrusted-payment account. The number format are as follows: • Fixed-line telephone number: area number + PSTN. • Mobile phone number: MSISDN.
201	SecondChargingPartyPayType	Char[1]	Indicates the payment mode of the entrusted-payment account, for example, the payment mode of a parent card. By default the parameter is the payment mode in the subscriber attribute.
202	SecondChargingPartyBillCycleID	Char[20]	Indicates the bill cycle of the entrusted-payment account, for example, the bill cycle of the parent account. The parameter Bill cycle ID is in the format YYYYMMTT, in which YYYYMM is the year and month of the bill cycle and TT is the bill cycle type.
203	ChargePartyIndicator	Char[1]	Charging party indicator. • 1: The calling party is charged. • 2: The called party is charged. • 3: charge of third party. • 4: charge of parent and child card. • 5: charge of calling party and VPN. The default value is **1**.

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Operation ID

Table 3-11 The description of operation ID

Operation ID	Description
4050000	Register individual subscriber
4050001	Subscribe to product for individual subscriber
4050002	Replace appendant product
4050003	Suspend service
4050004	Resume service
4050005	Claim missing
4050006	Disclaim missing
4050008	Change SIM card
4050009	Change MSISDN
4050014	Recharge account by cash
4050015	Recharge account by recharge card
4050018	Unsubscribe to product
4050019	Delete subscriber
4050022	Change subscriber password
4050023	Change customer information
4050025	Query customer interaction log
4050026	Set blacklist
4050042	Change subscriber information
4050043	Change appendant product
4050046	Lock
4050047	Unlock
4050050	Create customer information
4050051	Change bill cycle
4050056	Add group member
4050057	Delete group member
4050059	Set barring
4050061	cancel product
4050062	Change subscriber credit

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Operation ID	Description
4050063	Bind a child card
4051000	New subscriber in batches
4051001	Delete subscriber in batches
4051002	Change service in batches
4051003	Change product in batches
4051004	Activate subscribers in batches
4052001	Change subscriber state
4052002	Suspend services with the mobile number being reserved
4052003	query points
4052004	Modify points
4052100	Change account balance
4052101	Transfer account balance
4052102	Change account credit
4052103	First activation(PPS)
4052104	Release subscriber from blacklist
4052105	Change balance to validity period
4052106	Modify subscriber language
4052107	Add community
4052108	Delete community
4052109	Perform cash recharge rollback
4052110	Add subscriber to CUG
4052111	Delete subscriber from CUG
4052112	Change main product
4052113	Replace main package
4052114	Manage FN service
4052115	Change small balance notification threshold of subscriber
4052116	Query small balance notification threshold of subscriber
4052117	Switch free resources
4052118	Change postpaid subscriber state

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Operation ID	Description
4052119	Subscribe to product in batches
4052120	Unsubscribe to product in batches
4052121	Modify subscriber validity periods in batches
4052122	Change main product in batches
4052124	Batch service
4052126	One-off fee deduction
4052127	Integrated query
4052128	Change validity period of subscriber
4052129	Query card recharge log
4052130	Query cash recharge log
4052131	Query account transfer log
4052132	Query account adjustment log
4052133	Query account balance
4052134	Query accumulative item
4052136	Modify account information
4052144	Change initial account balances in batches
4052145	Change latest activation time in batches
4052146	Manage the call screening service
4052303	Unbind a child card
4052304	Change the bound child card
4052306	Reset password
4052309	Query the information about a child card
4052310	Query the consumption limit of a child card
4052313	Query a member of a CUG
4052346	Send a message to a subscriber with a child card when the child card is unbound from the parent card
4052347	Send a message to a subscriber with a parent card when a child card is unbound from the parent card
4052352	Query transfer information
4052353	Query last transfer information
4052361	Manage the international call forwarding (ICF)

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Operation ID	Description
	function
4052366	Add subscribers to a CUG in batches
4052369	Manage CUGs
4052370	Query the list of all CUG members
4052371	Query CUG types defined by the system
4052390	Enable and disable the cost notification function
4052401	Modify prepaid life cycle status
4052404	Query an accumulator
4052409	Penalty for point exchange
4053000	Create new group
4053001	Change group product
4053002	Modify group state
4053003	Query group information
4053004	Modify group information
4053005	Query sub group list
4053006	Apply for deleting group
4053007	Change orgnization relation
4053008	Modify top group quota
4053009	Modify allocated quota
4053010	Manage share relation
4053011	Join super group
4053012	Manage super group short number
4053013	Manage cross group
4053014	Manage group attendant
4053015	Query group attendant
4053016	Manage group CallScreen number
4053017	Query group CallScreen number
4053018	Manage group off-net Number
4053019	Query group off-net Number
4053020	Manage group hunting list
4053021	Manage group hunting number

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Operation ID	Description
4053022	Query group hunting list
4053023	Query group hunting number
4053024	Query group administrator
4053025	Add group member
4053026	Query group member list
4053027	Modify goup member state
4053028	Delete group member
4053029	Manage member off-net number
4053030	Query member off-net number
4053031	Query group member aasic information
4053032	Modify group member basic information
4053033	Modify group member number
4053034	Query group CUG
4053035	Manage group CUG
4053036	Query CUG member
4053037	Manage CUG member
4053038	Query member CUG
4053039	Query member number
4053040	Query member quota
4053041	Set member display mode
4053042	Set delete group task
4053043	Delete group
4053044	Query group quota information by IVR
4053045	Group Reward member
4053046	Add member in batches
4053047	Delete member in batches
4053048	Modify member state in batches

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3.1.10 clr CDR

Description

When the life cycles of subscribers change, clr CDRs record the information if the OCS clears the account balances or reward resources of the subscribers.

Generation Scenario

In the following scenario will generate the clr CDR.

- When the life cycles of subscribers change, the system clears the account balances or reward recourses of the subscribers. In this case, clr CDRs are generated. For example, when subscriber A enters the pool state, the system clears all resources of the subscriber's accounts and records a clr CDR.
- The CBP clears expired subaccounts during background scan, and the system generates a clr CDR.

CDR File Name Format

clrYYYYMMDD_CBPNODE_Seqno.unl

Example of a clr CDR file name: clr20071112_101_000151.unl

Format

Table 3-12 describes each field of clr CDR file.

Table 3-12 Format of clr CDR

SN.	Field Name	Field Type	Description
1	SerialNo	Integer[8]	Indicates the serial number. It is generated by the system and the serial number is exclusive. Not null The default value is **0**.
2	SubSequence	Integer[4]	Indicates the sequence number when the bill is split for mid-night and end of the bill cycle. The default value is **0**.
3	TimeStamp	Char[14]	Indicates the time stamp of the bill item generation. Format: YYYYMMDDhhmmss Not null The default value is **null**.
4	ChargingPartyNumber	Char[64]	Indicates the telephone number of the charging party. The number is same as calling party number in

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SN.	Field Name	Field Type	Description
			MO and called party number in MT. There are several special situations. Like Child-Parent card feature, the calling party number is the child, the charging party number is the parent's number. This field is null when the service is paid only by account. The number format are as follows: • **Area number + PSTN** The number format is the format of the fixed-line telephone number. • **MSISDN** The number format is the format of the mobile phone number. Value range: numerals within 0-9, *, # The default value is **null**.
5	ChargeEvent	Char[16]	Indicates the type of the current charge event.
6	SubChargeEvent	Char[16]	Indicates the type of the current sub charge event.
7	Reserved	Char[20]	-
8	Reserved	Char[20]	-
9	Reserved	Char[20]	-
10	Reserved	Char[20]	-
11	BrandID	Integer[4]	Indicates the brand ID of subscribers. Not null The default value is **0**.
12	SubCOSID	Integer[4]	Indicates the main product ID of subscribers. Not null The default value is **0**.
13	PayType	Char[1]	Indicates the pay type of subscribers. The options are as follows: • 0: Prepaid • 1: Postpaid • 2: Hybrid Not null
14	BillCycleID	Char[20]	Indicates the bill cycle ID. Format: YYYYMMTT The description of the format is as follows: • YYYY: indicates the year.

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SN.	Field Name	Field Type	Description
			• MM: indicates the month. • TT: indicates the type of a bill cycle.
15	TransitionID	Char[32]	Indicates a transaction ID.
16	SubscriberID	Char[20]	Indicates the subscriber ID. Value range: characters in [0-9, a-z, A-Z] Not null
17	CurrUserState	Char[10]	Indicates the life cycle state and the management state of a subscriber after operation. The **CurrUserState** parameter is a numeric string of seven digits, in the format CCMMMMM. The first and second digits C indicate the life cycle state of a subscriber. The first digit C indicates the state of a prepaid subscriber, and the second digit C indicates the state of a postpaid subscriber. The options are as follows: • 0: Idle • 1: Active • 2: Suspended • 3: Disable • 4: Pool • 5: Pool without activation The next five digits MMMMM indicate the management state of a subscriber. The options are as follows: Third digit: Whether the subscriber claims missing. • 0: no • 1: yes Fourth digit: Whether the subscriber suspends the service. • 0: non-suspend • 1: suspend • 2: suspend by operator (B1W) • 3: suspend by operator (B2W) Fifth digit: Whether the subscriber applies for suspension with the number being reserved. • 0: no • 1: yes Sixth digit: Whether the subscriber is blocked.

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SN.	Field Name	Field Type	Description
			• 0: no • 1: yes Seventh digit: Whether the subscriber is blacklisted. • 0: no • 1: yes Not null
18	ResultCode	Integer[4]	Indicates the internal result code. Value range: -2^{31}-2^{31}-1 The default value is **0**.
19	Reserved	Char[20]	-
20	Reserved	Char[20]	-
21	AccountID	Char[20]	Reserved
22	PrepaidBalance	Integer[8]	Indicates the balance of prepaid account. Value range: -2^{63}-2^{63}-1 The default value is **0**.
23	PostPaidBalance	Integer[8]	Indicates the balance of postpaid account. Value range: -2^{63}-2^{63}-1 The default value is **0**.
24	Reserved	Char[20]	-
25	Reserved	Char[20]	-
26	AccountType1	Integer[4]	Indicates the type of the first account that is cleared. The default value is **0**.
27	CleanedAmount1	Integer[8]	Indicates the change range of the first account that is cleared. Value range: -2^{63}-2^{63}-1 The default value is **0**.
28	AccountType2	Integer[4]	Indicates the type of the second account that is cleared. The default value is **0**.
29	CleanedAmount2	Integer[8]	Indicates the change range of the second account that is cleared. Value range: -2^{63}-2^{63}-1 The default value is **0**.
30	AccountType3	Integer[4]	Indicates the type of the third account that is cleared.

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SN.	Field Name	Field Type	Description
			The default value is **0**.
31	CleanedAmount3	Integer[8]	Indicates the change range of the third account that is cleared. Value range: -2^{63}-2^{63}-1 The default value is **0**.
32	AccountType4	Integer[4]	Indicates the type of the fourth account that is cleared. The default value is **0**.
33	CleanedAmount4	Integer[8]	Indicates the change range of the fourth account that is cleared. Value range: -2^{63}-2^{63}-1 The default value is **0**.
34	AccountType5	Integer[4]	Indicates the type of the fifth account that is cleared. The default value is **0**.
35	CleanedAmount5	Integer[8]	Indicates the change range of the fifth account that is cleared. Value range: -2^{63}-2^{63}-1 The default value is **0**.
36	AccountType6	Integer[4]	Indicates the type of the sixth account that is cleared. The default value is **0**.
37	CleanedAmount6	Integer[8]	Indicates the change range of the sixth account that is cleared. Value range: -2^{63}-2^{63}-1 The default value is **0**.
38	AccountType7	Integer[4]	Indicates the type of the seventh account that is cleared. The default value is **0**.
39	CleanedAmount7	Integer[8]	Indicates the change range of the seventh account that is cleared. Value range: -2^{63}-2^{63}-1 The default value is **0**.
40	AccountType8	Integer[4]	Indicates the type of the eighth account that is cleared. The default value is **0**.
41	CleanedAmount8	Integer[8]	Indicates the change range of the eighth account that is cleared. Value range: -2^{63}-2^{63}-1

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SN.	Field Name	Field Type	Description
			The default value is **0**.
42	AccountType9	Integer[4]	Indicates the type of the ninth account that is cleared. The default value is **0**.
43	CleanedAmount9	Integer[8]	Indicates the change range of the ninth account that is cleared. Value range: -2^{63}-2^{63}-1 The default value is **0**.
44	AccountType10	Integer[4]	Indicates the type of the tenth account that is cleared. The default value is **0**.
45	CleanedAmount10	Integer[8]	Indicates the change range of the tenth account that is cleared. Value range: -2^{63}-2^{63}-1 The default value is **0**.
46	IsTestNumber	Integer[4]	Reserved
47	MainProductStatus	Integer[4]	Reserved
48	ExpireTimeOfAccountType1	Char[14]	Indicates the expire time of the first account that is cleared. Value format: YYYYMMDDhhmmss
49	ExpireTimeOfAccountType2	Char[14]	Indicates the expire time of the second account that is cleared. Value format: YYYYMMDDhhmmss
50	ExpireTimeOfAccountType3	Char[14]	Indicates the expire time of the third account that is cleared. Value format: YYYYMMDDhhmmss
51	ExpireTimeOfAccountType4	Char[14]	Indicates the expire time of the fourth account that is cleared. Value format: YYYYMMDDhhmmss
52	ExpireTimeOfAccountType5	Char[14]	Indicates the expire time of the fifth account that is cleared. Value format: YYYYMMDDhhmmss
53	ExpireTimeOfAccountType6	Char[14]	Indicates the expire time of the sixth account that is cleared. Value format: YYYYMMDDhhmmss
54	ExpireTimeOfAccountType7	Char[14]	Indicates the expire time of the seventh account that is cleared. Value format: YYYYMMDDhhmmss

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SN.	Field Name	Field Type	Description
55	ExpireTimeOfAccountType8	Char[14]	Indicates the expire time of the eighth account that is cleared. Value format: YYYYMMDDhhmmss
56	ExpireTimeOfAccountType9	Char[14]	Indicates the expire time of the ninth account that is cleared. Value format: YYYYMMDDhhmmss
57	ExpireTimeOfAccountType10	Char[14]	Indicates the expire time of the tenth account that is cleared. Value format: YYYYMMDDhhmmss
58	subscriberkey	Integer[4]	Indicates the key of a subscriber. The default value is **0**.
59	accountkey	Integer[8]	Indicates the key of a account. The default value is **0**.

3.1.11 abvou CDR

Description

When the SCP fails to reset a recharge card and sends an exception to the CBP, the CBP generates an abvou CDR.

Generation Scenario

When the SCP fails to reset a recharge card and sends an exception to the CBP, the CBP generates an abvou CDR.

CDR File Name Format

abvouYYYYMMDD_CBPNODE_Seqno.unl

Example of an abvou CDR file name: abvou20071112_101_000151.unl

Format

Table 3-13 describes each field of abvou CDR file.

Table 3-13 Format of abvou CDR

SN	Field Name	Field Type	Description
1	CallingNumber	char[20]	Indicates the calling number that initiates a recharge request. Not null

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SN	Field Name	Field Type	Description
2	RechargeNumber	Char[20]	Indicates a number that is recharged. Not null
3	AccessMethod	Integer[4]	Indicates an access method. The options are as follows: • 1: IVR • 2: USSD • 3: SMS • 4: SMAP or WebService interface Not null.
4	Sequence	Char[20]	Indicates the sequence number of a recharge card. Not null.
5	TradeTime	Char[14]	Indicates the time when a recharge operation is performed. Format: YYYYMMDDhhmmss Not Null.
6	CardFaceValue	Integer[8]	Indicates the face value of a recharge card in the minimum currency unit. Not null.
7	CBPID	Integer[4]	Indicates the ID of a CBP. Not null.
8	CardPrefix	Char[24]	Indicates the prefix of a recharge card. Can be null.

3.1.12 BalExt CDR

Description

If any customer recharges or adjusts account balance and the balance is equal to more than upper limit, or when ever subscriber uses the OCS service, the balance will be deducted. After the deduction, if the balance is equal to or less than the lower limit, OCS generates "BALEXT" EDR in the same way like other CDRs at CBP.

Generation Scenario

Whenever This Balance Threshold is met the "BALEXT" EDR will be written for Voice, SMS, MMS, GPRS, OnceDeduct, AdjustAccount, Recharge and RentFlow.

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CDR File Name Format

BalExtYYYYMMDD_CBPNODE_Seqno.unl

Example of an BalExtCDR file name: BalExt20150130_101_000003.unl-500

Format

Table 3-14 describes each field of BalExt CDR file.

Table 3-14 Format of BalExt CDR

SN	Field Name	Field Type	Description
1	CallingPartyNumber	Char[64]	Indicates the calling number (CC + MSISDN). Not null.
2	Timestamp	Char[14]	Indicates the Timestamp. Format: YYYYMMDDhhmmss Not null.
3	PrepaidBalance	Integer[8]	Indicates the balance of prepaid account. Value range: -2^63 to 2^63-1. Not null.
4	Main ProductId	Integer[4]	Indicates the Main ProductId. Not null.

3.2 SCP-Generated CDR

The section describes the file name format of SCP-generated CDRs of each type, SCP-generated CDR storage directories and SCP-generated CDR generation scenarios.

3.2.1 General Description

The generated CDRs are stored in the SCP node. The directory for storing CDR files cannot be configured.

- Temporary directory: **${TELLIN_DIR}/sms/record/recbuf**
- Destination directory: **${TELLIN_DIR}/sms/record**
- Backup directory: **${TELLIN_DIR}/sms/record/backup**

Here, **${TELLIN_DIR}** refers to the installation directory of the SCP and can be configured in the **.cshrc** file. You are not advised to change the installation directory because the change affects the system running.

For description about the CDR storage directories, see 1.2 CDR Storage Directories.

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3.2.2 w_rec CDR

Description

w_rec indicates voice call CDR.

The w_rec CDR records the call information about subscribers.

Generation Scenario

In the OCS service,

- When subscriber A makes a voice call successfully, the SCP generates a w_rec CDR for the calling party after the call.
- When subscriber B answers a voice call successfully, the SCP generates a w_rec CDR for the called party after the call.

CDR File Name Format

w_recYYYYMMDD_SCPIDBBCC_Seqno.unl

Example of a w_rec CDR file name: w_rec20071012_1820001_0001.unl

Format

Table 3-15 describes each field of w_rec CDR file.

Table 3-15 Format of w_rec CDR

SN	Field Name	Field Type	Description
1	ServiceKey	Integer[4]	Indicates the service key. This CDR is recorded during multiple processes, and the service keys of different processes are not the same. Therefore, the field value is not unique. For example, when the value is **91**, it indicates the call process
2	SubCosID	Integer[4]	Indicates the sub brand ID of a subscriber.
3	CallingPartyNumber	Char[32]	Indicates the calling number. The number format are as follows: • **Country code + Area number + PSTN** The number format is the format of the fixed-line phone number. • **Country code + MSISDN** The number format is the format of the mobile phone number.
4	CalledPartyNumber	Char[32]	Indicates the called number. Records the original called number which is

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SN	Field Name	Field Type	Description
			reported by IDP.
5	Real-Called-Number	Char[32]	Indicates the real called number. The number format are as follows: • **Country code + Area number + PSTN** The number format is the format of the fixed-line phone number. • **Country code + MSISDN** The number format is the format of the mobile phone number.
6	Connect-Called-Number	Char[32]	Indicates the called number for connection. Records the number which is truly took in Connect.
7	Charge-Flow-Type	Integer[4]	Indicates the charge process type. The options are as follows: • 1: the calling process • 2: the called process • 3: the forwarding process
8	Calling-Partys-Category	Integer[4]	Indicates the category of the calling party. Example, a carrier, a public call office, or a user.
9	Calling-Vlr-Number	Char[32]	Indicates the VLR number of the visitor location of the calling party. The field can be null in the called process and forwarding process.
10	CallingCellID	Char[32]	Indicates the cell ID of the calling party. The field can be null in the called process and forwarding process.
11	Called-Vlr-Number	Char[32]	Indicates the VLR number of the visitor location of the called party. The field can be null in the calling process and forwarding process.
12	CalledCellID	Char[32]	Indicates the cell ID of the called party. The field can be null in the calling process and forwarding process.
13	Original-Called-Party-Id	Char[32]	Indicates the original called number. • In the forwarding process, the MSC reports the original called number. • In the calling process, the MSC does not report the original called number. Therefore, the field is null.

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SN	Field Name	Field Type	Description
14	Redirecting-Party-Id	Char[32]	Indicates the redirection number in the forwarding process. It is the calling number in the forwarding process. • In the forwarding process, the MSC reports the redirection number. • In the calling process, the MSC does not report the redirection number. Therefore, the field is null.
15	Redirection-Reason	Integer[4]	Indicates the forwarding reason. The options are as follows: • 0: not forward • 1: forward unconditionally • 2: forward on the busy mobile • 3: forward on no reply
16	MSC-Address	Char[32]	Indicates the MSC address.
17	CallBeginTime	Char[14]	The start time of call. Format: YYYYMMDDhhmmss
18	CallDuration	Integer[4]	The duration of a call, used as charge basis. Unit: second
19	TerminationReason	Integer[4]	Indicates the reason why the call cannot be connected. The options are as follows: • 0: success • 1: insufficient balance • 2: incorrect account status • 3: invalid called number • 4: barred call • 5: call authority restriction • 6: system error • 7: route selection failure • 8: the called having no answer • 9: the called being busy • 10: the caller abandoning the call • 12: wrong input of the subscriber • 13: recharge card resetting failure • 14: network connection failure • 15: over maximum call duration • 16: balance used up • 99: other

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SN	Field Name	Field Type	Description
20	CallReferenceNumber	Char[128]	Indicates the reference number of a call. It is used to uniquely identify a call. The options are as follows: • For the CDMA network, obtain the value of **BillID** in the ORREQ message for this field. • For the GSM network, obtain the value of **CallReferenceNumber** that is reported in the IDP signaling for this field. • For the NGN and IMS networks, obtain the value of **CallID** that is reported in the SIP signaling for this field. • If the system cannot obtain the required information (excluding the preceding conditions), the value of this field is in the format *YYYYMMDDhhmmss (14 digits) + SCPID (3 digits) + automaton number* (4 digits). **NOTE** If the length of the automaton number is less than four digits, prefix 0 for the automaton number. If the length of the automaton number is more than four digits, obtain the last four digits for the automaton number.
21	DiameterSessionID	Char[128]	Indicates a diameter session ID. Records this field only when the system alternated with the CBP and set up a conversation. The format is as follows: <DiameterIdentity>;<high 32 bits>;<low 32 bits>[;<optional value>] • <DiameterIdentity>: the same as the field in the Origin-Host AVP. • <high 32 bits>: indicates the current time of the system. • <low 32 bits>: indicates the cycle count-up. The initial value is **0**. When you restart the system, set the value to **0**. • <optional value>: reserved. Separate the preceding fields with the character ";". For example: SCP001.huawei.com;1876543210;523
22	NPPrefix	Char[10]	Indicates the NP prefix of a called number, that is, an RN prefix. **NOTE**

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SN	Field Name	Field Type	Description
			• NP: Number Portable • RN: Routing Number
23	CalledHomeNetWork	Char[10]	Indicates the home network code of a called number after the NP.
24	FirstActiveFlag	Char[1]	Indicates whether the account is activated during this call. • 0: No • 1: Yes

3.2.3 w_abr CDR

Description

w_abr indicates the CDR generated in case of CBP error.

Generation Scenario

When a subscriber uses the call service as a calling party, called party, or call forwarding party, if an error occurs during the interaction between the SCP and the CBP, and the system allows the subscriber to go on using the call service, the SCP generates an Abr CDR.

CDR File Name Format

w_abrYYYYMMDD_SCPIDBBCC_Seqno.unl

Example of a w_abr CDR file name: w_abr20071012_1820001_0001.unl

Format

Table 3-16 describes each field of w_abr CDR file.

Table 3-16 Format of w_abr CDR

SN	Field Name	Field Type	Description
1	ServiceKey	Integer[4]	Indicates the service key. This CDR is recorded during multiple processes, and the service keys of different processes are not the same. Therefore, the field value is not unique. For example, when the value is **91**, it indicates the call process
2	SubCosID	Integer[4]	Indicates the sub brand ID of a subscriber.
3	Calling-Party-Address	Char[20]	Indicates the calling number. Fixed-line telephone number with an area code, mobile phone number without an area

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SN	Field Name	Field Type	Description
			code (international number form).
4	Called-Party-Address	Char[20]	Indicates the called number (with prefix).
5	Calling-Party-NP	Char[16]	Indicates the number portability prefix of the calling number. It refers to the routing prefix of the calling number after the calling party transfers from one network to another.
6	Called-Party-NP	Char[16]	Indicates the number portability prefix of the called number. It refers to the routing prefix of the called number after the called party transfers from one network to another.
7	SSP-Time	Char[14]	Indicates the time reported by the SSP. Format: YYYYMMDDhhmmss
8	MF-Indicator	Integer[4]	Indicates the indicator of called forwarding. The options are as follows: • 0: not forward • 1: forward
9	Real-Called-Number	Char[20]	Indicates the real called number.
10	Connect-Called-Number	Char[20]	Indicates the called number for connection. For example, when an IP address is preset, the value of **Connect-Called-Number** is the IP prefix + the called number.
11	Charge-Flow-Type	Integer[4]	Indicates the charge process type. The options are as follows: • 0: the calling process • 1: the called process • 3: the forwarding process
12	Calling-Partys-Category	Integer[4]	Indicates the category of the calling party. Example: a carrier, a public call office, or a user. The value of the field **Calling-Partys-Category** depends on the value of the variable **CAP_InitialDP_callingPartysCategory**.
13	Calling-Vlr-Number	Char[10]	Indicates the VLR number of the visitor location of the calling party. If the number exists in the IDP message, enter that; otherwise, the value is null. The field can be null in the called process and forwarding process.

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SN	Field Name	Field Type	Description
14	Calling-CellID	Char[36]	Indicates the cell ID of the calling party. The field can be null in the called process and forwarding process.
15	Called-Vlr-Number	Char[10]	Indicates the VLR number of the visitor location of the called party. The field can be null in the called process and forwarding process.
16	Called-CellID	Char[36]	Indicates the cell ID of the called party. The field can be null in the called process and forwarding process.
17	Original-Called-Party-Id	Char[20]	Indicates the original called number. • In the forwarding process, the MSC reports the original called number. • In the calling process, the MSC does not report the original called number. Therefore, the field is null.
18	Additional-Calling-Party-Number	Char[20]	Indicates the additional calling number.
19	Bearer-Capability	Char[10]	Indicates the bearer capacity of subscriber connection.
20	EventType-BCSM	Integer[4]	Indicates the event report.
21	Redirecting-Party-Id	Char[20]	Indicates the redirected number in the forwarding process. It is the calling number in the forwarding process.
22	Redirection-Information	Char[36]	Indicates the redirection information. It is used in the forwarding process.
23	Redirection-Reason	Integer[4]	Indicates the forwarding reason. The options are as follows: • 0: not forward • 1: forward conditionally • 2: forward on the busy mobile • 3: forward unconditionally • 6: forward on no reply
24	UE-IMSI	Char[20]	Indicates the IMSI of the calling party or called party.
25	CallReferenceNumber	Char[128]	Indicates the reference number of a call. It is used to uniquely identify a call. The options are as follows: • For the CDMA network, obtain the value of **BillID** in the ORREQ message for this

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SN	Field Name	Field Type	Description
			field. • For the GSM network, obtain the value of **CallReferenceNumber** that is reported in the IDP signaling for this field. • For the NGN and IMS networks, obtain the value of **CallID** that is reported in the SIP signaling for this field. • If the system cannot obtain the required information (excluding the preceding conditions), the value of this field is in the format *YYYYMMDDhhmmss (14 digits) + SCPID (3 digits) + automaton number* (4 digits). **NOTE** If the length of the automaton number is less than four digits, prefix 0 for the automaton number. If the length of the automaton number is more than four digits, obtain the last four digits for the automaton number.
26	MSC-Address	Char[20]	Indicates the MSC address.
27	Time-Zone	Integer[4]	Indicates the time zone of the area where a subscriber is.
28	CallBeginTime	Char[14]	Indicates the time of the calling beginning. Format: YYYYMMDDhhmmss
29	CallDuration	Integer[4]	Indicates the call duration. Unit: second
30	SuccessACDuration	Integer[4]	Indicates the duration of which the fee has been deducted. Unit: second
31	FailACDuration	Integer[4]	Indicates the AC duration during which an error occurs. Unit: second
32	VPMNCallFlag	Integer[4]	Indicates the flag of calls in a VPN group.
33	CBEResultCode	Integer[4]	Indicates the error code returned by the CBE. In other cases, the value of this field is 0.
34	DDS-Discount	Integer[4]	Indicates the discount value of the MTN zone.
35	DiameterSessionID	Char[128]	Indicates a diameter session ID. The format is as follows: <DiameterIdentity>;<high 32 bits>;<low 32 bits>[;<optional value>] • <DiameterIdentity>: the same as the field

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SN	Field Name	Field Type	Description
			in the Origin-Host AVP. • <high 32 bits>: indicates the current time of the system. • <low 32 bits>: indicates the cycle count-up. The initial value is **0**. When you restart the system, set the value to **0**. • <optional value>: reserved. Separate the preceding fields with the character ";". For example: SCP001.huawei.com;1876543210;523
36	NPPrefix	Char[10]	Indicates the NP prefix of a called number, that is, an RN prefix. **NOTE** • NP: Number Portable • RN: Routing Number
37	CalledHomeNetWork	Char[10]	Indicates the home network code of a called number after the NP.
38	OpposeTenantID	Integer	Reserved.

3.2.4 w_vou CDR

Description

w_vou indicates the CDR generated when recharging accounts using recharge cards succeed.

Generation Scenario

In the OCS service, when subscriber A recharge accounts using ordinary recharge cards, the SCP generates a w_vou CDR after the successful recharging.

CDR File Name Format

w_vouYYYYMMDD_SCPIDBBCC_Seqno.unl

Example of a w_vou CDR file name: w_vou20071012_1820001_0001.unl

Format

Table 3-17 describes each field of w_vou CDR file.

Table 3-17 Format of w_vou CDR

SN	Field Name	Field Type	Description

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SN	Field Name	Field Type	Description
1	MSISDN	Char[16]	Indicates the recharged phone number. • The number format is as follows: **Country code + MSISDN** The number format is the format of the mobile phone number.
2	SubCosID	Integer[4]	Indicates the sub brand ID of a subscriber.
3	CallingPartyNumber	Char[32]	Indicates the phone number of a subscriber who recharges an account. • The number format is as follows: **Country code + MSISDN** The number format is the format of the mobile phone number.
4	CalledPartyNumber	Char[32]	Indicates the called party number. • During IVR recharge, the value is the access code of the IVR flow (with Country code). • During SMS recharge, the value is blank. • During USSD recharge, the value is the USSD string entered by a subscriber (in clear format).
5	Calling-Partys-Category	Integer[4]	Indicates the type of the calling party.
6	Calling-Vlr-Number	Char[32]	Indicates the VLR number of the calling party.
7	CallingCellID	Char[32]	Indicates the cell ID of the calling party.
8	UE-IMSI	Char[32]	Indicates the terminal of the IMSI of the calling party.
9	MSC-Address	Char[32]	Indicates the MSC address.
10	GMSC-Address	Char[32]	Indicates the GMSC address.
11	Time-Zone	Integer[4]	Indicates the time zone of the area where a subscriber is.
12	Charge-Number	Char[32]	Indicates the recharged phone number. • The number format is as follows: **Country code + MSISDN** The number format is the format of the mobile phone number.
13	CardCosID	Integer[4]	Indicates the ID of the card type.
14	Recharge Type	Integer[4]	Indicates the recharge type.

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SN	Field Name	Field Type	Description
			The options are as follows: • 0: the first activation • 1: Voucher recharge through IVR from other phone • 2: Voucher recharge through IVR from self phone • 3: Voucher recharge through SMS from other phone • 4: Voucher recharge through SMS from self phone • 5: Voucher recharge through USSD from other phone • 6: Voucher recharge through USSD from self phone • 7: Voucher recharge by operator/external system (manual recharge) • 999: EVC recharge
15	BatchNumber	Char[11]	Indicates the batch number of a recharge card.
16	SerialNumber	Char[16]	Indicates the serial number of a recharge card.
17	Card Number	Char[20]	Indicates the recharge card number in clear format.
18	FaceValue	Integer[4]	Indicates the face value of a recharge card.
19	ValidityPeriod	Integer[4]	Indicates the validity period of a recharge card. Unit: day
20	TradeTime	Char[14]	Indicates the recharge time. Format: YYYYMMDDhhmmss
21	OperatedBy	Char[32]	Indicates the mobile phone number used to perform the recharge operation. • When the recharge is performed through IVR, SM, or USSD, the value of the field is the mobile phone number used to perform the recharge. • When the recharge is performed at the customer service center, the value of the field is the ID of the operator at the customer service center.
22	RechargeResult	Integer[4]	Indicates the result of normal recharge. The options are as follows:

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SN	Field Name	Field Type	Description
			• 1: Resetting the recharge card and requesting recharge confirmation succeed. • 2: Resetting the recharge card succeeds but requesting recharge confirmation fails. The CBE returns the failure result. • 3: Resetting the recharge card fails. The VC returns the failure result. Requesting recharge cancellation succeeds. • 4: Resetting the recharge card fails. The VC returns the failure result. Requesting recharge cancellation fails. The CBE returns the failure result. • 5: The StreamWide VC resets the recharge card successfully, but the initial recharge request fails (the CBE returns the failure result). • 6: The StreamWide VC resets the recharge card successfully, but the recharge confirmation request fails (the CBE returns the failure result). • 7: Authenticating the recharge card fails. 0, Voucher card is not valid or not exist • 8: Authenticating the recharge card fails. 2, Voucher card is expired or used up • 9: Authenticating the recharge card fails. 3, Voucher card is inactive • 10: Authenticating the recharge card fails. 4, Voucher card is used up • 11: Authenticating the recharge card fails. 5, Voucher card is locked • 12: Authenticating the subscriber fails. The subscriber does not exist. • 13: Authenticating the subscriber fails. The subscriber state is incorrect.
23	UniformSerialNo	Char[32]	Indicates a unified sequence number. For the manual recharge initiated by the business system, it indicates the sequence number of the external system. For the card recharge initiated by a subscriber through self service, the value of this field is null.
24	DiameterSessionID	Char[128]	Indicate the session id that generated during the interaction between the SCP and the CBP. The field includes parts as follows: <DiameterIdentity>;<high 32 bits>;<low 32 bits>[;<optional value>].

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SN	Field Name	Field Type	Description
			• <DiameterIdentity>: same as Origin-Host; • <high 32 bits>: the current day • <low 32 bits>: increase by degrees circulatory, the start value is 0. Set the value is 0 when the system restarts. • <optional value>: reserved. Every fields are separated by a character ";". Example: SCP001.huawei.com;1876543210;523

3.2.5 w_abu CDR

Description

w_abu indicates the CDR generated when errors occur in recharging accounts using recharge cards.

Generation Scenario

In the OCS service, when subscriber A recharge accounts using ordinary recharge cards, the SCP generates a w_abu CDR after the failing recharging.

CDR File Name Format

w_abuYYYYMMDD_SCPIDBBCC_Seqno.unl

Example of a w_abu CDR file name: w_abu20071012_1820001_0001.unl

Format

Table 3-18 describes each field of w_abu CDR file.

Table 3-18 Format of w_abu CDR

SN	Field Name	Field Type	Description
1	MSISDN	Char[16]	Indicates the recharged phone number. • The number format is as follows: **Country code + MSISDN** The number format is the format of the mobile phone number.
2	SubCosID	Integer[4]	Indicates the sub brand ID of a subscriber.
3	CallingPartyNumber	Char[32]	Indicates the phone number of a subscriber who recharges an account. • The number format is as follows:

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SN	Field Name	Field Type	Description
			Country code + MSISDN The number format is the format of the mobile phone number.
4	CalledPartyNumber	Char[32]	Indicates the called party number. • During IVR recharge, the value is the access code of the IVR flow (with Country code). • During SMS recharge, the value is blank. • During USSD recharge, the value is the USSD string entered by a subscriber (in clear format).
5	Calling-Partys-Category	Integer[4]	Indicates the calling subscriber type.
6	Calling-Vlr-Number	Char[32]	Indicates the VLR number of the calling subscriber.
7	CallingCellID	Char[32]	Indicates the cell ID of the calling party.
8	UE-IMSI	Char[32]	Indicates the terminal of the IMSI of the calling party.
9	MSC-Address	Char[32]	Indicates the MSC address.
10	GMSC-Address	Char[32]	Indicates the GMSC address.
11	Time-Zone	Integer[4]	Indicates the time zone of the area where a subscriber is.
12	Charge-Number	Char[32]	Indicates the recharged phone number. • The number format is as follows: **Country code + MSISDN** The number format is the format of the mobile phone number.
13	CardCosID	Integer[4]	Indicates the ID of the card service class.
14	Recharge Type	Integer[4]	Indicates the recharge type. The options are as follows: • 0: first activation • 1: recharge with another number through the IVR system • 2: recharge with the personal number through the IVR system • 3: recharge with another number through SM • 4: recharge with the personal number through SM • 5: recharge with another number through

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SN	Field Name	Field Type	Description
			USSD • 6: recharge with the personal number through USSD • 7: recharge at the customer service center • 999: EVC recharge
15	BatchNumber	Char[11]	Indicates the batch number of a recharge card.
16	SerialNumber	Char[16]	Indicates the serial number of a recharge card.
17	Card Number	Char[20]	Indicates the recharge card number in clear format.
18	FaceValue	Integer[4]	Indicates the face value of a recharge card.
19	ValidityPeriod	Integer[4]	Indicates the validity period of a recharge card. Unit: day
20	TradeTime	Char[14]	Indicates the recharge time. Format: YYYYMMDDhhmmss
21	OperatedBy	Char[32]	Indicates the mobile phone number used to perform the recharge operation. • When the recharge is performed through IVR, SM, or USSD, the value of the field is the mobile phone number used to perform the recharge. • When the recharge is performed at the customer service center, the value of the field is the ID of the operator at the customer service center.
22	RechargeResult	Integer[4]	Indicates the recharge exception result. The options are as follows: • 1: Resetting the recharge card succeeds but requesting recharge confirmation fails. (An error occurs in CBE network communication.) • 2: Resetting the recharge card succeeds. (The interconnection for resetting the recharge card times out and the interconnection for retrieving data fails.) Requesting recharge confirmation fails. (An error occurs in CBE network communication.) • 3: Resetting the recharge card succeeds. (The interconnection for resetting the recharge card times out and the interconnection for retrieving data fails.)

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SN	Field Name	Field Type	Description
			Requesting recharge confirmation succeeds. • 4: Resetting the recharge card succeeds. (The interconnection for resetting the recharge card times out and the interconnection for retrieving data fails.) Requesting recharge confirmation fails. (An error occurs in CBE network communication.) • 5: Resetting the recharge card fails. (The VC returns the failure result.) Requesting recharge cancellation fails. (An error occurs in CBE network communication.) • 6: Resetting the recharge card fails. (An error occurs in VC network communication.) Requesting recharge cancellation succeeds. • 7: Resetting the recharge card fails. (An error occurs in VC network communication.) Requesting recharge cancellation fails. (The CBE returns the failure result.) • 8: Resetting the recharge card fails. (An error occurs in VC network communication.) Requesting recharge cancellation fails. (An error occurs in CBE network communication.) • 9: The StreamWide VC resets the recharge card successfully, but the initial recharge request fails (an error occurs in the network communications with the CBE). • 10: The StreamWide VC resets the recharge card successfully, but the recharge confirmation request fails (an error occurs in the network communications with the CBE).
23	UniformSerialNo	Char[32]	Indicates a unified sequence number. For the manual recharge initiated by the business system, it indicates the sequence number of the external system. For the card recharge initiated by a subscriber through self service, the value of this field is null.
24	DiameterSessionID	Char[128]	Indicate the session id that generated during the interaction between the SCP and the CBP. The field includes parts as follows: <DiameterIdentity>;<high 32 bits>;<low 32

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SN	Field Name	Field Type	Description
			bits>[;<optional value>]. • <DiameterIdentity>: same as Origin-Host; • <high 32 bits>: the current day • <low 32 bits>: increase by degrees circulatory, the start value is 0. Set the value is 0 when the system restarts. • <optional value>: reserved. Every fields are separated by a character ";". Example: SCP001.huawei.com;1876543210;523

3.2.6 w_smr CDR

Description

The w_smr CDR records the bills of short message.

Generation Scenario

In the OCS service, when a subscriber sends or receives a short message, the SCP generates a w_smr CDR.

CDR File Name Format

w_smrYYYYMMDD_SCPIDBBCC_Seqno.unl

Example of a w_smr CDR file name: w_smr20071012_1820001_0001.unl

Format

Table 3-19 describes each field of w_smr CDR file.

Table 3-19 Format of w_smr CDR

SN	Field Name	Field Type	Description
1	ServiceKey	Integer[4]	Indicates the service key.
2	SubCosID	Integer[4]	Indicates the sub brand ID of a subscriber.
3	SM Center flag	Char[21]	Indicates the short message service center (SMSC).
4	SM-Id	Char[64]	Indicates the ID of an SM.
5	SM-Length	Integer[4]	Indicates the length of SM contents, which is used for charging traffics.
6	MO/MTflag	Char	Indicates the flag of calling numbers and

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SN	Field Name	Field Type	Description
			called numbers. The options are as follows: • 1: MO process • 2: MT process
7	CallingPartyNumber	Char[32]	Indicates the sender number. • The number format is as follows: **Country code + MSISDN** The number format is the format of the mobile phone number.
8	CalledPartyNumber	Char[32]	Indicates the receiver number. • The number format is as follows: **Country code + MSISDN** The number format is the format of the mobile phone number.
9	status_report_request	Integer[4]	Whether a status report is needed after a request message is sent. The options are as follows: • 0: fee deduction request • 1: no (A status report is not needed.) • 2: yes (A status report is needed.)
10	SMS-Receive-Time	Char[14]	Indicates the time when the system receives short messages. After receiving the refund message, the CBS rates and refunds based on the time that the SM is received.
11	send_result	Integer[4]	Indicates the result sent by the CBP. The options are as follows: • 0: requesting fee deduction • 1: confirming the successful sending of a message • 2: refunding the fee if sending a message fails
12	SP sms Flag	Integer[4]	Indicates the flag of an SP SM. The options are as follows: • 0: non SP • 1: SP
13	mo_msc_addr	Char[32]	Indicates the MSC address of the roaming location of the sender. The number has the country code, but without the international prefix.

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SN	Field Name	Field Type	Description
14	mt_msc_addr	Char[32]	Indicates the MSC address of the roaming location of the receiver. The number has the country code, but without the international prefix.
15	ThirdPartyNumber	Char[32]	Indicates the number of third-party fee deduction, which is contained in RCOMM messages. • The number format is as follows: **Country code + MSISDN** The number format is the format of the mobile phone number.
16	TerminationReason	Integer[4]	Indicates the reason why the call cannot be connected. The options are as follows: • 0: normal • 2: MAP MO, system aborting when sends TERMINATION_REQUEST to the CBE • 3: MAP MO, network aborting when SCP interacts with the CBE • 5: MAP MT, system aborting when the SCP sends TERMINATION_REQUEST to the CBE • 6: MAP MT, network aborting when the SCP interacts with the CBE • 7: CAP3, system aborting when the SCP sends TERMINATION_REQUEST to the CBE • 8: CAP3, network aborting when the SCP interacts with the CBE • 9: SMPP+ ,System abort when SCP send TERMINATION_REQUEST to CBE • 10: SMPP+ ,Network abort when SCP interact with CBE • 11: CAP2 , System abort when SCP send TERMINATION_REQUEST to CBE • 12: CAP2 , Network abort when SCP interact with CBE
17	SubDataFlag	Integer[4]	Indicates the flag of service data. The options are as follows: • 0: GSM • 1: PPS
18	ORG_IMSI	Char[20]	Indicates the IMSI of the calling party.
19	DEST_IMSI	Char[20]	Indicates the IMSI of the called party.

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SN	Field Name	Field Type	Description
20	DiameterSessionID	Char[128]	Indicates a diameter session ID. The format is as follows: <DiameterIdentity>;<high 32 bits>;<low 32 bits>[;<optional value>] • <DiameterIdentity>: the same as the field in the Origin-Host AVP. • <high 32 bits>: indicates the current time of the system. • <low 32 bits>: indicates the cycle count-up. The initial value is **0**. When you restart the system, set the value to **0**. • <optional value>: indicates implementation specific but may include a modem's device Id, a layer 2 address, timestamp, etc. Separate the preceding fields with the character ";". For example: SCP001.huawei.com;1876543210;523
21	NPPrefix	Char[10]	Indicates the NP prefix of a called number, that is, an RN prefix. **NOTE** • NP: Number Portable • RN: Routing Number
22	CalledHomeNetWork	Char[10]	Indicates the home network code of a called number after the NP.

3.2.7 w_smo CDR

Description

w_smo indicates the CDR generated in case of SM fee deduction failure.

Generation Scenario

In the OCS service, when a subscriber succeeds in sending SMs, but the CBP fails to deduct fees due to system error, the SCP generates a w_smo CDR. The CBE rates and deducts fees based on the CDR after the CBE recovers.

The fields contained in the **w_smo** CDR are the same as the fields contained in the DCC interface between the SCP and CBP. For the fields of which the corresponding information cannot be obtained, the values are null.

CDR File Name Format

w_smoYYYYMMDD_SCPIDBBCC_Seqno.unl

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Example of a w_smo CDR file name: w_smo20071012_1820001_0001.unl

Format

Table 3-20 describes each field of w_smo CDR file.

Table 3-20 Format of w_smo CDR

SN	Field Name	Field Type	Description
1	SmsType	Char[1]	Indicates the type of an SM. The options are as follows: • 1: SMPP+ and MAP3 SM • 3: CAP2 and CAP3 SM
2	MessageType	Char[1]	Indicates the message type. The options are as follows: • 1: a message for requesting fee deduction • 2: a confirmation message
3	SMSCAddress	Char[21]	Indicates the address of the SMSC.
4	SMID	UTF8String	Indicates the ID of an SM.
5	SMLength	Integer[4]	Indicates the length of an SM, which is used for charging traffics.
6	MoMSCAddr	UTF8String	Indicates the MSC address of the roaming location of the sender. The number has the country code, but without the international prefix.
7	MtMSCAddr	UTF8String	Indicates the MSC address of the roaming location of the receiver. The number has the country code, but without the international prefix.
8	SourceAddr	Char[32]	Indicates the sender number. • The number format is as follows: **Country code + MSISDN** The number format is the format of the mobile phone number.
9	DestAddr	Char[32]	Indicates the receiver number. • The number format is as follows: **Country code + MSISDN** The number format is the format of the mobile phone number.
10	FeeAddr	Char[32]	Indicates the charge number. • The number format is as follows: **Country code + MSISDN**

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SN	Field Name	Field Type	Description
			The number format is the format of the mobile phone number.
11	FeeFlag	Integer[4]	Indicates the charge flag. The options are as follows: • 0: To charge by the original method. The value 0 indicates an exception. • 1: To charge the calling number. • 2: To charge the called number. • 3: To charge the charge number.
12	FeeType	Integer[4]	Indicates the fee type. The options are as follows: • 0: To charge in the default mode. (For the common point-to-point SMs, the fee is configured in the CBS. For other SMs, fees must be charged based on the charging policy sent by the SMSC.) • 1: To charge by the number of SMs by the SP. • 2: To charge by months (deducting fees) by the SP. • 3: To charge by months (checking). • 4: To charge by upper limit based on the number of SMs. • 5: To charge the fee of monthly SM package with the limited number of SMs (deducting fees). • 6: To charge the fee of monthly SM package with the limited number of SMs (checking).
13	FeeItdMsgNum	Integer[4]	Indicates the number of SMs in a monthly SM package, of which the number of SMs is limited. (It is not realized.)
14	FeeSingle	Integer[4]	Indicates the charge rate of a single SM when the SP charges the fee by the number of SMs.
15	FeeFixed	Integer[4]	Indicates the charge rate of each SM that is contained in a monthly SM package. If the number of SMs exceeds the limited number, the charge rates of the excess SMs are counted based on the value of the **FeeSingle** field. The field **FeeFixed** is used when SMs are charged by months.
16	MessagePID	Integer[4]	Indicates the message PID such as a picture and voice.

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SN	Field Name	Field Type	Description
17	StatusReportRequest	Integer[4]	Whether a status report is needed after a request message is sent. The options are as follows: • 0: no (The status report is not needed.) • 1: yes (The status report is needed.)
18	OriginalNetType	Integer[4]	Indicates the type of the source network. The options are as follows: • 0: GSM • 1: CDMA • 2: TDMA • 3: SMPP
19	SMSReceiveTime	Time	Indicates the time that an SM is received. After receiving the refund message, the CBS rates and refunds based on the time that the SM is received. The field **SMSReceiveTime** is valid to fee deduction request messages. For confirmation messages, the value is null.
20	CallingService	OctetString	Indicates the calling service, an array of 16 bytes with each byte representing a type of services. If the value of a byte is **0**, it indicates no service. If the value of a byte is **1**, it indicates that the current message contains the service. The options are as follows: • CallingServcie[0]: SM signature • CallingServcie[1]: enhanced return notification of MO SM • CallingServcie[2]: notification of power-on
21	CalledService	OctetString	Indicates the called service, an array of 16 bytes with each byte representing a type of services. If the value of a byte is **0**, it indicates no service. If the value of a byte is **1**, it indicates that the current message contains the service. The options are as follows: • CallingServcie[0]: SM transfer • CallingServcie[1]: automatic reply
22	OriginalMsgID	OctetString	Indicates the ID of the original SM that is associated with the current message.
23	SMDSendResult	Integer[4]	The value of this field varies with the

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SN	Field Name	Field Type	Description
			SMPP+ message.
24	MsgID	OctetString	Indicates the message ID that the gateway generates. The field **MsgID** is valid to inter-network SMs and Monternet SMs. For intra-network SMs, the value is null.
25	GWID	OctetString	Indicates the ID of the gateway that sends the message. The field **GWID** is valid to inter-network SMs and Monternet SMs. For intra-network SMs, the value is null.
26	MsgSeq	OctetString	Indicates the SM sequence number generated by the SMSC. The field **MsgSeq** is valid to inter-network SMs and Monternet SMs. For intra-network SMs, the value is null.
27	SPID	OctetString	Indicates the SPID. The field **SPID** is valid to inter-network SMs and Monternet SMs. For intra-network SMs, the value is null.
28	ServiceCode	OctetString	Indicates the code of the service type. The field **ServiceCode** is valid to inter-network SMs and Monternet SMs. For intra-network SMs, the value is null.
29	SourceGWID	OctetString	Indicates the ID of the gateway that forwards the SM. The field **SourceGWID** is valid to inter-network SMs and Monternet SMs. For intra-network SMs, the value is null.
30	PriorityFlag	Integer[4]	Indicates the priority level of an SM. The field **PriorityFlag** is valid to inter-network SMs and Monternet SMs. For intra-network SMs, the value is null.
31	ChargingRate	Integer[4]	Indicates the charge rate. The field **ChargingRate** is valid to inter-network SMs and Monternet SMs. For intra-network SMs, the value is null.
32	ChargingType	Integer[4]	Indicates the charge rate class, used in Monternet SMs and inter-network SMs. The options are as follows: • 0: No charge. Only the channel fee of the SP is deducted. • 1: To charge for free.

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SN	Field Name	Field Type	Description
			• 2: To charge by the number of SMs. • 3: To charge by months. • 4: To charge by upper limit. For inter-network SMs, the value is **2**. For intra-network SMs, the value is null.
33	CDRType	Integer[4]	Indicates the CDR type, used in Monternet SMs and inter-network SMs. For inter-network SMs, the options are as follows: • 11: the MO CDR when an inter-network SM interconnects with an SM of China Unicom • 12: the MT CDR when an inter-network SM interconnects with an SM of China Unicom For inter-network SMs, the value is **11**. For Monternet SMs, the options are as follows: • 1: an MO CDR • 2: an MT CDR • 3: monthly SM control For intra-network SMs, the value is null.
34	Discount	Integer[4]	Indicates the discount, a percent. The value **100** indicates no discount. The field **Discount** is valid to inter-network SMs and Monternet SMs. For intra-network SMs, the value is null.
35	GiftResult	Integer[4]	The default value of this field is null.
36	Subscription-Id-Data	Char[32]	Indicates the end user number.
37	DiameterSessionID	Char[128]	Indicates a diameter session ID. The format is as follows: <DiameterIdentity>;<high 32 bits>;<low 32 bits>[;<optional value>] • <DiameterIdentity>: the same as the field in the Origin-Host AVP. • <high 32 bits>: indicates the current time of the system. • <low 32 bits>: indicates the cycle count-up. The initial value is **0**. When you restart the system, set the value to **0**. • <optional value>: indicates implementation specific but may include

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SN	Field Name	Field Type	Description
			a modem's device Id, a layer 2 address, timestamp, etc. Separate the preceding fields with the character ";". For example: SCP001.huawei.com;1876543210;523
38	NPPrefix	Char[10]	Indicates the NP prefix of a called number, that is, an RN prefix. **NOTE** • NP: Number Portable • RN: Routing Number
39	CalledHomeNetWork	Char[10]	Indicates the home network code of a called number after the NP.
40	OpposeTenantID	Integer	Reserved

3.2.8 w_kpi CDR

Description

The w_kpi CDR records the information about various operations, and the information is the data source of the statistic feature. The w_kpi CDR is generated regardless of operation success or failure.

Generation Scenario

In the CBS service, when multiple operations or service processes are triggered, such as the calling and called processes, or query operations, the SCP generates a w_kpi CDR.

CDR File Name Format

w_kpiYYYYMMDD_SCPIDBBCC_Seqno.unl

Example of a w_kpi CDR file name: w_kpi20071112_1020001_0001.unl

Format

Table 3-21 describes each field of w_kpi CDR file.

Table 3-21 Format of w_kpi CDR

SN	Field Name	Field Type	Description
1	OperateType	Integer[4]	Indicates the type of operation. The options are as follows: • 1: MO call

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SN	Field Name	Field Type	Description
			• 2: MT call • 3: SM MO call (authentication) • 4: SM MT call (authentication) • 6: SM rolling back MT call • 7: IVR recharge from the main management flow • 8: IVR query • 10: Calls connecting SSP and Leg A and Leg B • 11: System short message • 12: Callscreen management processes • 13: Recharging through USSD • 14: Querying through USSD • 17: SM transfer • 19: Prompt of SMS in voice query • 20: MCN management processes • 21: IVR fn management processes • 22: SMS fn management processes • 23: USSD fn management processes • 24: USSD/IVR/SM transfer money management flow • 25: Homezone management processes • 29: IVR transfer money • 31: EVC recharge • 32: ICF number management processes • 33: USSD first activate • 34: SMS management process • 35: USSD management process • 39: SMS First Activate • 43: Call me back service • 47: Purchasing packages flexibly • 51: AOC • 52: Parent and children cards service management process • 55: Switching main packages through IVR • 61: USSD Loan • 62: IVR Loan • 65: Point exchange for other resources by IVR • 66: Point exchange for other resources by

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SN	Field Name	Field Type	Description
			USSD • 67: Query point by USSD • 79: Query accumulator by USSD • 80: Query accumulator by SMS • 83: IVR modify the subcos id • 84: SMS modify the subcos id • 85: USSD modify the subcos id • 99: Others (identify the abnormal process termination before an operation, including analyzing operation type failure during SMS access, and failure due to multiple input failure in the IVR process, etc.) • 143: SMS modify COS • 144: USSD modify COS • 151: Query accumulator by IVR • 156: Subscribe bucket by SMS • 157: Activate subscriber through EVC top up • 158: Activate subscriber through RCVS/VC top up • 159: Activate subscriber by making first call • 160: Other management flows • 161: Modify BrandID through IVR • 162: Modify BrandID through USSD • 163: Modify BrandID through SMS • 164: Activate subscriber through IVR RCVS/VC top up by another subscriber • 165: Activate subscriber through SMS RCVS/VC top up by another subscriber • 166: IVR Modify COS • 167: Enable AOC function by USSD • 168: Disable AOC function by USSD • 169: Enable AOC function by IVR • 170: Disable AOC function by IVR • 171: MCN Process • 172: Sponsor service • 173: Manage self-help IVR transfer function • 174: Point exchange for other resource by SMS

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SN	Field Name	Field Type	Description
			• 196: IVR recharge by dialing a direct access code **NOTE** If the system does not distinguish IVR recharge modes, value option 196 is invalid.
2	OperateResult	Integer[4]	Indicates the operation result. The options are as follows: • 0: Success • 1: Insufficient balance • 2: Incorrect account status • 3: Invalid called number • 4: Barred call • 5: Call authority restriction • 6: System error • 7: Route selection failure • 8: The called having no answer • 9: The called being busy • 10: The caller abandoning the call • 11: The called abandoning the call • 12: Wrong input of the subscriber • 13: Recharge card resetting failure • 14: Network connection error • 15: Over maximum call duration • 16: Balance used up • 17: First connection to the CBE being abnormal (service process ends) • 18: Failed service fee deduction on the CBE (service process continues) • 19: Failed service fee deduction on the CBE (service process ends) • 20: First connection to the CBE being abnormal (service is passed by) • 22: Signal exceptions • 23: Failed voice playing during the call process • 24: Failed CBP authentication or fee deduction which results in service logic interruption • 25: Key data error when the SCP is queried • 26: Default error of a service. • 27: The ACR is not reported during the

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SN	Field Name	Field Type	Description
			switch • 28: Failed to lock or unlock a number • 33: Lack of accumulation for SM transfer • 42: Operation before a sponsor announcement • 43: Operation after playing a sponsor announcement • 44: Cancelled by subscriber • 45: The subscriber inputs times out • 49: No matching record in 'Manage service information' [Table name: w_uvs_service] • 50: No matching record in 'Manage recharge service information' [Table name: w_uvs_rechgservice] • 51: No matching record in 'Manage main product switch information' [Table name: w_uvs_switchsubcos] • 52: The CBE has not received the recharge request • 53: The current location of the subscriber does not support this operation • 59: Incorrect IMSI received • 99: Others
3	OperateTime	Char[6]	Indicates the operation time. Format: hhmmss
4	CallingParty	Char[32]	Indicates the calling number.
5	CalledParty	Char[32]	Indicates the called number. **NOTE** • For the operation of recharging a personal account, the value of this parameter is the management process access code. • For the operation of recharging an account of another subscriber, the value of this parameter is the number of the recharged subscriber. • For the balance transfer operation, the value of this parameter is the number of the transfer-in subscriber. • For the missing claim operation, the value of this parameter is the number which is claimed missing.

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3.3 BMP-Generated CDR

The section describes the file name format of BMP-generated CDR of each type, BMP-generated CDR storage directories and BMP-generated CDR generation scenarios.

3.3.1 General Description

The generated CDR is stored in the BMP node. The directory for storing CDR files cannot be configured.

- Temporary directory: **${HOME}/cdr/output/temp**
- Destination directory: **${HOME}/cdr/output**
- Backup directory: **${HOME}/cdr/output/bak**

For description about the CDR storage directories, see 1.2 CDR Storage Directories.

3.3.2 fns CDR

Description

When a subscriber adds or modifies a familiarity number (FN) successfully, the Business Management Point (BMP) in the online charging system (OCS) generates an fns call detail record (CDR).

Generation Scenario

The BMP generate an fns CDR when:

- The subscriber changes the FN from A to B.
- The subscriber adds an FN.

CDR File Name Format

fnsYYYYMMDDhhmmss_BMPID_Seqno.unl

Example: fns20071012235959_182_000001.unl

Format

Table 3-22 describes the fields in an fns CDR file.

Table 3-22 Format of fns CDR

SN	Field Name	Field Type	Description
1	MSISDN	Char[20]	Indicates the mobile phone number of a subscriber.
2	SubCosID	Integer[4]	Indicates the main product of the subscriber.
3	NewFnNumber	Char[20]	Indicates the new FN. To add multiple FNs, separate the FNs with a comma (;).

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SN	Field Name	Field Type	Description
4	OldFnNumber	Char[20]	Indicates the old FN. This field can be blank if an FN is added. The new FN must match the old FN when a subscriber change the old FN.
5	Fee	Integer[4]	Indicates the fee for changing or adding an FN.
6	OperTime	Char[14]	Indicates the operation time.
7	FreeTimes	Integer[4]	Indicates the remaining number of free operation times for the free resource account.

3.3.3 prod CDR

Description

When a subscriber subscribes to a product, renews a product, or adjusts the validity period of a product, the BMP generates a prod CDR to record the operation result (success or failure).

Generation Scenario

The BMP generate a prod CDR when:

- Product subscription
- Product renewal
- Product validity period adjustment

CDR File Name Format

manproductYYYYMMDDhhmmss_BMPID_Seqno.unl

Example: manproduct20120711091910_1_002001.unl

Format

Table 3-23 describes the fields in a prod CDR file.

Table 3-23 Format of prod CDR

SN	Field Name	Field Type	Description
1	ISDN	Char[18]	Indicates the mobile phone number of a subscriber.
2	SubscriberKey	Integer[4]	Indicates a subscriber key.
3	ProductKey	Integer[8]	Indicates a product key.

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SN	Field Name	Field Type	Description
4	ProductOrderKey	Integer[8]	Indicates a product subscription key.
5	OpertorID	Char[18]	Indicates an operator ID.
6	TransactionID	Char[24]	Indicates a transaction ID.
7	OperTime	Char[14]	Indicates the time when an operation is performed. The format is YYYYMMDDhhmmss.
8	ApplyTime	Char[14]	Indicates the time when a subscription relationship takes effect. The format is YYYYMMDDhhmmss.
9	ExpireTime	Char[14]	Indicates the time when a subscription relationship expires. The format is YYYYMMDDhhmmss.
10	ServiceType	Char[32]	Indicates the service included in the product.
11	Status	Char[8]	Indicates an operation type in a CDR. The options are as follows: • 0: New Subscription • 1: Duplicate subscriber to extend validity • 2: Modify expire time directly
12	ResultCode	Integer[4]	Indicates an operation type in a CDR.
13	Description	Char[256]	Indicates the description for a subscription result.

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4 CDR Instances

About This Chapter

The section describes the CDRs with instances.

The following lists the contents of this chapter.

4.1 CBP-Generated CDR Instance-rec CDR

The section describes the rec CDR with an instance.

4.2 SCP-Generated CDR Instance-w_abr CDR

The section describes the w_abr CDR with an instance.

4.3 BMP-Generated CDR Instance fns CDR

The section describes the instance of a Generated fns CDR.

4.1 CBP-Generated CDR Instance-rec CDR

The section describes the rec CDR with an instance.

Service Usage Scenario

Assume that John in country A calls Jack in the same country. The mobile phone number of John is 8765432. The mobile phone number of Jack is 7654321. The country code of country A is 92. Jack receives the call from John at 2008-06-05 14:06:06.

CDR Details

- Original CDR record

706|0|20141030204652|1|2567829901105|2567829901106|7829901105123||0025678299
01106|002567829901106|1|00|0025677|0005623071200000|0025677||20111011152700|3
2|0|0|20141030204622|0|30|1|888888||30||||1|17612|7829901105|1|0|20141003|0|0|0|0||256
|77|1|256|77|1|256|77|1|256|77|1|20170769|10000000000000000000000000|0|0|0||0|0|1000
000|110162|2||0|0|3|3|899971|0|0|110171|110171|0||2000|1|3|899971|0|0|0|0|0|0|0|0|0|0|0|
0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|||||||||||||||||||||0||||||20141001000000||||0|0|0|0|0|0|0||
0|0||0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|0|||||-1|-1||0|0|7829901106|0||1000|scp102;14146732
11;6;1|0|0|-1|0|3|0||||1||30|110162|15,1414673182,3;3420,1414673182,30;|1111|0|0|99900

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0000000116268,20170769,1,0,3,2000,101,20141003;|0|0|0|0|0025677|69|000000000000000|

- CDR field description

Table 4-1 Format of rec CDR

SN	Value	Field Name	Description
1	10	SerialNo	Indicates that the serial number is **10**.
2	0	SubSequence	Indicates that the sequence number is **0** when the bill is split for mid-night and end of the bill cycle.
3	20080605140606	TimeStamp	Indicates that the timestamp of the bill item generation is **20080605140606**.
4	1	ServiceKey	Indicates that the service key is **1**.
5	928765432	CallingPartyNumber	Indicates that the calling number is **928765432**.
6	927654321	CalledPartyNumber	Indicates that the called number is **927654321**.
7	-	CallingPartyIMSI	Indicates that the relevant value cannot be obtained.
8	1	CalledPartyIMSI	Indicates that the IMSI of the called party is **1**.
9	-	DialedNumber	Indicates that the original called number does not exist. That is, the called party is not forwarded.
10	927654321	OriginalCalledParty	Indicates that the original called number is **927654321**.
11	2	ServiceFlow	Indicates that the call process is called process.
12	00	CallForwardIndicator	Indicates that the call forwarding flag is **00**.
13	92993909	CallingRoamInfo	Indicates that the information used to analyze the roaming state of calling party is **92993909**.
14	1050	CallingCellID	Indicates that the cell ID of the calling party is **1050**.
15	929675020	CalledRoamInfo	Indicates that the information used to analyze the roaming state of called party is **929675020**.
16	1010	CalledCellID	Indicates that the cell ID is **1010**.
17	20080605140623	TimeStampOfSSP	Indicates that the time stamp report from SSP in IDP is **20080605140623**.
18	0	TimeZoneOfSSP	Indicates that the time zone report from SSP in IDP is **0**.
19	0	BearerCapability	Indicates that the beared service is reserved.

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SN	Value	Field Name	Description
20	Reserved		
21	20080605 140829	ChargingTime	Indicates that the time stamp of the charging beginning is **20080605140829**.
22	0	WaitDuration	Indicates that the wait duration for calling party is **0**.
23	420	CallDuration	Indicates that the call duration is **420**.
24	0	TerminationReason	Indicates that the termination reason of the call is **unknown**.
25	3146	CallReference Number	Indicates that the call reference number is **3146**. It is reported by the SSP and is used to synchronized CDRs generated on both the IN and the SSP.
26	-	IMEI	Indicates that the IE containing the IMEISV of the ME in use by the served subscriber is null.
27	420	ChargeDuration	Indicates that the charge duration is **420**.
28	00928765 432	AccessPrefix	Indicates that the special prefix dialed by a subscriber for making calls is **00928765432**.
29	-	RoutingPrefix	Indicates that the route prefix for re-routing a call is null.
30	0	RedirectingPartyID	Indicates that the redirection number in the forwarding process is **0**.
31	108	BrandID	Indicates that the brand ID of subscribers is **108**.
32	10062	SubCOSID	Indicates the main product ID of subscribers is **10062**.
33	7654321	ChargingParty Number	Indicates that the telephone number of the charging party is **7654321**.
34	2	ChargePartyIndicator	Indicates that the called party is charged.
35	0	PayType	Indicates that the pay type of subscribers is **prepaid**.
36	20080502	BillCycleID	Indicates that the bill cycle ID is **20080502**.
37	3	ChargingType	Indicates that the charging type is **H-V**.
38	2	CallType	Indicates that the analyzed call type is toll call between charging area.
39	2	RoamState	Indicates that the roaming state of charging party is roaming out of charging area.
40	0	ResultCode	Indicates that the internal result code is **0**.

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SN	Value	Field Name	Description
41	-	FPHPrefix	Indicates the prefix for called party payment is unknown.
42	92	CallingHomeCountryCode	Indicates that the home country code information of calling party is **92**.
43	11	CallingHomeAreaNumber	Indicates that the home area number of the calling party is **11**.
44	2	CallingHomeNetworkCode	Indicates that the home network code of the calling party is **2**.
45	92	CallingRoamCountryCode	Indicates that the roam country code of the calling party is **92**.
46	11	CallingRoamAreaNumber	Indicates that the roam area number of the calling party is **11**.
47	2	CallingRoamNetworkCode	Indicates that the roam network code of the calling party is **2**.
48	92	CalledHomeCountryCode	Indicates that the home country code of the called party is **92**.
49	10	CalledHomeAreaNumber	Indicates that the home area number of the called party is **10**.
50	1	CalledHomeNetworkCode	Indicates that the home network code of the called party is **1**.
51	92	CalledRoamCountryCode	Indicates the roam country code of the called party is **92**.
52	11	CalledRoamAreaNumber	Indicates that the roam area number of the called party is **11**.
53	1	CalledRoamNetworkCode	Indicates that the roam network code of the called party is **1**.
54	10062	ProductID	Indicates that the last effect rating product is **10062**.
55	1000000000000000000	ServiceType	Indicates that the normal voice tariff.
56	0	HotLineIndicator	Indicates that the special number type is predefined hot line type.
57	0	HomeZoneID	Indicates that the home zone ID is **0**.
58	0	NPFlag	Indicates that the number portability is not enabled.
59	-	NPPrefix	Indicates that the calling number does not contain the number portability prefix. That is, the calling party is not enabled with the number portability function.

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SN	Value	Field Name	Description
60	0	CallingGroupNo	Indicates that the group number of the calling party is **0** when enjoying CUG service. It is the group number related to the rating.
61	0	CalledGroupNo	Indicates that the group number of the called party is **0** when enjoying CUG or VPN service.
62	1000000	UserState	Indicates the life cycle state and the management state of a subscriber is active.
63	66	SubscriberID	Indicates that the subscriber ID is **66**.
64	2	OpposeNetworkType	Indicates that the network type of oppose side is **Mobile**.
65	Reserved		
66	0	ChargeOfItemsAccounts	Indicates that the free call times used is **0**.
67	0	ChargeOfDurationAccounts	Indicates that the total call time charge using reword resource is **0**.
68	70	ChargeOfFundAccounts	Indicates that the total cost amount charging from fund account is **70**.
69	70	ChargeFromPrepaid	Indicates that the charge amount from prepaid account is **70**.
70	9930	PrepaidBalance	Indicates that the balance of prepaid account is **9930**.
71	0	ChargeFromPostpaid	Indicates that the charge amount from postpaid account is **0**.
72	0	PostPaidBalance	Indicates that the balance of postpaid account is **0**.
73	710101000	AccountID	Indicates that the ID of the main account is **710101000**.
74	2147483647	AccountKey	Indicates the key of the main account is **2147483647**.
75	0	CurrencyCode	Indicates that the system use local minimal currency as default.
76	Reserved		
77	2000	AccountType1	Indicates that the account type of the first account is **2000**.
78	68	FeeType1	Indicates that the fee type of the first account is **68**.
79	70	ChargeAmount1	Indicates that the changing amount of the first account is **70**.
80	9930	CurrentAcctA	Indicates that the current amount of the first

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SN	Value	Field Name	Description
		mount1	account is **9930**.
81	0	AccountType2	Indicates that the account is not used.
82	0	FeeType2	Indicates that the fee type is not used.
83	0	ChargeAmount 2	Indicates that the changing amount of the second account is **0**.
84	0	CurrentAcctA mount2	Indicates that the current amount of the second account is **0**.
85	0	AccountType3	Indicates that the account is not used.
86	0	FeeType3	Indicates that the fee type is not used.
87	0	ChargeAmount 3	Indicates that the changing amount of the third account has not been changed.
88	0	CurrentAcctA mount3	Indicates that the current amount of the third account is **0**.
89	0	AccountType4	Indicates that the account is not used.
90	0	FeeType4	Indicates that the fee type is not used.
91	0	ChargeAmount 4	Indicates that the changing amount of the fourth account has not been changed.
92	0	CurrentAcctA mount4	Indicates that the current amount of the fourth account is **0**.
93	0	AccountType5	Indicates that the account is not used.
94	0	FeeType5	Indicates that the fee type is not used.
95	0	ChargeAmount 5	Indicates that the changing amount of the fifth account has not been changed.
96	0	CurrentAcctA mount5	Indicates that the current amount of the fifth account is **0**.
97	0	AccountType6	Indicates that the account is not used.
98	0	FeeType6	Indicates that the fee type is not used.
99	0	ChargeAmount 6	Indicates that the changing amount of the sixth account has not been changed.
100	0	CurrentAcctA mount6	Indicates that the current amount of the sixth account is **0**.
101	0	AccountType7	Indicates that the account is not used.
102	0	FeeType7	Indicates that the fee type is not used.
103	0	ChargeAmount 7	Indicates that the changing amount of the seventh account has not been changed.
104	0	CurrentAcctA	Indicates that the current amount of the seventh

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SN	Value	Field Name	Description
		mount7	account is **0**.
105	0	AccountType8	Indicates that the account is not used.
106	0	FeeType8	Indicates that the fee type is not used.
107	0	ChargeAmount8	Indicates that the changing amount of the eighth account has not been changed.
108	0	CurrentAcctAmount8	Indicates that the current amount of the eighth account is **0**.
109	0	AccountType9	Indicates that the account is not used.
110	0	FeeType9	Indicates that the fee type is not used.
111	0	ChargeAmount9	Indicates that the changing amount of the ninth account has not been changed.
112	0	CurrentAcctAmount9	Indicates that the current amount of the ninth account is **0**.
113	0	AccountType10	Indicates that the account is not used.
114	0	FeeType10	Indicates that the fee type is not used.
115	0	ChargeAmount10	Indicates that the changing amount of the tenth account has not been changed.
116	0	CurrentAcctAmount10	Indicates that the current amount of the tenth account is **0**.
117	-	BonusValidity1	Indicates that the validity period of the first rewarded resource of the second party is null.
118	-	BonusValidity2	Indicates that the validity period of the second rewarded resource of the second party is null.
119	-	BonusValidity3	Indicates that the validity period of the third rewarded resource of the second party is null.
120	-	BonusValidity4	Indicates that the validity period of the fourth rewarded resource of the second party is null.
121	-	BonusValidity5	Indicates that the validity period of the fifth rewarded resource of the second party is null.
122	-	BonusValidity6	Indicates that the validity period of the sixth rewarded resource of the second party is null.
123	-	BonusValidity7	Indicates that the validity period of the seventh rewarded resource of the second party is null.
124	-	BonusValidity8	Indicates that the validity period of the eighth rewarded resource of the second party is null.
125	-	BonusValidity9	Indicates that the validity period of the ninth rewarded resource of the second party is null.

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SN	Value	Field Name	Description
126	-	BonusValidity10	Indicates that the validity period of the tenth rewarded resource of the second party is null.
127	-	CallingVPNTopGroupNumber	Indicates the top group number of the calling party.
128	-	CallingVPNGroupNumber	Indicates the group number of the calling party.
129	-	CallingVPNShortNumber	Indicates the short number of the calling party.
130	-	CalledVPNTopGroupNumber	Indicates the top group number of the called party.
131	-	CalledVPNGroupNumber	Indicates the group number of the called party.
132	-	CalledVPNShortNumber	Indicates the short number of the called party.
133	-	CallingNetworkType	Indicates the type of the network accessed by the calling party.
134	-	CalledNetworkType	Indicates the type of the network accessed by the called party.
135	-	GroupCallType	Indicates the call type when the charged subscriber is a VPN user.
136	-	GroupPayFlag	Indicates the payment mode of the group users.
137	-	OnlineType	Indicates online type.
138-141	Reserved		
142	-	AddtionalInfo	Indicates that the additional information about the service of the second party is null.
143	0	StartTimeOfBillCycle	Indicates that the start time of bill cycle of the second party is **0**.
144	-	SecondChargingPartyNumber	Indicates that the mobile number of the other charging party is null.
145	-	SecondChargingPartyPayType	Indicates that the payment type of the other charging party is null.
146	-	SecondChargingPartyBillCycleID	Indicates that the bill cycle ID of the other charging party is null.
147	-	ChargeOfItemsSecondAccounts	Indicates that the free call times used from others is null.

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SN	Value	Field Name	Description
148	-	ChargeOfDurationSecondAccounts	Indicates that the total call time charge using reword resource from others is null.
149	-	ChargeOfFundSecondAccounts	Indicates that the total cost amount charging from fund account of others is null.
150	12453	ChargeFromSecondPrepaidAccount	Indicates the charge amount from other account is **12453**.
151	0	SecondPrepaidAccountBalance	Reserved
152	12456	ChargeFromSecondPostpaidAccount	Indicates the charge amount from other account is **12456**.
153	-	SecondPostpaidAccountBalance	Indicates the charge amount from other postpaid account, such as the parent card.
154	-	SecondAccountID	Indicates that the main account ID of the other charging party that the system deducts fees from is null.
155	2147483623	SecondAccountKey	Indicates the main account key of the other charging party that the system deducts fees from is **2147483623**.
156	-	SecondAccountCurrencyCode	Indicates that the currency code of the second party is null.
157	Reserved		
158	0	SecondAccountType1	Indicates that the account type of the first account of the second party is **0**.
159	0	SecondFeeType1	Indicates that the fee type of the first account of the second party is **0**.
160	0	SecondChargeAmount1	Indicates that the changing amount of the first account of the second party is **0**.
161	0	SecondCurrentAcctAmount1	Indicates that the current amount of the first account of the second party is **0**.
162	0	SecondAccountType2	Indicates that the account type of the second account of the second party is **0**.
163	0	SecondFeeType2	Indicates that the fee type of the second account of the second party is **0**.
164	0	SecondChargeAmount2	Indicates that the changing amount of the second account of the second party is **0**.

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SN	Value	Field Name	Description
165	0	SecondCurrentAcctAmount2	Indicates that the current amount of the second account of the second party is **0**.
166	0	SecondAccountType3	Indicates that the account type of the third account of the second party is **0**.
167	0	SecondFeeType3	Indicates that the fee type of the third account of the second party is **0**.
168	0	SecondChargeAmount3	Indicates that the changing amount of the third account of the second party is **0**.
169	0	SecondCurrentAcctAmount3	Indicates that the current amount of the third account of the second party is **0**.
170	0	SecondAccountType4	Indicates that the account type of the fourth account of the second party is **0**.
171	-	SecondFeeType4	Indicates that the fee type of the fourth account of the second party is **0**.
172	0	SecondChargeAmount4	Indicates that the changing amount of the fourth account of the second party is **0**.
173	0	SecondCurrentAcctAmount4	Indicates that the current amount of the fourth account of the second party is **0**.
174	0	SecondAccountType5	Indicates that the account type of the fifth account of the second party is **0**.
175	0	SecondFeeType5	Indicates that the fee type of the fifth account of the second party is **0**.
176	0	SecondChargeAmount5	Indicates that the changing amount of the fifth account of the second party is **0**.
177	0	SecondCurrentAcctAmount5	Indicates that the current amount of the fifth account of the second party is **0**.
178	0	SecondBonusValidity1	Indicates that the validity period of the first rewarded resource of the second party is null.
179	0	SecondBonusValidity2	Indicates that the validity period of the second rewarded resource of the second party is null.
180	0	SecondBonusValidity3	Indicates that the validity period of the third rewarded resource of the second party is null.
181	0	SecondBonusValidity4	Indicates that the validity period of the fourth rewarded resource of the second party is null.
182	0	SecondBonusValidity5	Indicates that the validity period of the fifth rewarded resource of the second party is null.
183	0	CallingClusterNumber	Indicates that the calling cluster number is **0**.
184	0	CalledCluster	Indicates that the called cluster number is **0**.

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SN	Value	Field Name	Description
		Number	
185	0	OriginalCallingParty	Indicates that the called party is not forwarded.
186-187	Reserved		
188	212345678	OppositePartyNumber	Indicates the number of the opposite party is **212345678**.
189-191	Reserved		
192	0	DiameterSessionID	Indicates that the diameter session ID is null.
193	0	StartcountryGroup	Indicates the country group of a calling party is **0**.
194	0	EndcountryGroup	Indicates the country group of a called party is **0**.
195-203	Reserved		
204	100	DiscountOfLastEffProduct	Indicates the discount percentage for the last rated product that takes effect. For a product that does not provide discounts, the value is **100**.
205	0	subscriberkey	Indicates the key of a subscriber is **0**.
206	-	accumulationchange	Indicates the change value of an accumulative item is null.
207	1100	UsageServiceType	Indicates the type of a real-time service is **1100**.
208	0	SourceSerialNo	For the CDRs that are generated not for re-rating, the default value of this field is **0**.
209	0	SourceSubSequence	Indicates the sequence number when the bill is split is **0**.
210	-	AcctBalChgList	Indicates the change value of balance change is **-**.
211	0	SubscriberType	Indicates the subscriber type is **0**.
212	0	OriginCdrFilename	Indicates the name of a rated CDR is **0**.
213	212345678	CurCDRLineNo	Indicates the number of the line is **212345678** where the original CDR is located.
214	0	DDSDiscountOff	Indicates discount of dynamic tariff.

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SN	Value	Field Name	Description
215	DDSLocationNumber	Char[64]	Indicates to record the information of Location Number in DDS feature.
216	DTDiscount	String	Indicates dynamic discount.
217	DTCellId	String	Indicates the Cell ID considered for DT discount.

4.2 SCP-Generated CDR Instance-w_abr CDR

The section describes the w_abr CDR with an instance.

Service Application Scenario

Assume that John in country A calls Jack in the same country. The mobile phone number of John is 8765432. The mobile phone number of Jack is 7654321. The country code of country A is 92. The international number prefix of country A is 00. The number John calls is 00927654321. Jack answers the call at 2007-08-21-15: 00: 31. The call duration is 4 seconds.

CDR Details

- Original CDR record (CDR record for the calling party)

  91|1|009261167556|00927654321|||150025|0|00927654321|00927654321|1|10|00675756||009218765019||||0|17|||0|4605075601360|16776758464|009213900777|32|20070821150031|4|0|4|-1|0|100||01|02|00|

- CDR field description

Table 4-2 Format of w_abr CDR

SN	Value	Field Name	Description
1	91	ServiceKey	Indicates the service key. The value is **91** in this CDR.
2	1	SubCosID	Indicates the sub brand ID of a product.
3	0092711234567	Calling-Party-Address	Indicates the calling number. Fixed-line telephone number with an area code, mobile phone number without an area code (international number form). The country code of John is **92** and the mobile phone number is 711234567. Thus, the calling number is **0092711234567**.
4	00927654321	Called-Party-Address	Indicates the called number. The country code of Jack is **92** and the mobile phone number is 7654321. Thus, the called

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SN	Value	Field Name	Description
			number is **00927654321**.
5	-	Calling-Party-NP	Indicates the number portability prefix of the calling party. It is the route prefix of the number transfers from one network to another. The value of this field is null.
6	-	Called-Party-NP	Indicates the number portability prefix of the called party. It is the route prefix of the number transfers from one network to another. The value of this field is null.
7	150025	SSP-Time	Indicates the time reported by the SSP.
8	0	MF-Indicator	Indicates the call forwarding indicator.
9	00927654321	Real-Called-Number	Indicates the actual called number. The number John calls is **00927654321**, which is the actual called number.
10	00927654321	Connect-Called-Number	Indicates the called number for connection. It is the same as the actual called number **00927654321**.
11	1	Charge-Flow-Type	Indicates the charge process type. The options are as follows: • 0: the calling process • 1: the called process • 3: the forwarding process
12	10	Calling-Partys-Category	Indicates the calling party type indicating whether the call is from a carrier, a public number, or an ordinary subscriber. The value of the field **Calling-Partys-Category** depends on the value of the variable **CAP_InitialDP_ callingPartysCategory**.
13	00675756	Calling-Vlr-Number	Indicates the VLR number of the visitor location of the calling party. If the number exists in the IDP message, enter that; otherwise, the value is null.
14	-	Calling-CellID	Indicates the cell ID of the calling party.
15	009218765019	Called-Vlr-Number	Indicates the VLR number of the visitor location of the called party.
16	-	Called-CellID	Indicates the cell ID of the called party.
17	-	Original-Called-Party-Id	Indicates the original called number. In the charging process for calling party, the field is set to null because the MSC does not report the original called number.

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SN	Value	Field Name	Description
18	-	Additional-Calling-Party-Number	Indicates the additional calling number. In the calling process, the MSC does not report the number. Thus, the value is null.
19	0	Bearer-Capability	Indicates the bear-capability of the device that a subscriber connects to.
20	17	EventType-BCSM	Indicates an event report.
21	-	Redirecting-Party-Id	Indicates the ID of the redirecting party. It is used in the forwarding process. Thus, the value is null.
22	-	Redirection-Information	Indicates the redirection information. It is used in the forwarding process. Thus, the value is null.
23	0	Redirection-Reason	Indicates the forwarding reason. The call is not redirected. Thus, the field is set to **0**.
24	4605075601360	UE-IMSI	Indicates the IMSI of the calling party or called party.
25	16776758464	CallReferenceNumber	Indicates the reference number of a call. It is used to uniquely identify a call.
26	009213900777	MSC-Address	Indicates the MSC address.
27	32	Time-Zone	Indicates the time zone where a subscriber is.
28	20070821150031	CallBeginTime	Indicates the start time of the calling beginning. The call starts at **2007-08-21-15: 00: 31**.
29	4	CallDuration	Indicates the call duration. The call lasts for 4 seconds.
30	0	SuccesssACDuration	Indicates the duration of which the fee has been deducted.
31	4	FailACDuration	Indicates the AC duration during which an error occurs. The SCP and CBP interact abnormally when the call lasts for 4 seconds.
32	-1	VPMNCallFlag	Indicates the flag of calls in a VPN group.
33	0	CBEResultCode	Indicates the error code did not return by the CBE.
34	100	DDSDiscount	Indicates the discount of the MTN zone is **100%**.
35	-	DiameterSessionID	Indicates that the diameter session ID is null.

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SN	Value	Field Name	Description
36	01	NPPrefix	Indicates the NP prefix of a called number is **01**.
37	02	CalledHomeNetWork	Indicates the home network code of a called number after the NP is **02**.
38	00	OpposeTenantID	Reserved.

4.3 BMP-Generated CDR Instance fns CDR

The section describes the instance of a Generated fns CDR.

Service Application Scenario

This topic assumes that Bill asks an operator in a customer service center to add a familiarity number (FN) for him. When the FN is added, the online charging system (OCS) generates a fns CDR.

CDR Details

- CDR content
  8125101068|205|25481250003||0|20110831095217|6
- CDR field description

Table 4-3 Format of fns CDR

SN	Value	Field Name	Description
1	8125101068	MSISDN	Indicates the mobile phone number of the subscriber.
2	205	SubCosID	Indicates the main product of the subscriber.
3	25481250003	NewFnNumber	Indicates the new FN.
4	-	OldFnNumber	Indicates the old FN.
5	0	Fee	Indicates the fee for changing an FN.
6	20110831095217	OperTime	Indicates the date and time of the operation.
7	6	FreeTimes	Indicates the remaining number of times to modify or add an FN free of charge.

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